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SCHOLZ et al.(10) **Pub. No.: US 2025/0277199 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **NOVEL CRISPR-CAS NUCLEASES FROM
METAGENOMES**(71) Applicant: **Brain Biotech AG**, Zwingenberg (DE)(72) Inventors: **Paul SCHOLZ**, Köln (DE); **Christian
ZUREK**, Lorsch (DE); **Robin BACK**,
Bensheim (DE)(73) Assignee: **Brain Biotech AG**, Zwingenberg (DE)(21) Appl. No.: **18/548,429**(22) PCT Filed: **Mar. 2, 2022**(86) PCT No.: **PCT/EP2022/055257**

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(57)

ABSTRACT

The present invention relates to a nucleic acid molecule encoding an RNA-guided DNA endonuclease, which is (a) a nucleic acid molecule encoding the RNA-guided DNA endonuclease comprising or consisting of the amino acid sequence of any one of SEQ ID NOs 9, 1 to 5, 7, 8 and 10 to 15; (b) a nucleic acid molecule comprising or consisting of the nucleotide sequence of any one of SEQ ID NOs 24, 16 to 20, 22, 23 and 25 to 30; (c) a nucleic acid molecule encoding a RNA-guided DNA endonuclease the amino acid sequence of which is at least 70% identical to the amino acid sequence of (a), preferably at least 80% identical, more preferably at least 90% identical, and most preferred at least 95% identical; (d) a nucleic acid molecule comprising or consisting of a nucleotide sequence which is at least 70% identical to the nucleotide sequence of (b), preferably at least 80% identical, more preferably at least 90% identical, and most preferred at least 95% identical; (e) a nucleic acid molecule which is degenerate with respect to the nucleic acid molecule of (d); or (f) a nucleic acid molecule corresponding to the nucleic acid molecule of any one of (a) to (d) wherein T is replaced by U.

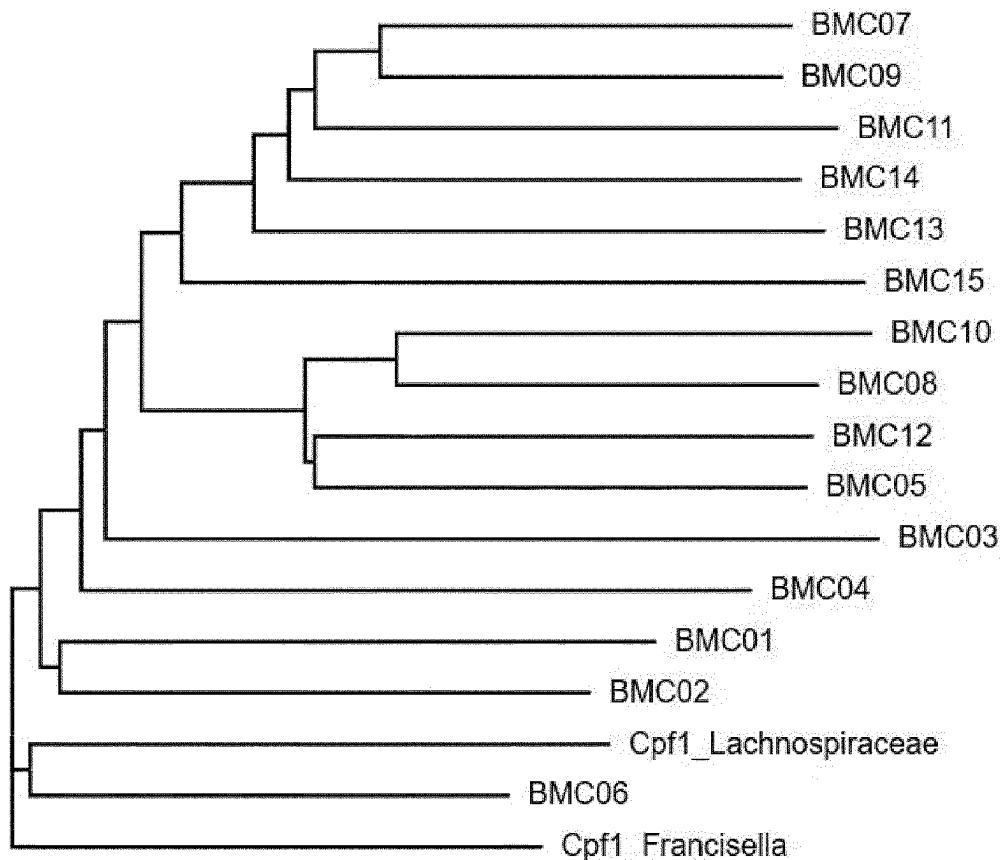
Specification includes a Sequence Listing.

Figure 1

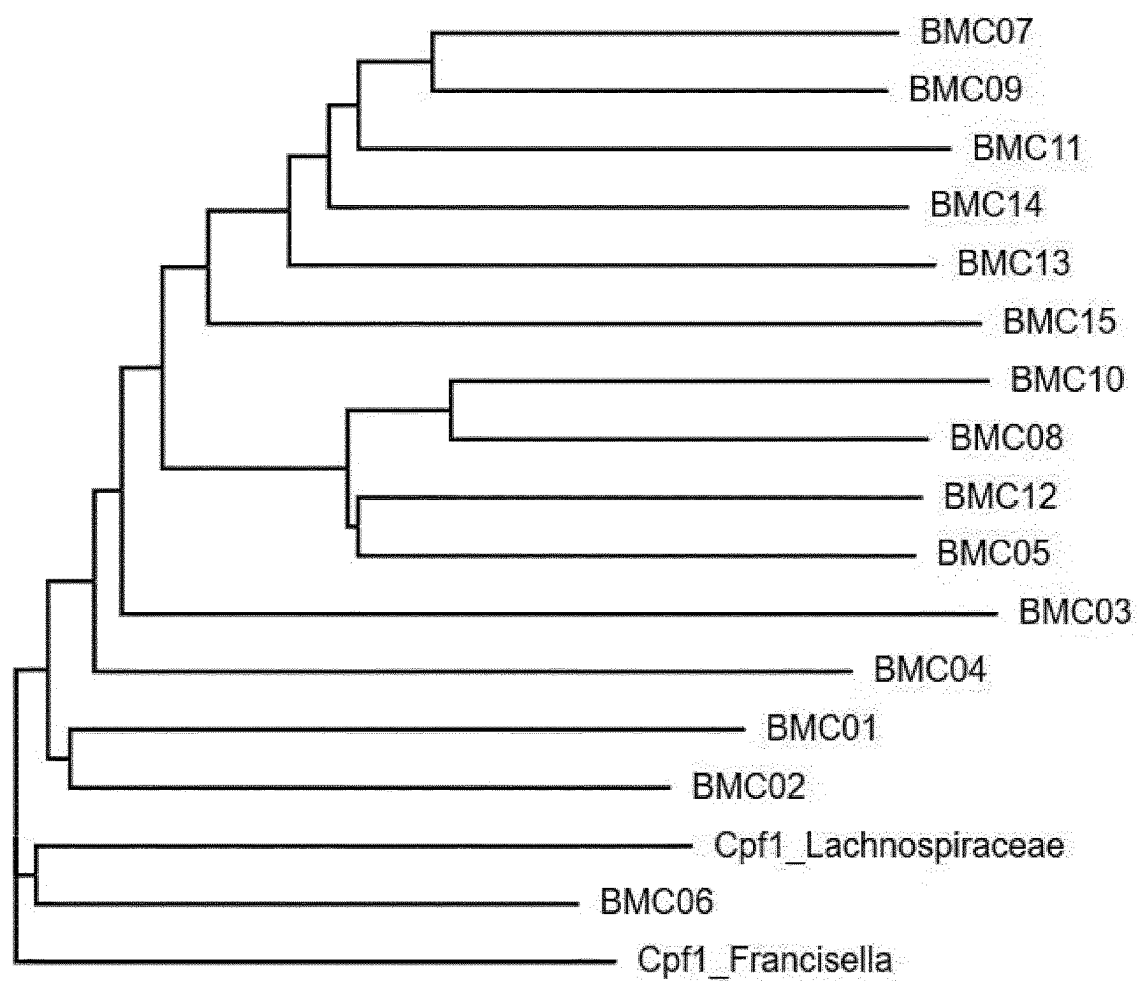


Figure 2

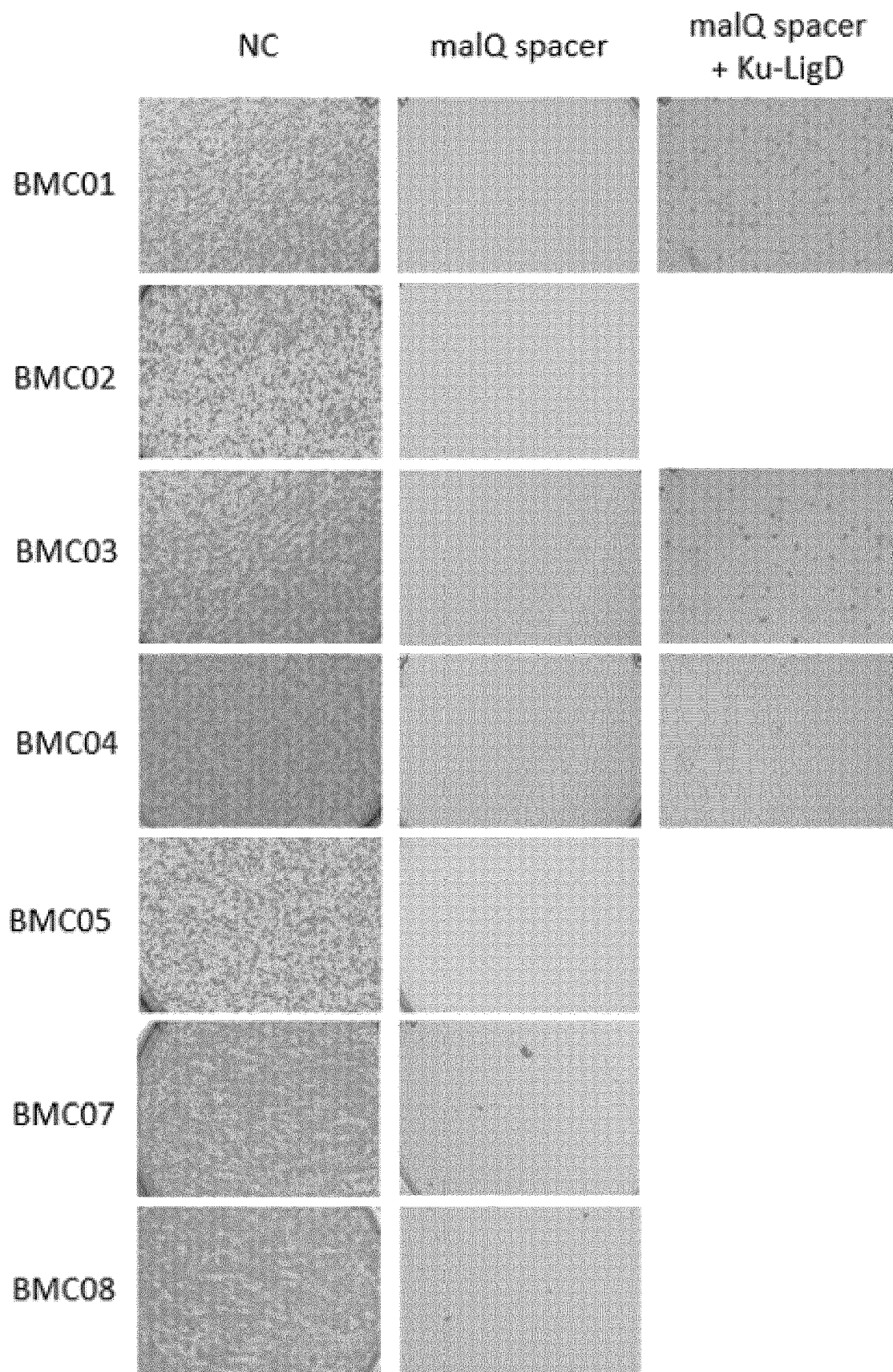


Figure 3

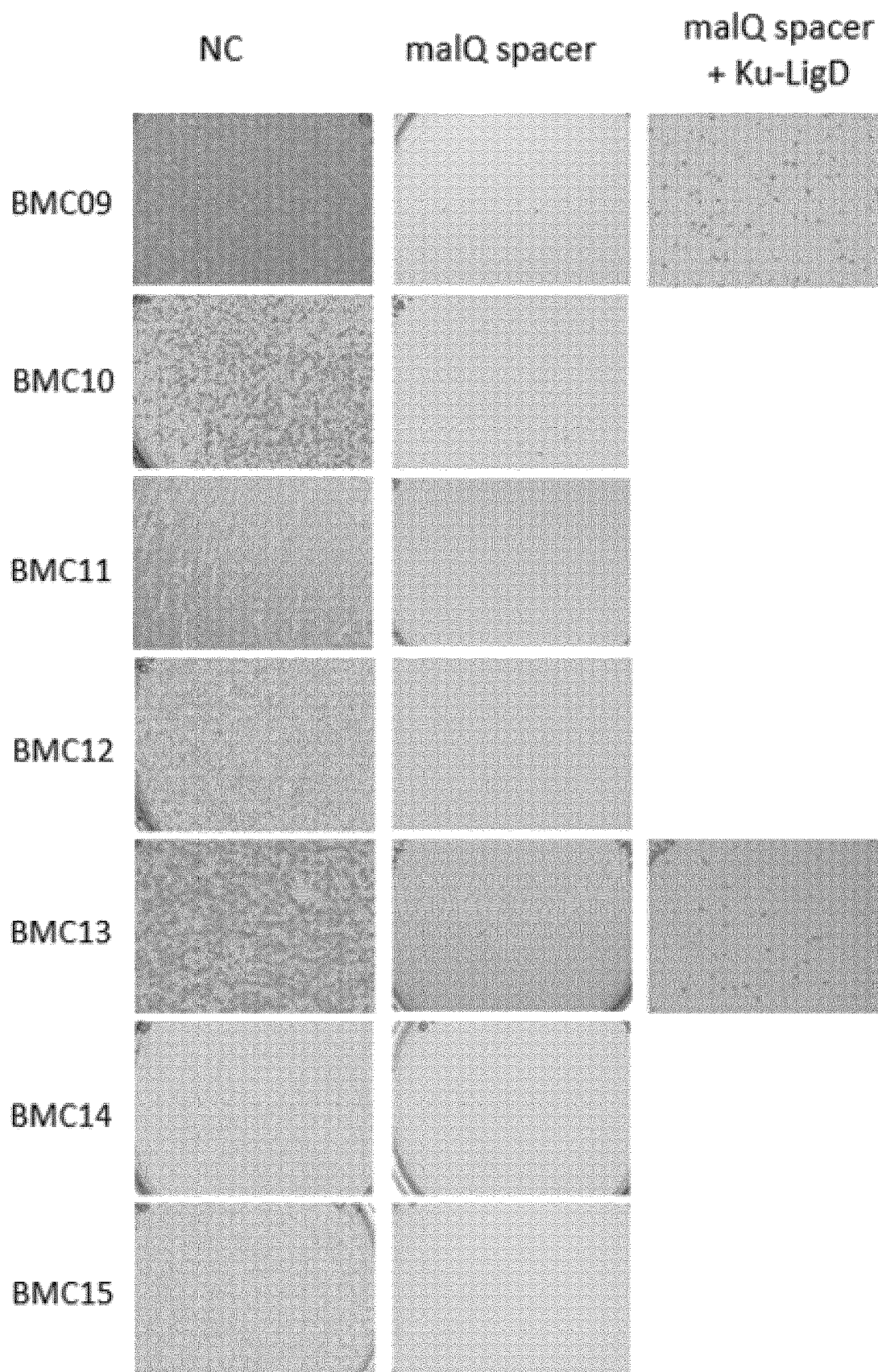
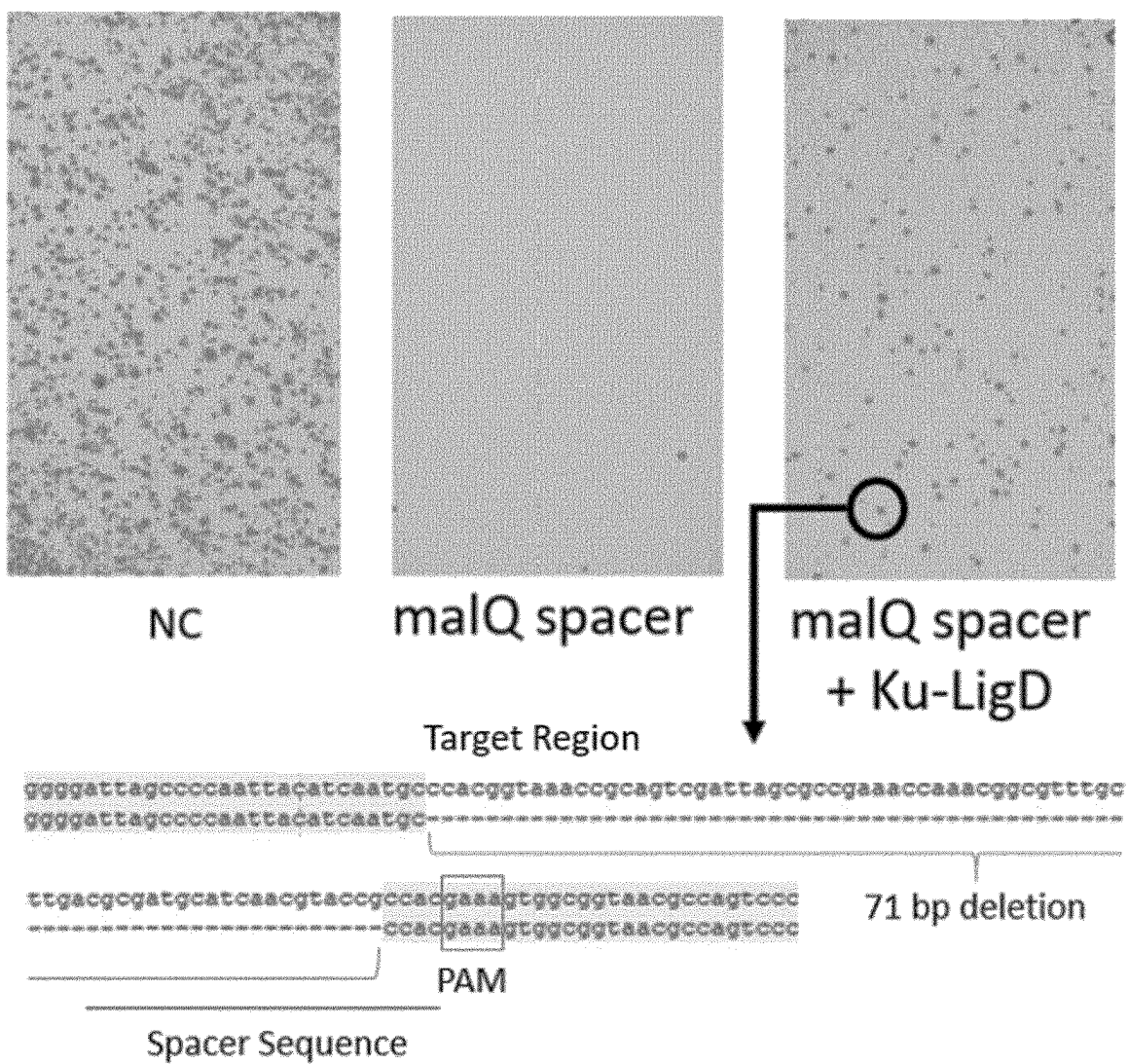


Figure 4



NOVEL CRISPR-CAS NUCLEASES FROM METAGENOMES

[0001] The present invention relates to a nucleic acid molecule encoding an RNA-guided DNA endonuclease, which is (a) a nucleic acid molecule encoding the RNA-guided DNA endonuclease comprising or consisting of the amino acid sequence of any one of SEQ ID NOs 9, 1 to 5, 7, 8 and 10 to 15; (b) a nucleic acid molecule comprising or consisting of the nucleotide sequence of any one of SEQ ID NOs 24, 16 to 20, 22, 23 and 25 to 30; (c) a nucleic acid molecule encoding a RNA-guided DNA endonuclease the amino acid sequence of which is at least 70% identical to the amino acid sequence of (a), preferably at least 80% identical, more preferably at least 90% identical, and most preferred at least 95% identical; (d) a nucleic acid molecule comprising or consisting of a nucleotide sequence which is at least 70% identical to the nucleotide sequence of (b), preferably at least 80% identical, more preferably at least 90% identical, and most preferred at least 95% identical; (e) a nucleic acid molecule which is degenerate with respect to the nucleic acid molecule of (d); or (f) a nucleic acid molecule corresponding to the nucleic acid molecule of any one of (a) to (d) wherein T is replaced by U.

[0002] In this specification, a number of documents including patent applications and manufacturer's manuals are cited. The disclosure of these documents, while not considered relevant for the patentability of this invention, is herewith incorporated by reference in its entirety. More specifically, all referenced documents are incorporated by reference to the same extent as if each individual document was specifically and individually indicated to be incorporated by reference.

[0003] CRISPR-Cas systems are widespread adaptive immunity systems of prokaryotes against invading foreign nucleic acids. So far, more than 30 different CRISPR-Cas systems have been identified that differ in their loci architecture, number, and identity of their genes encoding for the Cas (CRISPR-associated) proteins.

[0004] The typical signature of the CRISPR systems in prokaryotic genomes is the presence of short (30-45 bp) repetitive sequences (repeats) that are intervened by variable sequences (spacers) of similar lengths. The Cas proteins are located either upstream or downstream of the repeat-spacer cluster. According to their gene composition and mechanistic differences, the subtypes are classified into two CRISPR classes (Class 1 and 2). One of their major differences is that Class 1 CRISPR systems need a complex of multiple Cas proteins to degrade DNA, whereas Class 2 Cas proteins are single, large multidomain nucleases. The sequence-specificity of the Class 2 Cas proteins can simply be modified by synthetic CRISPR RNAs (crRNAs) in order to introduce targeted double-stranded DNA breaks. Most prominent members of such Class 2 Cas proteins are Cas9, Cpf1 (Cas12a) and Cms1, which are harnessed for genome-editing and successfully applied in many eukaryotic organisms including fungi, plants and mammalian cells. Whereas Cas9 and its orthologs are Class2 type II CRISPR nucleases, Cpf1 (WO2016/205711 BROAD Inst.; WO2017/141173 Benson Hill) and Cms1 (WO2019/030695 Benson Hill) belong to Class 2 type V nucleases. Cms1 and Cpf1 CRISPR nucleases are a class of CRISPR nucleases that have certain desirable properties compared to other CRISPR nucleases such as type II nucleases. For instance, in contrast to Cas9 nucleases Cms1 and Cpf1 do not require a trans-activating

crRNA (tracrRNA), which is partially complementary to the precursor crRNA (pre-crRNA) (Deltcheva et al. (2011), *Nature*, 471(7340):602-607). Base-pairing of tracrRNA and pre-crRNA forms a Cas9-bound RNA:RNA duplex, which gets processed by RNase III and other unidentified nucleases. This mature tracrRNA:crRNA duplex mediates the target DNA recognition and cleavage by Cas9. In contrast, type V nucleases can process the pre-crRNA without the need for tracrRNA or cellular nucleases (like RNase III), which significantly simplifies the application of type V nucleases for (multiplex) genome editing.

[0005] Several new Class 2 proteins, like C2c1 (Cas12b), C2c2 (Cas13a) and C2c3 (Cas12c) have been identified in the genomes of cultivated bacteria or public available metagenomics datasets, e.g. gut metagenome (Shmakov et al. (2015), *Mol Cell*, 60(3):385-97). According to the recent classification of CRISPR-Cas systems, Class 2 comprises 3 types and 17 subtypes (Makarova et al. (2020), *Nat Rev Microbiol*, 18(2):67-83).

[0006] Moreover, in a recent publication two new Class 2 proteins were discovered (CasX (Cas12a) and CasY (Cas12d)) in uncultivated prokaryotes by metagenome sequencing (Burststein et al. (2017), *Nature*, 542:237-241), indicating the presence of untapped Cas proteins from organisms which are not cultivated and/or identified yet.

[0007] A promising approach for the discovery of novel Class 2 Cas proteins is therefore to access the metagenomics resource by next generation sequencing selected environmental DNA (e.g. 1 cm³ forest soil contains ~2.5×10¹⁰ bp DNA or ~20 million genes) and computational identification of CRISPR-Cas systems (for example, Lei and Sun (2016), *Bioinformatics*, 32(17):1520-i528).

[0008] As discussed, the known CRISPR-Cas systems display certain differences as regards their mode of action. These molecular differences not only enlarge the possibilities of using the CRISPR-Cas system for genome editing in a broad range of different genetic backgrounds but also to circumvent issues of particular Cas nucleases when applied in certain organisms, e.g. pre-existing immune response to Cas9 in humans (Charlesworth et al. (2019), *Nat Med*, 25(2):249-254). Therefore, the identification of Cas nucleases from bacterial species with less direct contact to higher eukaryotes is of particular importance. It can be assumed that a number of yet undiscovered CRISPR-Cas systems exist with yet unknown characteristics, in particular in metagenome resources. Hence, although already several different CRISPR-Cas systems are known from the prior art there is an ongoing need to identify further CRISPR-Cas systems and in particular the RNA-guided DNA endonucleases of these systems. This need is addressed by the present invention

[0009] Accordingly, the present invention relates in first aspect to a nucleic acid molecule encoding an RNA-guided DNA endonuclease, which is (a) a nucleic acid molecule encoding the RNA-guided DNA endonuclease comprising or consisting of the amino acid sequence of any one of SEQ ID NOs 1-15, preferably of SEQ ID NOs 9, 1 to 5, 7, 8 and 10 to 15 or of SEQ ID NO: 6; (b) a nucleic acid molecule comprising or consisting of the nucleotide sequence of any one of SEQ ID NOs 16-30, preferably of SEQ ID NOs 9, 1 to 5, 7, 8 and 10 to 15 or of SEQ ID NO: 22; (c) a nucleic acid molecule encoding a RNA-guided DNA endonuclease the amino acid sequence of which is at least 70% identical to the amino acid sequence of (a), preferably at least 80%

identical, more preferably at least 90% identical, and most preferred at least 95% identical; (d) a nucleic acid molecule comprising or consisting of a nucleotide sequence which is at least 70% identical to the nucleotide sequence of (b), preferably at least 80% identical, more preferably at least 90% identical, and most preferred at least 95% identical; (e) a nucleic acid molecule which is degenerate with respect to the nucleic acid molecule of (d); or (f) a nucleic acid molecule corresponding to the nucleic acid molecule of any one of (a) to (d) wherein T is replaced by U.

[0010] SEQ ID NOs 1-15 are the amino acid sequences of the novel CRISPR-Cas endonucleases BMC01-BMC15, respectively, wherein BMC is an abbreviation of BRAIN Metagenome Cas. The novel CRISPR-Cas endonucleases BMC01-BMC15 are encoded by the nucleotide sequences of SEQ ID NOs 16-30, respectively. An unrooted tree of the amino acid sequences of BMC01-BMC15 is shown in FIG. 1 of the application. It can be seen that BMC01 to BMC05 and BMC07 to BMC15 are closer related to each other than to any prior art CRISPR-Cas endonuclease the application is aware of and are closer related to each other than to BMC06. For this reason SEQ ID NOs 1-15 are preferably SEQ ID NOs 9, 1 to 5, 7, 8 and 10 to 15 or is SEQ ID NO: 6, and SEQ ID NOs 16 to 30 are preferably SEQ ID NOs 24, 16 to 20, 22, 23 and 25 to 30 or is SEQ ID NO: 22. Among SEQ ID NOs 9, 1 to 5, 7, 8 and 10 to 15 SEQ ID NO: 9 is mentioned first because the corresponding novel CRISPR-Cas endonuclease BMC09 has been most extensively tested in the appended examples. For this reason the amino acid sequence of SEQ ID NO: 9 as well as the corresponding nucleotide sequence of SEQ ID NO: 24 are most preferred.

[0011] In accordance with the present invention the term “nucleic acid molecule” defines a linear molecular chain of nucleotides. The nucleic acid molecules according to the present invention consist of at least 2327 nucleotides. The group of molecules designated herein as “nucleic acid molecules” also comprises complete genes. The term “nucleic acid molecule” is interchangeably used herein with the term “polynucleotide”.

[0012] The term “nucleic acid molecule” in accordance with the present invention includes DNA, such as cDNA or double or single stranded genomic DNA and RNA. In this regard, “DNA” (deoxyribonucleic acid) means any chain or sequence of the chemical building blocks adenine (A), guanine (G), cytosine (C) and thymine (T), called nucleotide bases, that are linked together on a deoxyribose sugar backbone. DNA can have one strand of nucleotide bases, or two complementary strands which may form a double helix structure. “RNA” (ribonucleic acid) means any chain or sequence of the chemical building blocks adenine (A), guanine (G), cytosine (C) and uracil (U), called nucleotide bases, that are linked together on a ribose sugar backbone. RNA typically has one strand of nucleotide bases. Included are also single- and double-stranded hybrid molecules, i.e., DNA-DNA, DNA-RNA and RNA-RNA. The nucleic acid molecule may also be modified by many means known in the art. Non-limiting examples of such modifications include methylation, “caps”, substitution of one or more of the naturally occurring nucleotides with an analog, and inter-nucleotide modifications such as, for example, those with uncharged linkages (e.g., methyl phosphonates, phosphotriesters, phosphoramidates, carbamates, etc.) and with charged linkages (e.g., phosphorothioates, phosphorodithioates, etc.). Polynucleotides may contain one or more addi-

tional covalently linked moieties, such as, for example, proteins (e.g., nucleases, toxins, antibodies, signal peptides, poly-L-lysine, etc.), intercalators (e.g., acridine, psoralen, etc.), chelators (e.g., metals, radioactive metals, iron, oxidative metals, etc.), and alkylators. The polynucleotides may be derivatized by formation of a methyl or ethyl phosphotriester or an alkyl phosphoramidate linkage. Further included are nucleic acid mimicking molecules known in the art such as synthetic or semi-synthetic derivatives of DNA or RNA and mixed polymers. Such nucleic acid mimicking molecules or nucleic acid derivatives according to the invention include phosphorothioate nucleic acid, phosphoramidate nucleic acid, 2'-O-methoxyethyl ribonucleic acid, morpholino nucleic acid, hexitol nucleic acid (HNA), peptide nucleic acid (PNA) and locked nucleic acid (LNA) (see Braasch and Corey, Chem Biol 2001, 8:1). LNA is an RNA derivative in which the ribose ring is constrained by a methylene linkage between the 2'-oxygen and the 4'-carbon. Also included are nucleic acids containing modified bases, for example thio-uracil, thio-guanine and fluoro-uracil. A nucleic acid molecule typically carries genetic information, including the information used by cellular machinery to make proteins and/or polypeptides. The nucleic acid molecule of the invention may additionally comprise promoters, enhancers, response elements, signal sequences, polyadenylation sequences, introns, 5'- and 3'-non-coding regions, and the like.

[0013] The term “polypeptide” as used herein interchangeably with the term “protein” describes linear molecular chains of amino acids, including single chain proteins or their fragments. The polypeptides/proteins according to the present invention contain at least 775 amino acids. Polypeptides may further form oligomers consisting of at least two identical or different molecules. The corresponding higher order structures of such multimers are, correspondingly, termed homo- or heterodimers, homo- or heterotrimers etc. The polypeptides of the invention may form heteromultimers or homomultimers, such as heterodimers or homodimers. Furthermore, peptidomimetics of such proteins/polypeptides where amino acid(s) and/or peptide bond(s) have been replaced by functional analogues are also encompassed by the invention. Such functional analogues include all known amino acids other than the 20 gene-encoded amino acids, such as selenocysteine. The terms “polypeptide” and “protein” also refer to naturally modified polypeptides and proteins where the modification is affected e.g. by glycosylation, acetylation, phosphorylation, ubiquitinylation and similar modifications which are well known in the art.

[0014] The term “RNA-guided DNA endonuclease” or “CRISPR (-Cas) endonuclease” describes an enzyme having the capability of cleaving the phosphodiester bond within a deoxyribonucleotide (DNA) strand thereby producing a double-strand break (DSB). BMC01-15 are classified as novel type V class 2 CRISPR nucleases which are known to introduce a staggered cut with a 5' overhang. Hence, an RNA-guided DNA endonuclease comprises an endonuclease domain, in particular a RuvC domain. The RuvC domains of BMC01-15 is believed to comprise three split RuvC motifs. A RNA-guided DNA endonuclease also comprises a domain being capable of binding to a crRNA, also known as guide RNA (gRNA; also being designated DNA-targeting RNA herein).

[0015] The cleavage site of the RNA-guided DNA endonuclease is guided by a guide RNA. The gRNA confers the target sequence specificity to the RNA-guided DNA endonuclease. Such gRNAs are non-coding short RNA sequences which binds to the complementary target DNA sequences. The gRNA first binds to the RNA-guided DNA endonuclease by a binding domain that can interact with the RNA-guided DNA endonuclease. The binding domain that can interact with the RNA-guided DNA endonuclease typically comprises a region with a stem-loop structure. This stem-loop preferably comprises the sequence NATTTC-TACTNTTGTAGAT (SEQ ID NO: 31), wherein N denotes that any base may be present at this location. The stem-loop most preferably comprises the stem loop direct repeat sequence of BMC01-BMC15 (SEQ ID NO: 32), respectively, but in the form of RNA (i.e. wherein T is replaced by U). The gRNA sequence guides the complex (known as CRISPR ribonucleoprotein (RNP) complex of the gRNA and the RNA-guided DNA endonuclease) via pairing to a specific location on a DNA strand, where the RNA-guided DNA endonuclease performs its endonuclease activity by cutting the DNA strand at the target site. The genomic target site of the gRNA can be any about 20 (typically 17 to 26) nucleotide DNA sequence, provided it meets two conditions: (i) The sequence is unique compared to the rest of the genome, and (ii) the target is present immediately adjacent to a Protospacer Adjacent Motif (PAM).

[0016] The cleavage site of the RNA-guided DNA endonuclease is, thus, furthermore defined by a PAM. The PAM is a short DNA sequence (usually 2-6 base pairs in length) that follows the DNA region targeted for cleavage by the CRISPR system. The exact sequence depends on which CRISPR endonuclease is used. CRISPR endonucleases and their respective PAM sequences are known in the art (see <https://www.addgene.org/crispr/guide/#pam-table>). For instance, the PAM being recognized by the first identified RNA-guided DNA endonuclease Cas9 is 5'-NGG-3' (where "N" can be any nucleotide base). The PAM is required for a RNA-guided DNA endonuclease to cut. In Cas9 it is found about 2-6 nucleotides downstream of the DNA sequence targeted by the guide RNA and 3-6 nucleotides downstream from the cut site. In type V systems (including BMC01-BMC15) the PAM is located upstream of both, the target sequence and the cleavage site. The complex of the RNA-guided DNA endonuclease and the guide RNA comprises a so-called PAM interacting domain (Andres et al. (2014), Nature, 513(7519):569-573). Hence, the genomic locations that can be targeted for editing by an RNA-guided DNA endonuclease are limited by the presence and locations of the nuclease-specific PAM sequence. As BMC01-BMC15 belong to the group of Type V Class 2 CRISPR nucleases a T rich PAM site is predicted (TTTN, wherein N can be any nucleotide of A, T, C and G).

[0017] The term "percent (%) sequence identity" describes the number of matches ("hits") of identical nucleotides/amino acids of two or more aligned nucleic acid or amino acid sequences as compared to the number of nucleotides or amino acid residues making up the overall length of the template nucleic acid or amino acid sequences. In other terms, using an alignment, for two or more sequences or subsequences the percentage of amino acid residues or nucleotides that are the same (e.g. 70% identity) may be determined, when the (sub)sequences are compared and aligned for maximum correspondence over a window of

comparison, or over a designated region as measured using a sequence comparison algorithm as known in the art, or when manually aligned and visually inspected. This definition also applies to the complement of any sequence to be aligned.

[0018] Amino acid sequence as well as nucleotide sequence analysis and alignments in connection with the present invention are preferably carried out using the NCBI BLAST algorithm (Stephen F. Altschul, Thomas L. Madden, Alejandro A. Schäffer, Jinghui Zhang, Zheng Zhang, Webb Miller, and David J. Lipman (1997), "Gapped BLAST and PSI-BLAST: a new generation of protein database search programs", Nucleic Acids Res. 25:3389-3402). The skilled person is aware of additional suitable programs to align nucleic acid sequences.

[0019] As defined herein above, amino acid sequence and nucleotide sequence identities of at least 70%, at least 80%, at least 90% and at least 95% are envisaged by the invention. Furthermore, are envisaged with increasing preference amino acid sequence identities of at least 99.5%, at least 99.8%, and 99.9% identity by the invention.

[0020] With respect to these amino acid sequences and the amino acid sequences being encoded by these nucleotide sequences it is preferred that they maintain or essentially maintain the RNA-guided DNA endonuclease activity of SEQ ID NOs 1-16 of the invention. Hence, what is maintained or essentially maintained is the capability to bind to the gRNA to form a complex being capable of binding to the DNA target site of interest, where the endonuclease activity induces a DSB.

[0021] The term "degenerate" designates the degeneracy of the genetic code. As is well known, the codons encoding one amino acid may differ in any of their three positions; however, more often than not, this difference is in the second or third position. For instance, the amino acid glutamic acid is specified by GAA and GAG codons (difference in the third position); the amino acid leucine is specified by UUA, UUG, CUU, CUC, CUA, CUG codons (difference in the first or third position); and the amino acid serine is specified by UCA, UCG, UCC, UCU, AGU, AGC (difference in the first, second, or third position).

[0022] Just as the type V CRISPR endonucleases Cms1 and Cpf1, BMC01-BMC15 do not require a trans-activating crRNA (tracrRNA). Moreover, the BMC01-BMC15 containing CRISPR systems identified in the present application contain CRISPR repeat sequences with an RNA stem loop at 3'-ends of each repeat that is conserved in crRNAs of Cpf1 and Cms1 protein families and the "nearest neighbour" of BMC01-BMC15 among the known CRISPR-Cas endonucleases are members of the Cpf1 polypeptides. For this reason BMC01-BMC15 can be classified as novel Class 2 type V nucleases with overall no significant sequence identity to the known collection of Class 1 and Class 2 CRISPR-Cas endonucleases and with overall low sequence identity to individual Cpf1-type endonucleases.

[0023] Since BMC01-BMC15 are novel CRISPR-Cas endonucleases which are significantly distinct from the known collection of CRISPR-Cas endonucleases, BMC01-BMC15 expand the known collection of CRISPR-Cas endonucleases applicable for genome editing, gene regulation and nucleic acid enrichment/purification in different biotechnological and pharmaceutical sectors thereby advantageously expanding the options for molecular genome editing.

[0024] BMC01-BMC15 are significantly distinct from the known collection of known CRISPR-Cas endonucleases in at least two aspects.

[0025] First, they are structurally distinct. For instance, SEQ ID NO: 20 in WO 2021/154866 only shares a sequence similarity of 57.1% with CMS09 (SEQ ID NO: 9) and a sequence identity of 55.26% with the hypothetical protein US54 C0016G0015 from a *Candidatus roizmanbacteria* bacterium. Also the BMC endonucleases are structurally significantly distinct from known CRISPR-Cas endonucleases. The BMC endonucleases were found by the innovative screening approach as illustrated in Example 1, wherein metagenomics resources in terabase scale were accessed by Next Generation Sequencing and Candidata Phyla Radiation (CPR) species containing a nearly untouched space of novel sequences were enriched. It was surprising that despite the structurally significantly distinct sequences as compared to the known CRISPR-Cas endonucleases, BMC01-BMC15 still work very well for genome editing, which is demonstrated by Examples 3 to 5.

[0026] Second, BMC01-BMC15 were isolated from environmental samples, in particular from aquatic, sediment and soil habitats. On the other hand, most known CRISPR-Cas endonucleases were isolated from pathogenic bacteria. For instance, Cas9 was found in *Streptococcus pyogenes* M1, Cas12a (Cpf1) from *Prevotella* and *Francisella* 1, and Cas13a (C2c2) from *Leptotrichia shahii*. For this reason BMC01-BMC15 will be less immunogenic than most known CRISPR-Cas endonucleases when applied for genome editing in mammals, preferably in humans in particular for therapeutic applications.

[0027] In accordance with a preferred embodiment of the first aspect of the invention the nucleic acid molecule is operably linked to a promoter that is native or heterologous to the nucleic acid molecule.

[0028] A promoter is a region of DNA that leads to initiation of transcription of a particular gene. Promoters are generally located near the transcription start sites of genes, upstream on the DNA (towards the 5' region of the sense strand). Promoters are typically 100-1000 base pairs long. For transcription to take place, the enzyme that synthesizes RNA, known as RNA polymerase, must attach to the DNA near a gene. Promoters contain specific DNA sequences such as response elements that provide a secure initial binding site for RNA polymerase and for proteins called transcription factors that recruit RNA polymerase. Hence, the binding of the RNA polymerase and transcription factors to the promoter site ensures the transcription of the gene.

[0029] In this connection the term "operably linked" defines that the promoter is linked to the gene of the same DNA strand, such that upon binding of the RNA polymerase and transcription factors the transcription of the gene is initiated. Generally each gene is operably linked in its natural environment of the genome of a living organism to a promoter. This promoter is designated "natural promoter" or "wild-type promoter" herein. A heterologous promoter is distinct from the natural promoter or wild-type promoter. Hence, a nucleic acid molecule which is operably linked to a promoter that is heterologous to the nucleic acid molecule does not occur in nature.

[0030] Heterologous promoters that can be used to express a desired gene are known in the art and can, for example, be obtained from the EPD (eukaryotic promoter database) or

EDPnew (<https://epd.epfl.ch/index.php>). In this database eukaryotic promoters including animal, plant and yeast promoters can be found.

[0031] The promoter can, for example, be a constitutively active, inducible, tissue-specific, or developmental stage-specific promoter. By using such promoter the desired timing and site of expression can be regulated.

[0032] The AOX1 or GAL1 promoter in yeast or the CMV-(Cytomegalovirus), SV40-, RSV-promoter (Rous sarcoma virus), chicken beta-actin promoters, CAG-promoter (a combination of chicken beta-actin promoter and cytomegalovirus immediate-early enhancer), the *gai10* promoter, human elongation factor 1 α -promoter, CaM-kinase promoter, and the Autographa californica multiple nuclear polyhedrosis virus (AcMNPV) polyhedral promoter are examples of constitutively active promoters.

[0033] Examples of inducible promoters are the Adhl promoter which is inducible by hypoxia or cold stress, the Hsp70 promoter which is inducible by heat stress, the PPK promoter and the pepcarboxylase promoter which are both inducible by light. Also useful are promoters which are chemically inducible, such as the *ln2-2* promoter which is safener induced (U.S. Pat. No. 5,364,780), the ERE promoter which is estrogen induced, and the *Axigl* promoter which is auxin induced and tapetum specific but also active in callus (WO03060123 A1).

[0034] A tissue-specific promoter is a promoter that initiates transcription only in certain tissues. A developmental stage-specific promoter is a promoter that initiates transcription only in a certain developmental stage.

[0035] In accordance with a further preferred embodiment of the first aspect of the invention the nucleic acid molecule is linked to a nucleic acid sequence encoding a nuclear localization signal (NLS).

[0036] Further details on NLS will be provided herein below.

[0037] In accordance with another preferred embodiment of the first aspect of the invention said nucleic acid molecule is codon-optimized for expression in a eukaryotic cell, preferably a plant cell or an animal cell.

[0038] As discussed, BMC01-BMC15 were isolated from bacterial metagenome samples. Hence, if BMC01-BMC15 are to be expressed in eukaryotic cells they are expressed in a heterologous host cell or organism.

[0039] The gene encoding BMC01-BMC15 polypeptide, respectively, can be codon-optimized for expression in the target cell, and can optionally include a sequence encoding an NLS and/or a peptide tag, such a purification tag. Further details on the tag will be provided herein below.

[0040] Codon optimization is a process used to improve gene expression and increase the translational efficiency of a gene of interest by accommodating codon bias of the host cell. A "codon-optimized gene" is therefore a gene having its frequency of codon usage designed to mimic the frequency of preferred codon usage of the host cell. Nucleic acid molecules can be codon optimized, either wholly or in part. Because any one amino acid (except for methionine and tryptophan) is encoded by a number of codons, the sequence of the nucleic acid molecule may be changed without changing the encoded amino acid. Codon optimization is when one or more codons are altered at the nucleic acid level such that the amino acids are not changed but expression in a particular host organism is increased. Those having ordinary skill in the art will recognize that codon tables and other

references providing preference information for a wide range of organisms are available in the art (see, e.g., Zhang et al. (1991) *Gene* 105:61-72; Murray et al. (1989) *Nucl. Acids Res.* 17:477-508). Methodology for optimizing a nucleotide sequence for expression is provided, for example, in U.S. Pat. No. 6,015,891. Programs for codon optimization are available in the art (e.g., OPTIMIZER at genomes.urv.es/OPTIMIZER; OptimumGene.TM. from GenScript at: www.genscript.com/codon_opt.html).

[0041] The eukaryotic cell is preferably a Chinese hamster ovary (CHO) cell and accordingly the codon-optimization is preferably an optimization for the expression in CHO cells. CHO cells are of particular commercial interest since they are the most commonly used mammalian hosts for the industrial production of recombinant protein therapeutics.

[0042] Further details on suitable eukaryotic cells, including plant and animal cells will be provided herein below.

[0043] The present invention relates in a second aspect to a vector encoding the nucleic acid molecule of the first aspect.

[0044] The definitions and preferred embodiments as described herein above apply mutatis mutandis to the second aspect, if applicable.

[0045] A vector according to this invention is generally and preferably capable of directing the replication, and/or the expression of the nucleic acid molecule of the invention and/or the expression of the polypeptide encoded thereby.

[0046] Preferably, the vector is a plasmid, cosmid, virus, bacteriophage or another vector used conventionally e.g. in genetic engineering.

[0047] Exemplary plasmids and vectors are listed, for example, in Studier and coworkers (Studier, W. F.; Rosenberg A. H.; Dunn J. J.; Dubendorff J. W., 1990, Use of the T7 RNA polymerase to direct expression of cloned genes, *Methods Enzymol.* 185, 61-89) or the brochures supplied by the companies Novagen, Promega, New England Biolabs, Clontech and Gibco BRL. Other preferred plasmids and vectors can be found in: Glover, D.M., 1985, *DNA cloning: a practical approach*, Vol. I-III, IRL Press Ltd., Oxford; Rodriguez, R. L. and Denhardt, D. T. (eds), 1988, *Vectors: a survey of molecular cloning vectors and their uses*, 179-204, Butterworth, Stoneham; Goedeel, D.V., 1990, *Systems for heterologous gene expression*, *Methods Enzymol.* 185, 3-7; Sambrook, J.; Russell, D. W., 2001, *Molecular cloning: a laboratory manual*, 3rd ed., Cold Spring Harbor Laboratory Press, New York. Particularly preferred vectors are vectors that can be used for CRISPR genome editing, in particular vectors only expressing the nucleic acid molecule of the invention encoding the RNA-guided DNA endonuclease or vectors expressing both, the nucleic acid molecule of the invention encoding the RNA-guided DNA endonuclease and the guide RNA (so called "all-in one vectors"). In the former case a second vector is to be employed for the expression of the guide RNA. CRISPR genome editing vectors are commercially available, for example, from OriGene, Vector Builder or ThermoFisher.

[0048] The nucleic acid molecule of the present invention referred to above may also be inserted into vectors such that a translational fusion with another nucleic acid molecule is generated. To this aim, overlap extension PCR can be applied (e.g. Wurch, T., Lestienne, F., and Pauwels, P. J., A modified overlap extension PCR method to create chimeric genes in the absence of restriction enzymes, *Biotechn. Techn.* 12, 9, Sep. 1998, 653-657). The products arising

therefrom are termed fusion proteins and will be described further below. The other nucleic acid molecules may encode a protein which may e.g. increase the solubility and/or facilitate the purification of the protein encoded by the nucleic acid molecule of the invention. Non-limiting examples include pET32, pET41, pET43. The vectors may also contain an additional expressible nucleic acid coding for one or more chaperones to facilitate correct protein folding. Suitable bacterial expression hosts comprise e. g. strains derived from BL21 (such as BL21(DE3), BL21(DE3)PlysS, BL21(DE3)RIL, BL21(DE3)PRARE) or Rosetta®.

[0049] For vector modification techniques, see J. F. Sambrook and D. W. Russell, ed., Cold Spring Harbor Laboratory Press, 2001, ISBN-10 0-87969-577-3. Generally, vectors can contain one or more origins of replication (ori) and inheritance systems for cloning or expression, one or more markers for selection in the host, e.g., antibiotic resistance, and one or more expression cassettes. Suitable origins of replication include, for example, the Col E1, the SV40 viral and the M13 origins of replication.

[0050] The coding sequences inserted in the vector can e.g. be synthesized by standard methods, or isolated from natural sources. Ligation of the coding sequences to transcriptional regulatory elements and/or to other amino acid encoding sequences can be carried out using established methods. Transcriptional regulatory elements (parts of an expression cassette) ensuring expression in prokaryotes or eukaryotic cells are well known to those skilled in the art. These elements comprise regulatory sequences ensuring the initiation of the transcription (e. g., translation initiation codon, transcriptional termination sequences, promoters, enhancers, and/or insulators), internal ribosomal entry sites (IRES) (Owens et al., (2001), *PNAS.* 98 (4) 1471-1476) and optionally poly-A signals ensuring termination of transcription and stabilization of the transcript. Additional regulatory elements may include transcriptional as well as translational enhancers, and/or naturally-associated or heterologous promoter regions. The regulatory elements may be native to the endonuclease of the invention or heterologous regulatory elements. Preferably, the nucleic acid molecule of the invention is operably linked to such expression control sequences allowing expression in prokaryotes or eukaryotic cells. The vector may further comprise nucleotide sequences encoding secretion signals as further regulatory elements. Such sequences are well known to the person skilled in the art. Furthermore, depending on the expression system used, leader sequences capable of directing the expressed polypeptide to a cellular compartment may be added to the coding sequence of the nucleic acid molecule of the invention. Such leader sequences are well known in the art. Specifically designed vectors allow the shuttling of DNA between different hosts, such as bacteria-fungal cells or bacteria-animal cells.

[0051] Additionally, baculoviral systems or systems based on Vaccinia Virus or Semliki Forest Virus can be used as vectors in eukaryotic expression systems for the nucleic acid molecules of the invention. Expression vectors derived from viruses such as retroviruses, vaccinia virus, adeno-associated virus, herpes viruses, or bovine papilloma virus, may be used for delivery of the nucleic acids or vector into targeted cell populations. Methods which are well known to those skilled in the art can be used to construct recombinant viral

vectors; see, for example, the techniques described in Sambrook and D. W. Russell, ed., Cold Spring Harbor Laboratory Press, 2001.

[0052] Examples for regulatory elements permitting expression in eukaryotic host cells are promoters, including the promoters as described herein above. Besides elements which are responsible for the initiation of transcription such regulatory elements may also comprise transcription termination signals, such as the SV40-poly-A site or the tk-poly-A site or the SV40, lacZ and AcMNPV polyhedral polyadenylation signals, downstream of the nucleic acid.

[0053] The co-transfection with a selectable marker such as kanamycin or ampicillin resistance genes for culturing in *E. coli* and other bacteria allows the identification and isolation of the transfected cells. Selectable markers for mammalian cell culture are the dhfr, gpt, neomycin, hygromycin resistance genes. The transfected nucleic acid can also be amplified to express large amounts of the encoded (poly)peptide. The DHFR (dihydrofolate reductase) marker is useful to develop cell lines that carry several hundred or even several thousand copies of the gene of interest. Another useful selection marker is the enzyme glutamine synthase (GS). Using such markers, the cells are grown in selective medium and the cells with the highest resistance are selected.

[0054] However, the nucleic acid molecules of the invention as described herein above may also be designed for direct introduction or for introduction via liposomes, phage vectors or viral vectors (e.g. adenoviral or retroviral) into the cell.

[0055] The present invention relates in a third aspect to a host cell comprising the nucleic acid molecule of the first aspect or being transformed, transduced or transfected with the vector of the second aspect.

[0056] The definitions and preferred embodiments as described herein above apply mutatis mutandis to the third aspect, if applicable.

[0057] Large amounts of the RNA-guided DNA endonuclease may be produced by said host cell, wherein the isolated nucleotide sequence encoding the RNA-guided DNA endonuclease is inserted into an appropriate vector or expression vector before insertion into the host. The vector or expression vector is introduced into an appropriate host cell, which preferably can be grown in large quantities, and the RNA-guided DNA endonuclease is purified from the host cells or the culture media.

[0058] The host cells may also be used to supply the RNA-guided DNA endonuclease of the invention without requiring purification of the RNA-guided DNA endonuclease (see Yuan, Y.; Wang, S.; Song, Z.; and Gao, R., Immobilization of an L-aminoacylase-producing strain of *Aspergillus oryzae* into gelatin pellets and its application in the resolution of D,L-methionine, Biotechnol Appl. Biochem. (2002), 35:107-113). The RNA-guided DNA endonuclease of the invention may be secreted by host cells. Those skilled in the field of molecular biology will understand that any of a wide variety of expression systems may be used to provide the RNA-guided DNA endonuclease. The precise host cell used is not critical to the invention, so long as the host cells produce the RNA-guided DNA endonuclease when grown under suitable growth conditions.

[0059] Host cells into which vectors containing the nucleic acid molecule of the invention can be cloned are used for replicating and isolating a sufficient quantity of the recom-

binant enzyme. The methods used for this purpose are well known to the skilled person (Sambrook and D. W. Russell, ed., Cold Spring Harbor Laboratory Press, 2001).

[0060] The expression of the RNA-guided DNA endonuclease may not only be used to produce the RNA-guided DNA endonuclease in a host cell, but its expression may also be used to edit the genome of the host cell. In such a case the host cell also comprises a guide RNA. Vectors that can be used for CRISPR genome editing have been discussed herein above.

[0061] In accordance with a preferred aspect of the third aspect of the invention the host cell is a eukaryotic cell or a prokaryotic cell and is preferably a plant cell or an animal cell.

[0062] The host cell can be a eukaryotic cell that does not naturally include a gene encoding any one of BMC01-BMC15, and can be, for example, be the cell of a fungus, algae, plant, or animal, wherein the animal can be an avian, reptile, amphibian, fish, cephalopod, crustacean, insect, arachnid, marsupial, or mammalian. The gene encoding any one of BMC01-BMC15 that is non-native with respect to the host cell can be operably linked to a regulatory element, such as a promoter. The promoter can be native to the host organism or can be a promoter of another species. A construct for expressing any one of BMC01-BMC15 in a heterologous host cell, such as a eukaryotic cell, can optionally further include a transcriptional terminator. The gene encoding any one of BMC01-BMC15 can optionally be codon optimized for the host species, can optionally include one or more introns, and can optionally include one or more peptide tag sequences, one or more nuclear localisation sequences (NLSs) and/or one or more linkers or engineered cleavage sites (e.g. a 2a sequence). In various embodiments a host cell can include any of the engineered BMC01-BMC15 CRISPR systems disclosed above, where the nucleic acid sequence encoding the effector is present in the cell prior to introduction of a guide RNA. In other embodiments, the cell that is engineered to include a gene for expressing BMC01-BMC15 polypeptide can further include a polynucleotide encoding a guide RNA (e.g., a guide RNA) that is operably linked to a regulatory element.

[0063] The cell or organism can be a prokaryotic cell that does not naturally include a gene encoding any one of BMC01-BMC15. Suitable prokaryotic host cells comprise e.g. bacteria of the species *Escherichia*, such as strains derived from *E. coli* BL21 (e.g. BL21(DE3), BL21(DE3) PlyS, BL21(DE3)RIL, BL21(DE3)PRARE, BL21 codon plus, BL21(DE3) codon plus), Rosetta®, XL1 Blue, NM522, JM101, JM109, JM105, RR1, DH5α, TOP 10, HB101 or MM294. Further suitable bacterial host cells are, but not limited to, *Streptomyces*, *Pseudomonas*, such as *Pseudomonas putida*, *Corynebacterium*, such as *C. glutamicum*, *Lactobacillus*, such as *L. salivarius* or *Bacillus* such as *Bacillus subtilis*.

[0064] In general, a eukaryotic host cell is preferred over a prokaryotic host cell.

[0065] The eukaryotic cell can be a yeast, fungus, amoebae, insect, vertebrate (e.g. mammalian) or plant cells.

[0066] Yeasts cells can be, for example, from *Saccharomyces cerevisiae*, *Ogataea angusta*, *Kluyveromyces* sp. such as *K. marxianus* or *K. lactis* or *Pichia* sp. such as *P. pastoris*, insect cells such as *Drosophila* S2 or *Spodoptera* Sf9 cells, plant cells, or fungi cells, preferably of the family Trichoco-

maceae, more preferably of the genus *Aspergillus*, *Penicillium* or *Trichoderma*, or of the family Ustilaginaceae, preferably *Ustilago* sp.

[0067] Plant host cells that may be used include monocots and dicots (i.e., monocotyledonous and dicotyledonous, respectively), preferably crop plant cells and tobacco cells. Crop plants include cereals (e.g. maize, wheat, rice), legumes (e.g. soybean, pea, and alfalfa), fruit and vegetables (e.g. lettuce, tomato, and potato). Examples of plant species of interest include, but are not limited to, corn (*Zea mays*), *Brassica* sp. (e.g., *B. napus*, *B. rapa*, *B. juncea*), particularly those *Brassica* species useful as sources of seed oil, alfalfa (*Medicago sativa*), rice (*Oryza sativa*), rye (*Secale cereale*), sorghum (*Sorghum bicolor*, *Sorghum vulgare*), camelina (*Camelina sativa*), millet (e.g., pearl millet (*Pennisetum glaucum*), proso millet (*Panicum miliaceum*), foxtail millet (*Setaria italica*), finger millet (*Eleusine coracana*)), sunflower (*Helianthus annuus*), quinoa (*Chenopodium quinoa*), chicory (*Cichorium intybus*), lettuce (*Lactuca sativa*), safflower (*Carthamus tinctorius*), wheat (*Triticum aestivum*), soybean (*Glycine max*), tobacco (*Nicotiana tabacum*), potato (*Solanum tuberosum*), peanuts (*Arachis hypogaea*), cotton (*Gossypium barbadense*, *Gossypium hirsutum*), sweet potato (*Ipomoea batatas*), cassava (*Manihot esculenta*), coffee (*Coffea* spp.), coconut (*Cocos nucifera*), pineapple (*Ananas comosus*), citrus trees (*Citrus* spp.), cocoa (*Theobroma cacao*), tea (*Camellia sinensis*), banana (*Musa* spp.), avocado (*Persea americana*), fig (*Ficus casica*), guava (*Psidium guajava*), mango (*Mangifera indica*), olive (*Olea europaea*), papaya (*Carica papaya*), cashew (*Anacardium occidentale*), macadamia (*Macadamia integrifolia*), almond (*Prunus amygdalus*), sugar beets (*Beta vulgaris*), sugarcane (*Saccharum* spp.), oil palm (*Elaeis guineensis*), poplar (*Populus* spp.), eucalyptus (*Eucalyptus* spp.), oats (*Avena sativa*), barley (*Hordeum vulgare*), vegetables, ornamentals, and conifers.

[0068] Mammalian host cells that could be used include human Hela, HEK293, H9 and Jurkat cells, mouse NIH3T3 and C127 cells, COS 1, COS 7 and CV1, quail QC1-3 cells, mouse L cells, Bowes melanoma cells and Chinese hamster ovary (CHO) cells.

[0069] The present invention relates in a fourth aspect to a plant, seed or a part of a plant, said part of a plant no being a single plant cell, or an animal comprising the nucleic acid molecule of the first aspect or being transformed, transduced or transfected with the vector of the second aspect.

[0070] The definitions and preferred embodiments as described herein above apply mutatis mutandis to the fourth aspect, if applicable.

[0071] The plant or a part thereof may be a monocotyledonous plant or part thereof or a dicotyledonous plant or part thereof. The dicotyledonous plant is preferably a tobacco plant or part thereof. Likewise the seed may be a monocotyledonous or dicotyledonous plant seed and is preferably tobacco plant seed.

[0072] Preferred examples of crop plants have been described herein above. Non-limiting examples of a part of a plant is a leave, stem, or root.

[0073] The animal is preferably a mammal and most preferably a non-human mammal. The mammal can be, for example, a mouse, rat, hamster, cat, dog, horse, swine, cattle, monkey, ape, etc.

[0074] By the expression of the nucleic acid molecule of the first aspect in a plant, seed or a part of a plant or an

animal along with a guide RNA the genome of the host may be edited. The genome may be edited, for example, in order to introduce a targeted gene mutation, for gene therapy, for a creating chromosome rearrangement, for studying gene function, for the production of a transgenic organism, for endogenous gene labeling or for targeted transgene addition.

[0075] The present invention relates in a fifth aspect to a method of producing an RNA-guided DNA endonuclease comprising culturing the host cell of the third aspect and isolating the RNA-guided DNA endonuclease produced.

[0076] The definitions and preferred embodiments as described herein above apply mutatis mutandis to the fifth aspect, if applicable.

[0077] Suitable conditions for culturing a prokaryotic or eukaryotic host are well known to the person skilled in the art. In general, suitable conditions for culturing bacteria are growing them under aeration in Luria Bertani (LB) medium. To increase the yield and the solubility of the expression product, the medium can be buffered or supplemented with suitable additives known to enhance or facilitate both. *E. coli* can be cultured from 4 to about 37° C., the exact temperature or sequence of temperatures depends on the molecule to be overexpressed.

[0078] In general, *Aspergillus* sp. may be grown on Sabouraud dextrose agar, or potato dextrose agar at about 10° C. to about 40° C., and preferably at about 25° C. Suitable conditions for yeast cultures are known, for example from Guthrie and Fink, "Guide to Yeast Genetics and Molecular Cell Biology" (2002); Academic Press Inc. The skilled person is also aware of all these conditions and may further adapt these conditions to the needs of a particular host species and the requirements of the polypeptide expressed. In case an inducible promoter controls the nucleic acid of the invention in the vector present in the host cell, expression of the polypeptide can be induced by addition of an appropriate inducing agent. Suitable expression protocols and strategies are known to the skilled person.

[0079] Depending on the cell type and its specific requirements, mammalian cell culture can e.g. be carried out in RPMI or DMEM medium containing 10% (v/v) FCS, 2 mM L-glutamine and 100 U/ml penicillin/streptomycin. The cells can be kept at 37° C. in a 5% CO₂, water saturated atmosphere. Suitable expression protocols for eukaryotic cells are well known to the skilled person and can be retrieved e.g. from Sambrook, 2001.

[0080] Methods for the isolation of the produced RNA-guided DNA endonuclease are well-known in the art and comprise without limitation method steps such as ion exchange chromatography, gel filtration chromatography (size exclusion chromatography), affinity chromatography, high pressure liquid chromatography (HPLC), reversed phase HPLC, disc gel electrophoresis or immunoprecipitation, see, for example, in Sambrook, 2001.

[0081] The step of protein isolation is preferably a step of protein purification. Protein purification in accordance with the invention specifies a process or a series of processes intended to further isolate the polypeptide of the invention from a complex mixture, preferably to homogeneity. Purification steps, for example, exploit differences in protein size, physico-chemical properties and binding affinity. For example, proteins may be purified according to their isoelectric points by running them through a pH graded gel or an ion exchange column. Further, proteins may be separated according to their size or molecular weight via size exclu-

sion chromatography or by SDS-PAGE (sodium dodecyl sulfate-polyacrylamide gel electrophoresis) analysis. In the art, proteins are often purified by using 2D-PAGE and are then further analysed by peptide mass fingerprinting to establish the protein identity. This is useful for scientific purposes and the detection limits for protein are very low and nanogram amounts of protein are sufficient for their analysis. Proteins may also be purified by polarity/hydrophobicity via high performance liquid chromatography or reversed-phase chromatography. Thus, methods for protein purification are well known to the skilled person.

[0082] The present invention relates in a sixth aspect to an RNA-guided DNA endonuclease encoded by the nucleic acid molecule of the first aspect.

[0083] The definitions and preferred embodiments as described herein above apply mutatis mutandis to the sixth aspect, if applicable.

[0084] The amino acid sequences of SEQ ID NOs 1-15 are particularly preferred examples of the RNA-guided DNA endonuclease of the invention.

[0085] The RNA-guided DNA endonuclease of the sixth aspect of the invention may also be a fusion protein, wherein the amino acid sequence of the RNA-guided DNA endonuclease is fused to a fusion partner. The fusion may be a direct fusion or a fusion via a linker. The linker is preferably a peptide, such as a GS-linker.

[0086] The fusion partner can be located at the N-terminus, the C-terminus, at both termini, or in an internal location of the RNA-guided DNA endonuclease polypeptide, preferably at the N- or C-terminus.

[0087] The fusion partner is preferably a nuclear localization signal (NLS), a cell-penetrating domain, a plastid targeting signal, a mitochondrial targeting signal peptide, a signal peptide targeting both plastids and mitochondria, a marker domain, a tag (such as a purification tag), a DNA modifying enzyme or a transactivation domain.

[0088] DNA modifying enzymes may modify the DNA by phosphorylation, dephosphorylation of blunting DNA, wherein blunting refers the digestion single-stranded overhangs. Non-limiting examples of dephosphorylation enzymes are the Shrimp Alkaline Phosphatase (rSAP), Quick CIP Phosphatase and Antarctic Phosphatase. Non-limiting examples of phosphorylation enzymes are polynucleotide kinases, such as T4 PNK. Non-limiting examples of blunting enzymes are the DNA Polymerase I Large (Klenow) Fragment, T4 DNA Polymerase or Mung Bean Nuclease.

[0089] Transactivation domains (or trans-activating domains (TADs)) are transcription factor scaffold domains which contain binding sites for other proteins such as transcription co-regulators. Non-limiting examples are nine-amino-acid transactivation domains (9aaTADs) and Glutamine (Q)-rich TADs.

[0090] In general, an NLS comprises a stretch of basic amino acids. Nuclear localization signals are known in the art. The NLS can be at the N-terminus, the C-terminus or both the RNA-guided DNA endonuclease polypeptide according to the invention. For instance, the RNA-guided DNA endonuclease polypeptide according to the invention may comprise about or more than about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more NLSs at or near the amino-terminus, about or more than about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more NLSs at or near the carboxy-terminus, or a combination of these (e.g., zero or at least one or more NLS at the amino-terminus

and zero or at one or more NLS at the carboxy-terminus). When more than one NLS is present, each may be selected independently of the others, such that a single NLS may be present in more than one copy and/or in combination with one or more other NLSs present in one or more copies. In some embodiments, an NLS is considered near the N- or C-terminus when the nearest amino acid of the NLS is within about 1, 2, 3, 4, 5, 10, 15, 20, 25, 30, 40, 50, or more amino acids along the polypeptide chain from the N- or C-terminus. The RNA-guided DNA endonuclease polypeptide sequence and the NLS may in some embodiments be fused with a linker between 1 to about 20 amino acids in length.

[0091] Non-limiting examples of NLSs include an NLS sequence derived from: the NLS of the SV40 virus large T-antigen; the NLS from nucleoplasmin (e.g., the nucleoplasmin bipartite NLS); the c-myc NLS; the hRNP A1 M9 NLS; the IBB domain from importin- α ; the NLS sequences of the myoma T protein, the p53 protein; the c-abl IV protein, or influenza virus NS 1; the NLS of the Hepatitis virus delta antigen, the Mx1 protein; the poly(ADP-ribose) polymerase; and the steroid hormone receptors (human) glucocorticoid. In general, the one or more NLSs are of sufficient strength to drive accumulation of the RNA-guided DNA endonuclease polypeptide according to the invention in a detectable amount in the nucleus of a eukaryotic cell.

[0092] Plastid, mitochondrial, and dual-targeting signal peptide localization signals are also known in the art (see, e.g., Nassoury and Morse (2005) *Biochim Biophys Acta* 1743:5-19; Kunze and Berger (2015) *Front Physiol* 6:259; Herrmann and Neupert (2003) *IUBMB Life* 55:219-225; Soil (2002) *Curr Opin Plant Biol* 5:529-535; Carrie and Small (2013) *Biochim Biophys Acta* 1833:253-259; Carrie et al. (2009) *FEBS J* 276:1187-1195; Silva-Filho (2003) *Curr Opin Plant Biol* 6:589-595; Peeters and Small (2001) *Biochim Biophys Acta* 1541:54-63; Murcha et al. (2014) *Exp Bot* 65:6301-6335; Mackenzie (2005) *Trends Cell Biol* 15:548-554; Glaser et al. (1998) *Plant Mol Biol* 38:311-338).

[0093] Non-limiting examples of marker domains include fluorescent proteins, purification tags, and epitope tags. In certain embodiments, the marker domain can be a fluorescent protein. Non-limiting examples of suitable fluorescent proteins include green fluorescent proteins (e.g., GFP, GFP-2, tagGFP, turboGFP, EGFP, Emerald, Azami Green, Monomeric Azami Green, CopGFP, AceGFP, ZsGreen1), yellow fluorescent proteins (e.g. YFP, EYFP, Citrine, Venus, YPet, PhiYFP, ZsYellow1), blue fluorescent proteins (e.g. EBFP, EBFP2, Azurite, mKalamal, GFPuv, Sapphire, T-sapphire), cyan fluorescent proteins (e.g. ECFP, Cerulean, CyPet, AmCyan1, Midoriishi-Cyan), red fluorescent proteins (mKate, mKate2, mPlum, DsRed monomer, mCherry, mRFPI, DsRed-Express, DsRed2, DsRed-Monomer, HcRed-Tandem, HcRed1, AsRed2, eqFP611, mRasberry, mStrawberry, Jred), and orange fluorescent proteins (mOrange, mKO, Kusabira-Orange).

[0094] A tag is short amino acid sequence that allows the identification of the RNA-guided DNA endonuclease polypeptide according to the invention in a mixture of polypeptides. Hence, the tag is preferably a purification tag. Non-limiting examples of a purification tag are a His-tag (e.g. His-6-tag), a GST-tag, DHFR-tag and a CBP-tag. A review of known purification tags can be found in Kimple et al. (2015), *Curr Protoc Protein Sci.* 2013; 73: Unit-9.9.

[0095] The present invention relates in a seventh aspect to a composition comprising the nucleic acid molecule of the first aspect, the vector of the second aspect, the host cell of the third aspect, the plant, seed, part of a cell or animal of the fourth aspect, the RNA-guided DNA endonuclease of the sixth aspect or a combination thereof.

[0096] The definitions and preferred embodiments as described herein above apply *mutatis mutandis* to the seventh aspect, if applicable.

[0097] The term “composition” as used herein refers to a composition comprising at least one of the nucleic acid molecule of the first aspect, the vector of the second aspect, the host cell of the third aspect, the plant, seed, part of a cell or animal of the fourth aspect, the RNA-guided DNA endonuclease of the sixth aspect or a combination thereof which are also collectively referred in the following as compounds.

[0098] In accordance with a preferred embodiment of the seventh aspect the composition is a pharmaceutical composition or a diagnostic composition.

[0099] In accordance with the present invention, the term “pharmaceutical composition” relates to a composition for administration to a patient, preferably a human patient. The pharmaceutical composition of the invention comprises at least one of the compounds recited above. It may, optionally, comprise further molecules capable of altering the characteristics of the compounds of the invention thereby, for example, stabilizing, modulating and/or activating their function. The composition may be in solid, liquid or gaseous form and may be, *inter alia*, in the form of (a) powder(s), (a) tablet(s), (a) solution(s) or (an) aerosol(s). The pharmaceutical composition of the present invention may, optionally and additionally, comprise a pharmaceutically acceptable carrier. Examples of suitable pharmaceutical carriers are well known in the art and include phosphate buffered saline solutions, water, emulsions, such as oil/water emulsions, various types of wetting agents, sterile solutions, organic solvents including DMSO etc. Compositions comprising such carriers can be formulated by conventional methods. These pharmaceutical compositions may be administered to the subject at a suitable dose. The dosage regimen will be determined by the attending physician and clinical factors. As is well known in the medical arts, dosages for any one patient depends upon many factors, including the patient's size, body surface area, age, the particular compound to be administered, sex, time and route of administration, general health, and other drugs being administered concurrently. The therapeutically effective amount for a given situation will readily be determined by routine experimentation and is within the skills and judgement of the ordinary clinician or physician. Generally, the regimen as a regular administration of the pharmaceutical composition should be in the range of 1 µg to 5 g of the active compound per day. However, a more preferred dosage might be in the range of 0.01 mg to 100 mg, even more preferably 0.01 mg to 50 mg and most preferably 0.01 mg to 10 mg per day. The length of treatment needed to observe changes and the interval following treatment for responses to occur vary depending on the desired effect. The particular amounts may be determined by conventional tests which are well known to the person skilled in the art.

[0100] The pharmaceutical composition may be used, for example, to treat or prevent a pathogenic disease, such as a viral or bacterial disease. For instance, the RNA-guided DNA endonuclease of the sixth aspect may be used together

with gRNA targeting the genome of the pathogen thereby modifying the genome of the pathogen, such that the disease being caused by the pathogen is prevented or treated.

[0101] The pharmaceutical composition may also be used, for example, to treat or prevent a microbiome imbalance. An imbalance in the microbiome can occur, for example, because of an overuse of antibiotics, which may cause an overgrowth of pathogenic bacteria and yeast.

[0102] A “diagnostic composition” relates to composition which is suitable to detect a disease in subject, both infectious and non-infectious disease. The diagnostic composition may in particular comprise a marker portion as described herein above in connection with the fusion protein of the invention being attached to ssDNA strands, so that when RNA-guided DNA endonuclease polypeptide according to the invention cuts the ssDNA, it activates the reporter, causing it to fluoresce or change colour, thus enabling visual detection of the specific disease nucleic marker. The diagnostic composition may be applied to a body fluid sample, such as a blood, urine, or saliva.

[0103] The present invention relates in a eighth aspect to the nucleic acid molecule of the first aspect, the vector of the second aspect, the host cell of the third aspect, the plant, seed, part of a cell or animal of the fourth aspect, the RNA-guided DNA endonuclease of the sixth aspect or a combination thereof for use in the treatment of a disease in a subject or a plant by modifying a nucleotide sequence at a target site in the genome of the subject or plant.

[0104] Also described is method of treating or preventing a disease in a subject or a plant comprising modifying a nucleotide sequence at a target site in the genome of the subject or plant by the nucleic acid molecule of the first aspect, the vector of the second aspect, the host cell of the third aspect, the plant, seed, part of a cell or animal of the fourth aspect, the RNA-guided DNA endonuclease of the sixth aspect or a combination thereof.

[0105] The definitions and preferred embodiments as described herein above apply *mutatis mutandis* to the eighth aspect, if applicable.

[0106] The modification of a nucleotide sequence at a target site in the genome of the subject or plant is in accordance with the invention a genome editing by the CRISPR technology and in particular by the novel RNA-guided DNA endonucleases as provided herewith which is to be used in combination with a proper gRNA and optionally a repair substrate as described herein below in order to determine the target side of the genome modification.

[0107] Genome editing (also known as genome engineering) is a type of genetic engineering in which a target site, preferably a gene of interest is inserted, deleted, modified or replaced in the genome of the cell. The target site, preferably the gene of interest can be in the genome but may also be in the mitochondrial DNA (animal cells) or chloroplast DNA (plant cells). Genome editing may result in a loss-of-function mutation or a gain-of-function mutation in the genome of the cell. A loss-of-function mutation (also called inactivating mutation) results in the gene of interest having less or no function (being partially or wholly inactivated). When the allele has a complete loss of function (wholly inactivated) this is also called herein a (gene) knock-out. A gene knock-out may be achieved by inserting, deleting, modifying or replacing one or more nucleotides of a gene. A gain-of-function mutation (also called activating mutation) may change the gene of interest such that its effect becomes

stronger (enhanced activation) or even is superseded by a different (e.g. abnormal) function. A gain-of-function mutation may also introduce a new function or effect into a cell which the cell did not have before. In this context the new gene may be added to the genome of the cell (insertion) or may replace a gene within the genome. A gain-of-function mutation introducing such a new function or effect is also called gene knock-in. Genome editing may also result in the up- or down-regulation of one or more genes. By targeting DNA sites which are responsible for the regulation of the expression of a gene (e.g. a promoter region or a gene encoding transcription factor) the expression of genes can be up- or down-regulated by CRISPR technology. Further details on the mode of action of the CRISPR technology will be provided herein below.

[0108] Since its discovery, the CRISPR technology has been increasingly applied to therapeutic genome editing. Employment of several viral and non-viral vectors has enabled efficient delivery of the CRISPR system to target cells or tissues. In addition, the CRISPR system is able to modulate the target gene's expression in various ways, such as mutagenesis, gene integration, epigenome regulation, chromosomal rearrangement, base editing and mRNA editing (for review Le and Kim (2019), Hum Genet; 138(6): 563-590).

[0109] The modification of a nucleotide sequence at a target site in the genome of the subject is preferably a gene therapy. Gene therapy is based on the principle of the genetic manipulation of nucleotide sequence at a target site for treating and preventing a disease, in particular a human disease.

[0110] In clinical trials of the CRISPR technology, scientists are using the CRISPR technology to combat cancer and blood disorders in humans. In these trials, some cells are removed from the subject to be treated, the DNA is genome-edited and then the genome edited cells are put back into the subject, said cells now being armed to fight the disease to be treated.

[0111] The present invention relates in a ninth aspect to a method of modifying a nucleotide sequence at a target site in the genome of a cell comprising introducing into said cell (i) a DNA-targeting RNA or a DNA polynucleotide encoding a DNA-targeting RNA, wherein the DNA-targeting RNA comprises: (a) a first segment comprising a nucleotide sequence that is complementary to a sequence in the target DNA; and (b) a second segment that interacts with the RNA-guided DNA endonuclease of the sixth aspect; and (ii) the RNA-guided DNA endonuclease of the sixth aspect, or the nucleic acid molecule encoding an RNA-guided DNA endonuclease of the first aspect, or the vector of the second aspect, wherein the RNA-guided DNA endonuclease comprises (a) an RNA-binding portion that interacts with the DNA-targeting RNA and (b) an activity portion that exhibits site-directed enzymatic activity.

[0112] Accordingly, the present invention also relates to a composition (e.g. a pharmaceutical or a diagnostic composition) comprising (i) a DNA-targeting RNA or a DNA polynucleotide encoding a DNA-targeting RNA, wherein the DNA-targeting RNA comprises: (a) a first segment comprising a nucleotide sequence that is complementary to a sequence in the target DNA; and (b) a second segment that interacts with the RNA-guided DNA endonuclease of the sixth aspect; and (ii) the RNA-guided DNA endonuclease of the sixth aspect, or the nucleic acid molecule encoding an

RNA-guided DNA endonuclease of the first aspect, or the vector of the second aspect, wherein the RNA-guided DNA endonuclease comprises (a) an RNA-binding portion that interacts with the DNA-targeting RNA and (b) an activity portion that exhibits site-directed enzymatic activity.

[0113] The definitions and preferred embodiments as described herein above apply mutatis mutandis to the ninth aspect, if applicable.

[0114] The DNA-targeting RNA comprises a first segment comprising a nucleotide sequence that is complementary to a sequence in the target DNA and a second segment that interacts with the RNA-guided DNA endonuclease. As discussed herein above, the nucleotide sequence that is complementary to a sequence in the target DNA defines the target specificity of the RNA-guided DNA endonuclease. As also discussed herein above, the DNA-targeting RNA binds to the RNA-guided DNA endonuclease, whereby a complex is formed. The second segment interacts with the RNA-guided DNA endonuclease and is responsible for the formation of the complex. The second segment that interacts with the RNA-guided DNA endonuclease of the sixth aspect preferably comprises or consists of SEQ ID NO: 31 and more preferably of SEQ ID NO: 32. SEQ ID NO: 31 is a consensus sequence of the second segment of Type V Class 2 CRISPR nucleases. In Type V Class 2 CRISPR nucleases the second segment is also known as 5' handle. SEQ ID NO: 32 is the second segment of any one of BMC01-BMC15.

[0115] The RNA-guided DNA endonuclease comprises as first segment being an RNA-binding portion that interacts with the DNA-targeting RNA and as a second segment being an activity portion that exhibits site-directed enzymatic activity. The first segment interacts with the DNA-targeting RNA and is responsible for the formation of the discussed complex. The second segment harbours the endonuclease domain, which preferably comprises a RuvC domain.

[0116] As also discussed herein above, the DNA-targeting RNA is the guide RNA. The guide RNA may either be directly introduced into the cells or as a DNA polynucleotide encoding the DNA-targeting RNA. In the latter case, the DNA encoding the guide RNA is generally operably linked to one or more promoter sequences for expression of the guide RNA. For example, the RNA coding sequence can be operably linked to a promoter sequence that is recognized by RNA polymerase III (Pol III) or RNA polymerase II (Pol II). The DNA polynucleotide encoding the DNA-targeting RNA is preferably a vector. Many single gRNA empty vectors (with and without the CRISPR endonuclease) are available in the art. Also several empty multiplex gRNA vectors are available that can be used to express multiple gRNAs from a single plasmid (with or without the expression of the CRISPR endonuclease). The DNA polynucleotide encodes the DNA-targeting RNA in expressible form.

[0117] Likewise, the RNA-guided DNA endonuclease may either be directly introduced into the cells or as a nucleic acid molecule encoding the RNA-guided DNA endonuclease, the latter being preferably a vector of the second aspect. The DNA polynucleotide encodes the RNA-guided DNA endonuclease in expressible form.

[0118] As is discussed in greater detail herein above, the RNA-guided DNA endonuclease and the DNA-targeting RNA may also be encoded by the same DNA polynucleotide, such as an all-in one CRISPR-cas vector.

[0119] The term "in expressible form" means that the one or more DNA polynucleotides encoding the RNA-guided

DNA endonuclease and the DNA-targeting RNA are in a form that ensures that the DNA-targeting RNA is transcribed and that the RNA-guided DNA endonuclease is transcribed and translated into the active enzyme in the cells.

[0120] In accordance with a preferred embodiment of the ninth aspect of the invention in case the RNA-guided DNA endonuclease and the DNA-targeting RNA are directly introduced into the cell they are introduced in the form of a ribonucleoprotein complex (RNP).

[0121] RNPs are assembled in vitro and can be delivered to the cell by methods known in the art, for example, electroporation or lipofection. RNPs are capable to cleave the target site with comparable efficacy as nucleic acid-based (e.g. vector-based) RNA-guided DNA endonucleases (Kim et al. (2014), *Genome Research* 24(6):1012-1019).

[0122] Means for introducing proteins (or peptides) or RNPs into living cells are known in the art and comprise but are not limited to microinjection, electroporation, lipofection (using liposomes), nanoparticle-based delivery, and protein transduction. Any one of these methods may be used.

[0123] A liposome used for lipofection is a small vesicle, composed of the same material as a cell membrane (i.e., normally a lipid bilayer e.g. made of phospholipids), which can be filled with one or more protein(s) (e.g. Torchilin VP. (2006), *Adv Drug Deliv Rev.*, 58(14):1532-55). To deliver a protein or RNP into a cell, the lipid bilayer of the liposome can fuse with the lipid bilayer of the cell membrane, thereby delivering the contained protein into the cell. It is preferred that the liposomes used in accordance with invention are composed of cationic lipids. The cationic liposome strategy has been applied successfully to protein delivery (Zelphati et al. (2001), *J. Biol. Chem.* 276, 35103-35110). As known in the art, the exact composition and/or mixture of cationic lipids used can be altered, depending upon the protein(s) of interest and the cell type used (Felgner et al. (1994), *J. Biol. Chem.* 269, 2550-2561). Nanoparticle-based delivery of Cas9 ribonucleoprotein and donor DNA for the induction of homology-directed DNA repair is, for example, described in Lee et al. (2017), *Nature Biomedical Engineering*, 1:889-90.

[0124] Protein transduction specifies the internalisation of proteins into the cell from the external environment (Ford et al (2001), *Gene Therapy*, 8:1-4). This method relies on the inherent property of a small number of proteins and peptides (preferably 10 to 16 amino acids long) to penetrate the cell membrane. The transducing property of these molecules can be conferred upon proteins which are expressed as fusions with them and thus offer, for example, an alternative to gene therapy for the delivery of therapeutic proteins into target cells. Commonly used proteins or peptides being able to penetrate the cell membrane are, for example; the antennapedia peptide, the herpes simplex virus VP22 protein, HIV TAT protein transduction domain, peptides derived from neurotransmitters or hormones, or a 9xArg-tag.

[0125] Microinjection and electroporation are well known in the art and the skilled person knows how to perform these methods. Microinjection refers to the process of using a glass micropipette to introduce substances at a microscopic or borderline macroscopic level into a single living cell. Electroporation is a significant increase in the electrical conductivity and permeability of the cell plasma membrane caused by an externally applied electrical field. By increasing permeability, protein (or peptides or nucleic acid sequences) can be introduced into the living cell.

[0126] The RNA-guided DNA endonuclease may be introduced into the cells as an active enzyme or as a proenzyme. In the latter case the RNA-guided DNA endonuclease is biochemically changed within the cells (for example by a hydrolysis reaction revealing the active site, or changing the configuration to reveal the active site), so that the proenzyme becomes an active enzyme.

[0127] Means and methods for the introduction of nucleic acid molecule(s) and DNA-targeting RNA into cells are likewise known in the art and these methods encompass transducing or transfecting cells.

[0128] Transduction is the process by which foreign DNA is introduced into a cell by a virus or viral vector. Transduction is a common tool used by molecular biologists to stably introduce a foreign gene into a host cell's genome. Generally, a plasmid is constructed in which the genes to be transferred are flanked by viral sequences that are used by viral proteins to recognize and package the viral genome into viral particles. This plasmid is inserted (usually by transfection) into a producer cell together with other plasmids (DNA constructs) that carry the viral genes required for formation of infectious virions. In these producer cells, the viral proteins expressed by these packaging constructs bind the sequences on the DNA/RNA (depending on the type of viral vector) to be transferred and insert it into viral particles. For safety, none of the plasmids used contains all the sequences required for virus formation, so that simultaneous transfection of multiple plasmids is required to get infectious virions. Moreover, only the plasmid carrying the sequences to be transferred contains signals that allow the genetic materials to be packaged in virions, so that none of the genes encoding viral proteins are packaged.

[0129] Viruses collected from these cells are then applied to the cells to be altered. The initial stages of these infections mimic infection with natural viruses and lead to expression of the genes transferred and (in the case of lentivirus/retrovirus vectors) insertion of the DNA to be transferred into the cellular genome. However, since the transferred genetic material does not encode any of the viral genes, these infections do not generate new viruses (the viruses are "replication-deficient"). In the present case transduction may be used to generate cells that comprise the RNA-guided DNA endonuclease in their genome in expressible form.

[0130] Transfection is the process of deliberately introducing naked or purified nucleic acids or purified proteins or assembled ribonucleoprotein complexes into cells. Transfection is generally a non-viral based method.

[0131] Transfection may be a chemical-based transfection. Chemical-based transfection can be divided into several kinds: transfection using cyclodextrin, polymers, liposomes, or nanoparticles. One of the cheapest methods uses calcium phosphate. HEPES-buffered saline solution (HeBS) containing phosphate ions are combined with a calcium chloride solution containing the DNA to be transfected. When the two are combined, a fine precipitate of the positively charged calcium and the negatively charged phosphate will form, binding the DNA to be transfected on its surface. The suspension of the precipitate is then added to the cells to be transfected (usually a cell culture grown in a monolayer). By a process not entirely understood, the cells take up some of the precipitate, and with it, the DNA. This process has been a preferred method of identifying many oncogenes. Other methods use highly branched organic compounds, so-called dendrimers, to bind the DNA and transfer it into the cell.

Another method is the use of cationic polymers such as DEAE-dextran or polyethylenimine (PEI). The negatively charged DNA binds to the polycation and the complex is taken up by the cell via endocytosis. Lipofection (or liposome transfection) is a technique used to inject genetic material into a cell by means of liposomes, which are vesicles that can easily merge with the cell membrane since they are both made of a phospholipid bilayer, as mentioned above. Lipofection generally uses a positively charged (cationic) lipid (cationic liposomes or mixtures) to form an aggregate with the negatively charged (anionic) genetic material. This transfection technology performs the same tasks in terms of transfer into cells as other biochemical procedures utilizing polymers, DEAE-dextran, calcium phosphate, and electroporation. The efficiency of lipofection can be improved by treating transfected cells with a mild heat shock. Eugene is a series of widely used proprietary non-liposomal transfection reagents capable of directly transfecting a wide variety of cells with high efficiency and low toxicity.

[0132] Transfection may also be a non-chemical method. Electroporation (gene electrotransfer) is a popular method, where transient increase in the permeability of cell membrane is achieved when the cells are exposed to short pulses of an intense electric field. Cell squeezing enables delivery of molecules into cells via cell membrane deformation. Sonoporation uses high-intensity ultrasound to induce pore formation in cell membranes. This pore formation is attributed mainly to the cavitation of gas bubbles interacting with nearby cell membranes since it is enhanced by the addition of ultrasound contrast agent, a source of cavitation nuclei. Optical transfection is a method where a tiny (~1 μm diameter) hole is transiently generated in the plasma membrane of a cell using a highly focused laser. Protoplast fusion is a technique in which transformed bacterial cells are treated with lysozyme in order to remove the cell wall. Following this, fusogenic agents (e.g., Sendai virus, PEG, electroporation) are used in order to fuse the protoplast carrying the gene of interest with the recipient target cell.

[0133] Finally, transfection may be a particle-based method. A direct approach to transfection is the gene gun, where the DNA is coupled to a nanoparticle of an inert solid (commonly gold), which is then “shot” (or particle bombardment) directly into the target cell’s nucleus. Hence, the nucleic acid is delivered through membrane penetration at a high velocity, usually connected to microprojectiles. Magnetofection, or magnet-assisted transfection, is a transfection method that uses magnetic force to deliver DNA into target cells. Impalefection is carried out by impaling cells by elongated nanostructures and arrays of such nanostructures such as carbon nanofibers or silicon nanowires which have been functionalized with plasmid DNA.

[0134] The method of the ninth aspect of the invention relates to a method for editing (i.e. “mutating”) with the RNA-guided DNA endonuclease of the invention a nucleotide sequence at a target site in the genome of a cell. This requires essentially three sequential preconditions: (1) Efficient delivery of the RNA-guided DNA endonuclease-encoding genes or the RNA-guided DNA endonuclease itself into the target cell; (2) efficient expression or presence of the CRISPR-components in the target cell (DNA-targeting RNA and the RNA-guided DNA endonuclease of the sixth aspect); and (3) targeting of the genomic site of interest by CRISPR ribonucleoprotein complexes and repair of the

DNA by cell’s own repair pathways. Step (3) is automatically carried out in the cell upon the expression of the CRISPR-components in the cell the genome of which is to be edited.

[0135] By genome editing a target site may be inserted, deleted, modified (including single nucleotide polymorphisms (SNPs)) or replaced in the genome of the cell. The target site can be in the coding region of a gene, in an intron of a gene, in a control region of a gene, in a non-coding region between genes, etc. The gene can be a protein coding gene or an RNA coding gene. The gene can be any gene of interest.

[0136] In this connection genome editing uses the cell’s own repair pathways, including the non-homologous end-joining (NHEJ) or homology directed recombination (HDR) pathway. Once the DNA is cut by the RNA-guided DNA endonuclease, the cell’s own DNA repair machinery (NHEJ or HDR) adds or deletes pieces of genetic material, or makes changes to the DNA by replacing an existing segment with a customized DNA sequence. Hence, in the CRISPR-Cas system, the CRISPR nuclease makes a double-stranded break in DNA at a site determined by the short (~20 nucleotide) gRNA which break is then repaired within the cell by NHEJ or HDR. It is preferred that genome editing uses NHEJ. In a different embodiment, it is preferred that genome editing uses HDR.

[0137] NHEJ uses a variety of enzymes to directly join the DNA ends in a double-strand break. In contrast, in HDR, a homologous sequence is utilized as a template for the regeneration of missing DNA sequence at the break point. NHEJ is the canonical homology-independent pathway as it involves the alignment of only one to a few complementary bases at most for the re-ligation of two ends, whereas HDR uses longer stretches of sequence homology to repair DNA lesions.

[0138] The natural properties of these pathways form the very basis of RNA-guided DNA endonuclease-based genome editing. NHEJ is error-prone, and has been shown to cause mutations at the repair site. Thus, if one is able to create a double strand break (DSB) at a desired gene in multiple samples, it is very likely that mutations will be generated at that site in some of the treatments because of errors created by the NHEJ infidelity. On the other hand, the dependency of HDR on a homologous sequence to repair DSBs can be exploited by inserting a desired sequence within a sequence that is homologous to the flanking sequences of a DSB which, when used as a template by the HDR system, would lead to the creation of the desired change within the genomic region of interest. Despite the distinct mechanisms, the concept of the HDR based gene editing is in a way similar to that of homologous recombination based gene targeting. So based on these principles if one is able to create a DSB at a specific location within the genome, then the cell’s own repair systems will help in creating the desired mutations.

[0139] The homologous sequence template for HDR is also referred to herein as “repair template”.

[0140] Hence, by modifying a nucleotide sequence at a target site in the genome of a cell according to the ninth aspect of the invention a gene may be knocked-out (by introducing per-mature stop codon) or knocked-in (via the repair substrate). It is likewise possible to alter the expression of a gene by the method of the ninth aspect of the invention. For instance, the target site in the genome may be

a promoter region changing the promoter region may increase or decrease the expression of the gene being controlled via the target promoter region.

[0141] Hence, in accordance with a preferred embodiment of the ninth aspect the method further comprises the introduction of a repair substrate into said cell.

[0142] The repair substrate may either be directly introduced into the cell or as a nucleic acid molecule, preferably a vector encoding the repair substrate in expressible form. The use of a repair substrate for correcting a mutant beta-glucuronidase in wheat embryo is illustrated in Example 3.

[0143] The designs and structures of repair templates being suitable for HDR are known in the art. HDR is error-free if the repair template is identical to the original DNA sequence at the double-strand break (DSB), or it can introduce very specific mutations into DNA. The three central steps of the HDR pathways are: (1) The 5'-ended DNA strand is resected at the break to create a 3' overhang. This will serve as both a substrate for proteins required for strand invasion and a primer for DNA repair synthesis. (2) The invasive strand can then displace one strand of the homologous DNA duplex and pair with the other. This results in the formation of the hybrid DNA, referred to as the displacement loop (D loop). (3) The recombination intermediates can then be resolved to complete the DNA repair process.

[0144] HDR templates used, for example, to introduce mutations or insert new nucleotides or nucleotide sequences into a gene require a certain amount of homology surrounding the target sequence that will be modified. Homology arms can be used that start at the CRISPR-induced DSB. In general, the insertion sites of the modification should be very close to the DSB, ideally less than 10 bp away, if possible. One important point to note is that the CRISPR enzymes may continue to cleave DNA once a DSB is introduced and repaired. As long as the gRNA target site/PAM site remain intact, the CRISPR nuclease will keep cutting and repairing the DNA. This repeated editing may be problematic if a very specific mutation or sequence is to be introduced into a gene of interest. To get around this, the repair template can be designed in such a way that it will ultimately block further CRISPR nuclease targeting after the initial DSB is repaired. Two common ways to block further editing are mutating the PAM sequence or the gRNA seed sequence. When designing a repair template, the size of the intended edit is to be taken into consideration. ssDNA templates (also referred to as ssODNs) are commonly used for smaller modifications. Small insertions/edits may require as little as 30-50 bases for each homology arm, and the best exact number may vary based on the gene of interest. 50-80 base homology arms are commonly used. For example, Richardson et al. (2016). *Nat Biotechnol.* 34(3):339-44) found that asymmetric homology arms (36 bases distal to the PAM and 91 bases proximal to the PAM) supported

[0145] HDR efficiencies up to 60%. Due to difficulties that might be associated with creating ssODNs longer than 200 bases, it is preferred to use dsDNA plasmid repair templates for larger insertions such as fluorescent proteins or selection cassettes into a gene of interest. These templates can have homology arms of at least 800 bp. To increase the frequency of HDR edits based on plasmid repair templates, self-cleaving plasmids can be used that contain gRNA target sites flanking the template. When the CRISPR nuclease and the appropriate gRNA(s) are present, the template is liberated

from the vector. To avoid plasmid cloning, it is possible to use PCR-generated long dsDNA templates. Moreover, Quadros et al. (2017) *Genome Biol.* 17;18(1):92) developed Easi-CRISPR, a technique that allows making large mutations and to take advantage of the benefits of ssODNs. To create ssODNs longer than 200 bases, RNA encoding the repair template are in vitro transcribed and then reverse transcriptase is used to create the complementary ssDNA. Easi-CRISPR works well in mouse knock-in models, increasing editing efficiency from 1-10% with dsDNA to 25-50% with ssODNs. Although HDR efficiency varies across loci and experimental systems, ssODN templates generally provide the highest frequency of HDR edits.

[0146] In accordance with a preferred embodiment of the ninth aspect the cell is not the natural host of a gene encoding said RNA-guided DNA endonuclease.

[0147] As discussed herein above, the RNA-guided DNA endonucleases of SEQ ID NOs 1-15 were isolated from a metagenome library comprising bacteria that cannot be cultured applying standard microbiological conditions. For this reason the exact nature of the bacteria from which SEQ ID NOs 1-15 were isolated is not known.

[0148] Hence, the cell that is not the natural host of any one of SEQ ID NOs 1-15 may be any bacterial cell from a bacterial strain that can be cultured or any eukaryotic cell. Such manipulated cells do not exist in nature.

[0149] In accordance with another preferred embodiment of the ninth aspect the cell is a eukaryotic cell, preferably a plant cell or animal cell.

[0150] Eukaryotic cells, plant cells and animal cells as well as eukaryotes, plants and animals from which cells may be obtained including preferred examples thereof have been described herein above in connection with the third and fourth aspect of the invention.

[0151] These cells may likewise be used in connection with the ninth aspect of the invention.

[0152] In accordance with a more preferred embodiment of the ninth aspect the method further comprises culturing the plant cell or animal cell to produce a plant or animal under conditions in which the RNA-guided DNA endonuclease is expressed and cleaves the nucleotide sequence at the target site to produce a modified nucleotide sequence; and selecting a plant or animal comprising said modified nucleotide sequence.

[0153] In this connection the cell(s) into which the components of the CRISPR-Cas system are to be introduced has/have to be a totipotent stem cell or a germ line cell (oocyte and/or sperm) or a collection of stem cells being capable of developing into a complete plant or animal. Means and method for culturing such cell(s) in order to produce a plant or animal are known in the art (see for example, <https://www.stembook.org/node/720>).

[0154] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. In case of conflict, the patent specification including definitions, will prevail.

[0155] The present invention relates in a tenth aspect to modified cells that have been produced by the method according to the ninth aspect of the invention for use in treatment of a disease in a subject.

[0156] The modified cells are preferably modified T lymphocytes and the disease to be treated is preferably cancer (Stadtmauer et al., Science 28 Feb. 2020: Vol. 367, Issue 6481, eaba7365).

[0157] The cells to be modified by the method of the ninth aspect of the invention are preferably obtained from the subject to be treatment and then the modified cells are used in accordance with the tenth aspect of the invention.

[0158] Regarding the embodiments characterized in this specification, in particular in the claims, it is intended that each embodiment mentioned in a dependent claim is combined with each embodiment of each claim (independent or dependent) said dependent claim depends from. For example, in case of an independent claim 1 reciting 3 alternatives A, B and C, a dependent claim 2 reciting 3 alternatives D, E and F and a claim 3 depending from claims 1 and 2 and reciting 3 alternatives G, H and I, it is to be understood that the specification unambiguously discloses embodiments corresponding to combinations A, D, G; A, D, H; A, D, I; A, E, G; A, E, H; A, E, I; A, F, G; A, F, H; A, F, I; B, D, G; B, D, H; B, D, I; B, E, G; B, E, H; B, E, I; B, F, G; B, F, H; B, F, I; C, D, G; C, D, H; C, D, I; C, E, G; C, E, H; C, E, I; C, F, G; C, F, H; C, F, I, unless specifically mentioned otherwise.

[0159] Similarly, and also in those cases where independent and/or dependent claims do not recite alternatives, it is understood that if dependent claims refer back to a plurality of preceding claims, any combination of subject-matter covered thereby is considered to be explicitly disclosed. For example, in case of an independent claim 1, a dependent claim 2 referring back to claim 1, and a dependent claim 3 referring back to both claims 2 and 1, it follows that the combination of the subject-matter of claims 3 and 1 is clearly and unambiguously disclosed as is the combination of the subject-matter of claims 3, 2 and 1. In case a further dependent claim 4 is present which refers to any one of claims 1 to 3, it follows that the combination of the subject-matter of claims 4 and 1, of claims 4, 2 and 1, of claims 4, 3 and 1, as well as of claims 4, 3, 2 and 1 is clearly and unambiguously disclosed.

[0160] The figures show.

[0161] FIG. 1: Unrooted tree of the amino acid sequences BMC01-BMC15

[0162] FIG. 2: Exemplary culture plates showing *E. coli* colonies 48 h after transformation (incubated at 30° C.) to visualize the colony depletion after the guided DNA targeting of the malQ gene using the BMC01, BMC02 BMC03, BMC04, BMC05, BMC07 or BMC08 nucleases and the partial recovery of colony growth (for BMC01, BMC03 and BMC04) induced by the co-transformation of the Ku-LigD proteins leading to a NHEJ mediated knock-out of the malQ gene

[0163] FIG. 3: Exemplary culture plates showing *E. coli* colonies 48 h after transformation (incubated at 30° C.) to visualize the colony depletion after the guided DNA targeting of the malQ gene using the BMC09, BMC10 BMC11, BMC12, BMC13, BMC14 or BMC15 nucleases and the partial recovery of colony growth (for BMC09 and BMC13) induced by the co-transformation of the Ku-LigD proteins leading to a NHEJ mediated knock-out of the malQ gene

[0164] FIG. 4: BMC characterization pipeline

[0165] Cell depletion and malQ gene knock-out in *E. coli* using the BMC09 nuclease and a Ku-LigD NHEJ strategy.

An excerpt of the malQ gene shows the deletion of 71 bp directly related to the target region of the BMC09 nuclease.

[0166] The examples illustrate the invention.

EXAMPLES

Example 1—Habitat Selection and DNA Preparation

[0167] An approach for the discovery of novel Cas proteins is to access metagenomics resources in terabase scale by Next Generation Sequencing of selected environmental DNA (e.g. 1 cm³ forest soil contains ~2.5×10¹⁰ bp DNA or ~20 million genes) and computational identification of CRISPR-Cas systems.

[0168] To this aim, metagenome habitats were selected from different locations in Germany. Aquatic, sediment and soil habitats were sampled and preprocessed before the DNA isolation. A staggered filtration process using a variety of filters of different pore sizes (0.1-20 µm) were used to enrich

[0169] Candidata Phyla Radiation (CPR) species containing a nearly untouched space of novel sequences (Hug et al. (2016) Nature Microbiology 1, 16048).

[0170] DNA was extracted from 0.2 and 0.1 µm filters and isolated using the PowerWater DNA isolation Kit (Qiagen).

Example 2—Next Generation Sequencing and Sequence Evaluation

[0171] DNA libraries for Next Generation Sequencing were prepared using the TruSeq PCR free Library Prep Kit (Illumina) and the metagenomics libraries were sequenced using a HiSeq 2500 (Illumina) in Rapid Mode (2×250 Bp) generating an average output of 50 Gbp per sample.

[0172] The generated paired end metagenome sequence libraries were assembled, annotated and evaluated to identify potential CRISPR operons by the identification of CRISPR repeats. The contigs containing the identified CRISPR repeat-spacer units were used to analyze the surrounding regions for the presence of Cas proteins. Open reading frames in close proximity to the CRISPR repeats were analyzed for their patterns and sequences by comparing them to genome and protein databases to identify potential novel Cas proteins.

[0173] BLAST nucleotide searches can be performed with the BLASTN program, score=100, wordlength=12, to obtain nucleotide sequences homologous to a nucleotide sequence encoding a protein of the invention. BLAST protein searches can be performed with the BLASTX program, score=50, wordlength=3, to obtain amino acid sequences homologous to a protein or polypeptide of the invention. To obtain gapped alignments for comparison purposes, Gapped BLAST (in BLAST 2.0) can be utilized as described in Altschul et al. (1997) Nucleic Acids Res. 25:3389. Alternatively, PSI-BLAST (in BLAST 2.0) can be used to perform an iterated search that detects distant relationships between molecules. See Altschul et al. (1997) supra. When utilizing BLAST, Gapped BLAST, PSI-BLAST, the default parameters of the respective programs (e.g., BLASTN for nucleotide sequences, BLASTX for proteins) can be used. See the website at www.ncbi.nlm.nih.gov. Alignment may also be performed manually by inspection.

Example 3: Construction of a Functional Genome
Editing System for *E. coli* Using Selected BMC
Sequences

3.1 CRISPR/BMC-Ec Vector Systems for Genome
Editing in *E. coli* BW25113

[0174] The necessary genetic elements for inducible expression of the BMC01, BMC02 BMC03, BMC04, BMC05, BMC07, BMC08, BMC09, BMC10, BMC11, BMC12, BMC13, BMC14 or BMC15 nuclease, for the constitutive expression of the guide RNA (gRNA) transcription and for the expression of the Ku-LigD proteins were provided on 16 separate vectors (CRISPR/BMC01-Ec, CRISPR/BMC02-Ec, CRISPR/BMC03-Ec, CRISPR/BMC04-Ec, CRISPR/BMC05-Ec, CRISPR/BMC07-Ec, CRISPR/BMC08-Ec, CRISPR/BMC09-Ec, CRISPR/BMC10-Ec, CRISPR/BMC11-Ec, CRISPR/BMC12-Ec, CRISPR/BMC13-Ec, CRISPR/BMC14-Ec, CRISPR/BMC15-Ec, CRISPR/gRNA-Ec and CRISPR/Ku-LigD-Ec)

[0175] In the following, the construction of the CRISPR/BMC01-Ec, CRISPR/gRNA-Ec and the CRISPR/Ku-LigD-Ec vector systems are described. The CRISPR/BMC02-Ec, CRISPR/BMC03-Ec, CRISPR/BMC04-Ec, CRISPR/BMC05-Ec, CRISPR/BMC07-Ec, CRISPR/BMC08-Ec, CRISPR/BMC09-Ec, CRISPR/BMC10-Ec, CRISPR/BMC11-Ec, CRISPR/BMC12-Ec, CRISPR/BMC13-Ec, CRISPR/BMC14-Ec and CRISPR/BMC15-Ec vector systems were constructed in an analogous approach as the CRISPR/BMC01-Ec vector system.

Design of the BMC01 *E. coli* Protein Expression
Vector

[0176] The synthetic 3888 bps BMC01 nucleotide sequence was codon optimized for expression in *E. coli* BW25113, using a bioinformatics application provided by the gene synthesis provider GeneArt (Thermo Fisher Scientific, Regensburg, Germany), SEQ ID NO: 33. For protein expression, the resulting synthetic gene was fused to the inducible araC-ParaBAD inducible promoter system (SEQ ID NO: 34) and the fdT terminator (SEQ ID NO: 35) (Otsuka & Kunisawa, Journal of Theoretical Biology 97 (1982), 415-436) The final BMC01 *E. coli* protein expression cassette was inserted by Gibson Assembly Cloning (NEB, Frankfurt, Germany) into an *E. coli* shuttle vector, containing all necessary genetic elements for episomal propagation and selection of recombinant *E. coli* cells.

CRISPR/BMC01-Ec Vector System

[0177] The complete nucleotide sequence of the constructed CRISPR/BMC01-Ec vector system is provided as SEQ ID NO: 36.

CRISPR/BMC02-Ec Vector System

[0178] The complete nucleotide sequence of the constructed CRISPR/BMC02-Ec vector system is provided as SEQ ID NO: 57.

CRISPR/BMC03-Ec Vector System

[0179] The complete nucleotide sequence of the constructed CRISPR/BMC03-Ec vector system is provided as SEQ ID NO: 37.

CRISPR/BMC04-Ec Vector System

[0180] The complete nucleotide sequence of the constructed CRISPR/BMC04-Ec vector system is provided as SEQ ID NO: 38.

CRISPR/BMC05-Ec Vector System

[0181] The complete nucleotide sequence of the constructed CRISPR/BMC05-Ec vector system is provided as SEQ ID NO: 58.

CRISPR/BMC07-Ec Vector System

[0182] The complete nucleotide sequence of the constructed CRISPR/BMC07-Ec vector system is provided as SEQ ID NO: 59.

CRISPR/BMC08-Ec Vector System

[0183] The complete nucleotide sequence of the constructed CRISPR/BMC08-Ec vector system is provided as SEQ ID NO: 60.

CRISPR/BMC09-Ec Vector System

[0184] The complete nucleotide sequence of the constructed CRISPR/BMC09-Ec vector system is provided as SEQ ID NO: 39.

CRISPR/BMC10-Ec Vector System

[0185] The complete nucleotide sequence of the constructed CRISPR/BMC10-Ec vector system is provided as SEQ ID NO: 61.

CRISPR/BMC11-Ec Vector System

[0186] The complete nucleotide sequence of the constructed CRISPR/BMC11-Ec vector system is provided as SEQ ID NO: 62.

CRISPR/BMC12-Ec Vector System

[0187] The complete nucleotide sequence of the constructed CRISPR/BMC12-Ec vector system is provided as SEQ ID NO: 63.

CRISPR/BMC13-Ec Vector System

[0188] The complete nucleotide sequence of the constructed CRISPR/BMC13-Ec vector system is provided as SEQ ID NO: 40.

CRISPR/BMC14-Ec Vector System

[0189] The complete nucleotide sequence of the constructed CRISPR/BMC14-Ec vector system is provided as SEQ ID NO: 64.

CRISPR/BMC15-Ec Vector System

[0190] The complete nucleotide sequence of the constructed CRISPR/BMC15-Ec vector system is provided as SEQ ID NO: 65.

Design of the Guide RNA (gRNA) Expression
Vector

[0191] The expression of the chimeric gRNA for specific malQ gene targeting by the BMC01, BMC02, BMC03,

BMC04, BMC05, BMC07, BMC08, BMC09, BMC10, BMC11, BMC12, BMC13, BMC14 or BMC15 nuclease was driven by the SacB RNA polymerase II promoter from *Bacillus megaterium* (SEQ ID NO: 41) (Richhardt et al., Applied Microbiology Biotechnology 86 (2010), 1959-1965) and terminated using the transcription T1 and T2 termination region of the *E. coli* *rrnB* gene (SEQ ID NO: 42) (Orosz et al., European Journal of Biochemistry 201 (1991), 653-659). The chimeric gRNA was composed of a constant 19 bps BMC family Stem-Loop sequence (SEQ ID NO: 32) fused to the malQ target-specific 24 bps spacer sequence (SEQ ID NO: 43) located inside the malQ gene of the *E. coli* genome.

[0192] The final gRNA expression cassette was inserted by Gibson Assembly Cloning (NEB, Frankfurt, Germany) into an *E. coli* shuttle vector, containing all necessary genetic elements for episomal propagation and selection of recombinant *E. coli* cells.

[0193] The construction of the final CRISPR/gRNA-Ec vector system was mediated by Gibson Assembly Cloning (NEB, Frankfurt, Germany).

[0194] The complete nucleotide sequence of the constructed CRISPR/gRNA-Ec vector system is provided as SEQ ID NO: 44.

Design of the Ku-LigD Expression Vector

[0195] CRISPR/BMC introduced double-strand breaks in the genome of recombinant *E. coli* cells were repaired by co-expression of the bacterial proteins Ku and LigD from *Mycobacterium tuberculosis* (Della et al. Science 306 (2004), 683-685), enabling marker-free gene knockout by error-prone non-homologous end joining (NHEJ) of targeted DNA in absence of homologous repair template (Yang et al., Biotechnology Letters 43 (2021), 2273-2281). The mycobacterial two-component NHEJ repair machine was ectopically provided on a plasmid containing different genetic elements for replication initiation and marker selection in comparison to the generated episomal BMC-expression vectors. The compatibility between the used plasmids for Ku-LigD and BMC-expression led to stable coexistence in the recombinant *E. coli* cells. For this, the Ku-LigD-expression plasmid possessed a pUC-derived high copy origin of replication (SEQ ID NO: 45) and additionally a nucleotide fragment from pRO1614, (SEQ ID NO: 46), that allowed stable maintenance of pUC-based cloning vectors in the genus *Pseudomonas* (West et al., Gene 148 (1) (1994), 81-86). The *Escherichia/Pseudomonas* shuttle vector contained the nourseothricin acetyltransferase gene (*nat1*) (SEQ ID NO: 47) as a dominant marker for selection of transformed cells (Krügel et al., Gene 127 (1) (1993), 127-131). At its N-terminus, the expressed acetyltransferase protein was fused to residues 1 to 30 of the Veg family protein from *Bacillus subtilis* (SEQ ID NO: 48) as generated in plasmid pHN15 (Kück and Hoff, Fungal Genetics Reports 53 (2006), article 3. <https://doi.org/10.4148/1941-4765.1106>). The gene expression of the fusion protein (SEQ ID NO: 49) was under the control of the constitutive veg Promoter from *Bacillus subtilis* (SEQ ID NO: 50). Both, the 822 bps nucleotide sequence of the ku gene (SEQ ID NO: 51) and the 2280 bps nucleotide sequence of ligD (SEQ ID NO: 52) from *Mycobacterium tuberculosis* were provided as synthetic DNA fragments by GeneArt (Thermo Fisher Scientific, Regensburg, Germany). The transcription and translation of both protein coding sequences were independently

controlled by use of two different constitutive promoters. Genexpression of ligD was regulated by the Veg Promoter RNA polymerase II promoter from *Bacillus subtilis* (SEQ ID NO: 53). The T7 terminator sequence (SEQ ID NO: 54) was used for transcription termination. Gene expression of ku was under the control of the SacB RNA polymerase II promoter from *Bacillus megaterium*. (SEQ ID NO: 55). For transcription termination the fdT terminator (SEQ ID NO: 35) (Otsuka & Kunisawa, Journal of Theoretical Biology 97 (1982), 415-436) was fused downstream to the coding sequence.

[0196] The complete nucleotide sequence of the constructed CRISPR/Ku-LigD-Ec vector system is provided as SEQ ID NO: 56.

[0197] The identity of all cloned DNA elements was confirmed by Sanger-Sequencing at LGC Genomics (Berlin, Germany).

3.2 *E. coli* Cultivation and Transformation

Transformation of Competent *E. coli* BW25113 Cells

[0198] In brief, a single colony of recombinant *E. coli* BW25113 host cells harboring the (CRISPR/BMC (NN)-Ec (Example 4) expression plasmid or carrying the CRISPR/BMC (NN)-Ec and CRISPR/Ku-LigD-Ec (Example 5)) plasmids for coupled expression was inoculated into 5 ml LB-Kanamycin (25 µg/ml) medium (Example 4) or LB-Kanamycin (25 µg/ml)/Nourseothricin (50 µg/ml) medium (Example 5) and incubated for 12 to 14 h at 37° C. on a horizontal shaker at 250 rpm. Overnight grown pre-cultures were diluted into fresh 60 ml LB-Kanamycin (25 µg/ml) (Example 4) or LB-Kanamycin (25 µg/ml)/Nourseothricin (50 µg/ml) medium (Example 5) to obtain an optical density at 600 nm (OD600) of 0.06. The inoculated medium was incubated at 30° C. on a horizontal shaker at 250 rpm until the culture reached an optical density at OD600 of 0.2. For induced BMC (NN) nuclease expression, 600 µl 20% L-(+)-arabinose (w/v) was added to the culture at a final concentration of 0.2% (v/v). The cultivation was continued until the culture reached an optical density at OD600 of 0.5. Subsequently 50 ml culture was transferred into one 50 ml conical tube and harvested by centrifugation at 4° C. for 5 min and 4000xg. Pelleted cells were resuspended in 50 ml ultrapure water (ice-cold) and centrifuged at 4° C. for 5 min and 4000xg.

[0199] This washing procedure was repeated twice. Washed cells were resuspended in 25 ml 10% (w/w) glycerin and re-centrifuged at 4° C. for 5 min and 4000xg. The pelleted cells were resuspended in 5 ml 10% (w/w) glycerin. After the final centrifugation step at 4° C. for 5 min and 4000xg, the cells were resuspended in 125 µl 10% (w/w) glycerin. Aliquots of 25 µl competent cells were stored at -80° C. until use. For the transformation procedure, aliquots of competent cells were thawed on ice and 50-100 ng plasmid DNA (CRISPR/gRNA-Ec) was added. Prepared cells were electroporated at 1800 V, 25 µF and 200 Ω in 1 mm gap size electroporation cuvettes, using the Bio-Rad Gene Pulser Xcell Electroporation System. Subsequently, 975 µL of NEB® 10-beta/Stable Outgrowth Medium was added immediately after the pulse to the transformed cells. Regeneration was done at 30° C. on a horizontal shaker at 250 rpm. Finally, 25-100 µl of the cell suspension was plated onto selective M9-Glucose agar plates supplemented with

Kanamycin (25 µg/ml)/Ampicillin (100 µg/ml) (Example 4) or Kanamycin (25 µg/ml)/Ampicillin (100 µg/ml)/Nourseothrin (50 µg/ml) (Example 5). Growth of transformed *E. coli* cells on selective agar plates was screened for 48 h at 30° C.

Example 4: *E. coli* Depletion Assay to Demonstrate the DNA Targeting Activity of the Novel BMC Nucleases

[0200] To evaluate and visualize the DNA targeting activity of the BMC family nucleases a so-called depletion assays was carried out wherein the survival rate of the *E. coli* cells after the nuclease targeting is monitored in comparison to a negative control. A lower survival rate means better nuclease activity as *E. coli* cells are not able to repair DNA double strand breaks using non homologues end joining (NHEJ) the targeting of the genomic DNA using a CRISPR nuclease leads to cell death.

[0201] For this experimental approach, the CRISPR/BMC01-Ec, CRISPR/BMC02-Ec, CRISPR/BMC03-Ec, CRISPR/BMC04-Ec, CRISPR/BMC05-Ec, CRISPR/BMC07-Ec, CRISPR/BMC08-Ec, CRISPR/BMC09-Ec, CRISPR/BMC10-Ec, CRISPR/BMC11-Ec, CRISPR/BMC12-Ec, CRISPR/BMC13-Ec, CRISPR/BMC14-Ec, or CRISPR/BMC15-Ec vector system was co-transformed together with the CRISPR/gRNA-Ec vector system containing a spacer sequence to target the *E. coli* malQ gene MalQ is coding for 4- α -glucanotransferase and is an essential gene for starch metabolism, yet not essential for survival of *E. coli* cells when cultivated on a medium containing glucose as a carbon source.

[0202] In parallel, negative control experiments using the CRISPR/BMC01-Ec, CRISPR/BMC02-Ec, CRISPR/BMC03-Ec, CRISPR/BMC04-Ec, CRISPR/BMC05-Ec, CRISPR/BMC07-Ec, CRISPR/BMC08-Ec,

[0203] CRISPR/BMC09-Ec, CRISPR/BMC10-Ec, CRISPR/BMC11-Ec, CRISPR/BMC12-Ec, CRISPR/BMC13-Ec, CRISPR/BMC14-Ec, or CRISPR/BMC15-Ec vector system co-transformed together with the CRISPR/gRNA-Ec vector lacking a spacer sequence targeting the *E. coli* genome were performed to demonstrate the dependency of the Cas proteins to be guided to the target DNA region by a specific spacer.

[0204] After transformation and 48 h incubation at 30° C. the culture plates were analyzed by counting the number of grown colonies.

Results

[0205] All experiments were carried out in 5 biological replicates and the results obtained from these replicates were combined to evaluate the DNA targeting of the BMC01, BMC02, BMC03, BMC04, BMC05, BMC07, BMC08, BMC09, BMC10, BMC11, BMC12, BMC13, BMC14, and BMC15 nucleases (Exemplary plates are shown in FIG. 2 and FIG. 3).

[0206] As described above, each of the BMC01, BMC02, BMC03, BMC04, BMC05, BMC07, BMC08, BMC09, BMC10, BMC11, BMC12, BMC13, BMC14, or BMC15 nucleases was co-transformed into *E. coli* with a gRNA targeting the *E. coli* malQ gene to visualize a) the activity and b) the DNA targeting efficiency of the BMC nucleases. After co-transformation and incubation of the plates at 30° C. for 48 h, all BMC nucleases (BMC01, BMC02, BMC03, BMC04, BMC05, BMC07, BMC08, BMC09, BMC10,

BMC11, BMC12, BMC13, BMC14, and BMC15) showed a strong colony reduction (>99.9%) in comparison to the negative control proving the highly efficient gRNA dependent DNA targeting of the BMC nucleases.

Example 5: NHEJ Mediated Genome Editing in *E. coli* Using Selected BMC Nucleases and a Ku-LigD Mediated Strategy

[0207] To further evaluate and visualize the genome editing activity of the BMC nucleases, five representative nucleases (BMC01, BMC03, BMC04, BMC09 and BMC13) were selected and a Ku-LigD mediated strategy was used to knock-out the malQ gene located on the *E. coli* genome. As *E. coli* cells are natively not able to repair DNA double strand breaks using nonhomologous end joining (NHEJ), the targeting of the genomic DNA using a CRISPR nuclease leads to cell death.

[0208] To prevent cell death and to monitor NHEJ derived knock-out of the malQ gene (malQ is a non-essential gene and a knockout of malQ does not have any effect on the phenotype of the *E. coli* cells when cultured under the conditions used in this experimental approach), the proteins Ku-LigD (both play an important role in the DNA repair machinery) were co-transformed into the *E. coli* cells providing the cells with the ability to repair DNA double strand breaks using NHEJ (see e.g. WO 2017/109167).

[0209] For this experimental approach, the CRISPR/BMC01-Ec, CRISPR/BMC03-Ec, CRISPR/BMC04-Ec, CRISPR/BMC09-Ec or CRISPR/BMC13-Ec vector, respectively, were co-transformed together with the CRISPR/Ku-LigD-Ec vector system and the CRISPR/gRNA-Ec vector system containing a spacer sequence to target the *E. coli* malQ gene.

[0210] In parallel, negative control experiments using the CRISPR/BMC01-Ec, CRISPR/BMC03-Ec, CRISPR/BMC04-Ec, CRISPR/BMC09-Ec or CRISPR/BMC13-Ec vectors, respectively, were co-transformed together with the CRISPR/Ku-LigD-Ec vector system and the CRISPR/gRNA-Ec vector system lacking a spacer sequence targeting the *E. coli* genome were performed to demonstrate the dependency of the Cas proteins to be guided to the target DNA region by a specific spacer.

[0211] After transformation and 48 h incubation at 30° C. the culture plates were analyzed by counting the number of grown colonies.

Results

[0212] All experiments were carried out in 5 biological replicates and the results obtained from these replicates were combined to evaluate the genome editing activity of the BMC01, BMC03, BMC04, BMC09 and BMC13 nucleases. Exemplary plates are shown in FIG. 2 and FIG. 3.

[0213] To visualize the genome editing activity of the BMC nucleases experiments were carried out using a Ku-LigD mediated NHEJ strategy described above. First, 48 h after co-transformation of the respective BMC nuclease, the gRNA (targeting the malQ gene) and the Ku-LigD expression system the plates were evaluated based on the grown colonies in comparison to the negative control and the results obtained in the cell depletion assay (Example 4). Overall, all tested BMC nucleases showed a strong colony reduction ($\approx$ 98%) in comparison of the negative control but a weaker colony reduction in comparison to the results

obtained in Example 4 (>99.9%). These results can be explained by the fact, that the expression of the Ku-LigD proteins enable *E. coli* to repair DNA double strand breaks using the NHEJ DNA repair mechanism. Despite from the repair of the DNA double strand break, it is known that NHEJ mediated DNA repair can be faulty, leading to indels (insertions/deletions) and frame shift mutations at the targeted position inside genome. Those cells, where the BMC mediated DNA double strand break was repaired by the NHEJ mechanism but the genomic sequence was altered due to introduced indels were able to survive the BMC treatment, as the PAM and/or spacer sequences were altered due to the indel mutation and the BMC nucleases are not able to target the genome again.

[0214] To proof the knock-out of the malQ gene, two colonies from every plate (10 colonies for every BMC nuclease) were isolated and the genomic locus of interest (inside of the malQ gene) was sequenced using Sanger sequencing.

[0215] The results of this experimental approach showed that a NHEJ mediated knock-out of the malQ gene was detectable on the sequence level for all selected BMC nucleases

[0216] BMC01: 9 out of 10 evaluated colonies showed a NHEJ mediated knock-out of the malQ gene
[0217] BMC03: 7 out of 10 evaluated colonies showed a NHEJ mediated knock-out of the malQ gene
[0218] BMC04: 7 out of 10 evaluated colonies showed a NHEJ mediated knock-out of the malQ gene
[0219] BMC09: 10 out of 10 evaluated colonies showed a NHEJ mediated knock-out of the malQ gene
[0220] BMC13: 8 out of 10 evaluated colonies showed a NHEJ mediated knock-out of the malQ gene
[0221] To demonstrate the complete workflow for the characterization of the BMC nucleases FIG. 4 shows the characterization of the BMC09 nuclease: Negative control, cell depletion assay, colony growth with co-expression of Ku-LigD and the sequencing of the targeted locus (71 bp deletion inside of the malQ gene) of one *E. coli* colony.

Summary

[0222] Taken together, the results obtained in Examples 4 and 5 demonstrate that all BMC proteins disclosed here are novel CRISPR nucleases with a high DNA targeting and genome editing efficiency and with low similarity to hitherto described CRISPR nucleases.

SEQUENCE LISTING

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Leu Thr Ile Phe Glu Glu Thr Ile Lys Asn Asp Gln Lys Arg Tyr Glu
35             40             45

Ala Tyr Leu Ala Ile Lys Pro Leu Val Asp Glu Thr His Lys His Phe
50             55             60

Ile Gln Thr Val Leu Ser Gly Leu Thr Asp Leu Ile Lys Ser Asp Glu
65             70             75             80

Met Lys Asn Tyr Leu Glu His Lys Asn Leu Ile Arg Gln Lys Asp Val
85             90             95

Glu Glu Lys Val Lys Thr Lys Ser Ile Asp Val Ile Asn Lys Ile Glu
100            105            110

Lys Asp Trp Arg Lys Arg Val Ser Asp Ser Phe Thr Lys His Pro Gln
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Lys	Asp	Tyr	Ser	Gln	Asn	Lys	Leu	Lys	Ile	Asn	Phe	Asp	Lys	Gly	Thr	
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Arg	His	Lys	Lys	Gly	Asp	Asn	Phe	Asn	Ile	Ser	Asp	Cys	His	Tyr	Leu	660	665	670
Ile	Asp	Phe	Tyr	Lys	Thr	Ala	Ile	Asn	Lys	His	Pro	Glu	Trp	Ser	Gln	675	680	685
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Tyr	Tyr	Cys	Val	Ile	Asn	Val	Ala	Ser	Gly	Lys	Ile	Val	Glu	Gln	Gly	885	890	895
Ser	Phe	Asn	Gln	Ile	Ser	Thr	Asn	Tyr	Thr	Pro	Glu	Gln	Ile	Thr	Asp	900	905	910
Asp	Gly	Glu	Ile	Ile	Lys	Gly	Glu	Thr	Val	Asn	Lys	Thr	Thr	Asp	Tyr	915	920	925
His	Asn	Leu	Leu	Asn	Thr	Lys	Glu	Gly	Asp	Arg	Gln	Lys	Ala	Arg	Lys	930	935	940
Asn	Trp	Gln	Thr	Ile	Glu	Asn	Ile	Lys	Glu	Leu	Lys	Ala	Gly	Tyr	Leu	945	950	955
Ser	Asn	Val	Ile	His	Lys	Ile	Ser	Gln	Leu	Met	Val	Lys	Tyr	Asn	Ala	965	970	975
Phe	Val	Val	Leu	Glu	Glu	Leu	Lys	Tyr	Gly	Phe	Lys	Arg	Gly	Arg	Phe	980	985	990

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Lys	Val	Glu	Lys	Gln	Val	Tyr	Gln	Lys	Phe	Glu	Lys	Ala	Leu	Ile	Asp	995	1000	1005
Lys	Leu	Asn	Tyr	Leu	Val	Phe	Lys	Asp	Arg	Ala	Pro	Ala	Glu	Val		1010	1015	1020
Gly	Gly	Val	Leu	Asn	Ala	Leu	Gln	Leu	Ala	Pro	Pro	Val	Ala	Ser		1025	1030	1035
Tyr	Ile	Asp	Ile	Gly	Lys	Gln	Ala	Gly	Phe	Leu	Phe	Tyr	Val	Pro		1040	1045	1050
Ala	His	His	Thr	Ser	Lys	Ile	Cys	Pro	Trp	Thr	Gly	Phe	Val	Asp		1055	1060	1065
Trp	Leu	Lys	Pro	Arg	Tyr	Asp	Gly	Ile	Asp	Lys	Ala	Lys	Ala	Phe		1070	1075	1080
Phe	Thr	Cys	Phe	Glu	Ser	Ile	His	Phe	Asn	Thr	Gln	Lys	Asn	Tyr		1085	1090	1095
Phe	Glu	Phe	Ala	Phe	Asp	Tyr	Glu	Lys	Phe	Arg	Gly	Asn	Ile	Asn		1100	1105	1110
His	Leu	Pro	Glu	Gly	Leu	Lys	Arg	Thr	Ser	Trp	Thr	Leu	Cys	Ser		1115	1120	1125
His	Asn	Ser	Leu	Arg	Asp	Ile	Ala	Thr	Lys	Asp	Lys	Asn	Gly	Asn		1130	1135	1140
Trp	Pro	Tyr	Lys	Gln	Ile	Asn	Leu	Thr	Ala	Glu	Leu	Leu	Glu	Ile		1145	1150	1155
Leu	Lys	Thr	Leu	Asn	Pro	Arg	Asn	Gly	Glu	Asn	Leu	Val	Glu	Arg		1160	1165	1170
Ile	Ile	Glu	Met	Asn	Asp	Lys	Lys	Phe	Phe	Glu	Ser	Leu	Met	Trp		1175	1180	1185
Ala	Leu	Arg	Val	Leu	Leu	Gln	Leu	Arg	Tyr	Gly	Tyr	Ile	Lys	Arg		1190	1195	1200
Asn	Asn	Glu	Gly	Ile	Ile	Ile	Glu	Glu	Val	Asp	Tyr	Ile	Leu	Ser		1205	1210	1215
Pro	Val	Ala	Asn	Glu	Asn	Gly	Glu	Phe	Phe	Asp	Ser	Arg	Asn	Phe		1220	1225	1230
Val	Asn	Ile	Glu	Lys	Ala	Asp	Phe	Pro	Lys	Asp	Ala	Asp	Ala	Asn		1235	1240	1245
Gly	Ala	Tyr	Asn	Ile	Ala	Arg	Lys	Gly	Leu	Leu	Leu	Ile	Ala	Gln		1250	1255	1260
Asn	Ile	Asn	Asn	Ala	Lys	Ile	Asn	Asp	Lys	Gly	Glu	Val	Lys	Cys		1265	1270	1275
Asp	Leu	Gln	Ile	Asp	Lys	Thr	Thr	Trp	Phe	Asn	Trp	Val	Gln	Ser		1280	1285	1290
Lys	Ser															1295		

<210> SEQ ID NO 2

<211> LENGTH: 1107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 2

Met	Pro	Lys	Glu	Asp	Phe	Asp	Lys	Ala	Cys	Ile	Tyr	Leu	Ser	Asn	Phe
1			5				10					15			

Asp	Lys	Phe	Ser	Thr	Tyr	Phe	Val	Gly	Phe	Asn	Gln	Asn	Arg	Glu	Asn	
			20				25						30			
Leu	Tyr	Thr	Asp	Glu	Glu	Gln	Ala	Thr	Ala	Ile	Pro	Tyr	Arg	Ile	Ile	
			35				40						45			
Asn	Asp	Asn	Met	Val	Arg	His	Phe	Asp	Asn	Cys	Arg	Lys	Phe	Glu	Lys	
			50				55						60			
Ile	Val	Lys	Lys	Tyr	Gly	Asp	Ile	Ser	Asn	Val	Leu	Ser	Thr	Tyr	Lys	
			65				70						75			
Glu	Phe	Phe	Ala	Pro	Asp	Cys	Phe	Lys	Asn	Lys	Leu	Asn	Gln	Ser	Gln	
			85				90						95			
Ile	Asp	His	Tyr	Asn	Asn	Thr	Ile	Gly	His	Thr	Ala	Asp	Asp	Ile	Tyr	
			100				105						110			
Gly	Val	Gly	Ile	Asn	Gln	Ile	Leu	Ser	Lys	Tyr	Lys	Gln	Asp	Asn	Lys	
			115				120						125			
Leu	Asn	Ser	Ser	Asp	Leu	Pro	Leu	Ile	Ser	Lys	Leu	Tyr	Lys	Gln	Ile	
			130				135						140			
Leu	Ser	Asp	Thr	Glu	Ser	Tyr	Ala	Ile	Glu	Asn	Phe	Ala	Asp	Asp	Lys	
			145				150						155			
Met	Met	Leu	Asn	Ala	Val	Asp	Lys	Glu	Tyr	Ser	Arg	Ile	Lys	Glu	Asn	
			165				170						175			
Asp	Val	Phe	Ile	Asn	Ile	Glu	Thr	Cys	Met	Asn	Glu	Tyr	Leu	Thr	Leu	
			180				185						190			
Glu	Asn	Ser	His	Met	Ile	Tyr	Leu	Lys	Asn	Asp	Ser	Ser	Leu	Thr	Asp	
			195				200						205			
Ile	Ser	Asn	Lys	Leu	Trp	Glu	Asp	Trp	Ala	Phe	Val	Lys	Asn	Ala	Ile	
			210				215						220			
Gln	Lys	Tyr	Ser	Lys	Glu	Ile	Leu	Cys	Leu	Ser	Asp	Lys	Lys	Ile	Glu	
			225				230						235			
Asp	Met	Leu	Lys	Met	Ser	His	Tyr	Ser	Ile	Ser	Phe	Val	Gln	Asn	Ser	
			245				250						255			
Val	Tyr	Tyr	Tyr	Val	Asp	Asn	Tyr	Met	Glu	Ser	Cys	Glu	Asp	Lys	Arg	
			260				265						270			
Lys	Ser	Ile	Ile	Asp	Tyr	Ile	Lys	Thr	Phe	Tyr	Ser	Ile	Lys	Tyr	Asn	
			275				280						285			
Asn	Val	Phe	Ser	Cys	Tyr	Lys	Glu	Ala	Glu	Ala	Val	Leu	Arg	Leu	Asp	
			290				295						300			
Ser	Ile	His	Lys	Asn	Arg	Arg	Ser	Pro	Val	Asp	Lys	Asn	Gly	Ile	Gly	
			305				310						315			
Gly	Glu	Gly	Phe	Ala	Gln	Ile	Glu	Lys	Ile	Lys	Asn	Phe	Leu	Asp	Ser	
			325				330						335			
Ile	Leu	Glu	Val	Lys	Asn	Phe	Leu	Asn	Pro	Leu	Tyr	Leu	Ile	Lys	Ser	
			340				345						350			
Gly	Lys	Met	Ala	Glu	Ile	Glu	Asp	Lys	Ser	Glu	Glu	Phe	Tyr	Asn	Arg	
			355				360						365			
Phe	Asn	Glu	Leu	Tyr	Asn	Ser	Leu	Ser	Asp	Thr	Thr	Tyr	Leu	Tyr	Asn	
			370				375						380			
Lys	Val	Arg	Asn	Tyr	Leu	Thr	Lys	Lys	Pro	Tyr	Lys	Lys	Glu	Lys	Phe	
			385				390						395			
Lys	Met	Asn	Phe	Glu	Asn	Ser	Thr	Leu	Leu	Ser	Gly	Trp	Asp	Val	Asn	
			405				410						415			
Lys	Glu	Asn	Cys	Ser	Asn	Ser	Ile	Ile	Leu	Ile	Arg	Asn	Gly	Lys	Tyr	

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420							425					430				
Tyr	Leu	Gly	Ile	Ile	Asp	Lys	Gln	Cys	Gly	Asn	Met	Phe	Asn	Phe	Lys	
		435					440					445				
Ile	Asp	Ala	Glu	Asp	Asn	Glu	Lys	Lys	Arg	Lys	Glu	Lys	Glu	Asp	Leu	
	450					455					460					
Ala	Glu	Asp	Ile	Leu	Ser	Asp	Gly	Ser	Asp	Ser	Tyr	Tyr	Glu	Lys	Met	
465					470					475					480	
Val	Tyr	Lys	Leu	Leu	Pro	Asp	Pro	Ser	Lys	Met	Leu	Pro	Lys	Val	Phe	
			485						490					495		
Phe	Ser	Asn	Lys	Ser	Ile	Asp	Phe	Tyr	Ala	Pro	Ser	Glu	Asp	Ile	Lys	
		500						505					510			
Tyr	Ile	Arg	Glu	Asn	Gly	Leu	Phe	Lys	Lys	Asp	Ala	Lys	Asn	Lys	Lys	
	515						520					525				
Ala	Leu	Tyr	Ile	Trp	Ile	Glu	Phe	Met	Gln	Asn	Ser	Leu	Lys	Lys	His	
530						535					540					
Pro	Glu	Trp	Ser	Asn	Tyr	Phe	Asn	Phe	Asn	Phe	Lys	Pro	Ser	Thr	Glu	
545					550					555					560	
Tyr	Ala	Asp	Val	Ser	Glu	Phe	Tyr	Lys	Gln	Val	Ser	Asp	Gln	Gly	Tyr	
			565						570					575		
Ser	Leu	Ser	Phe	Asp	Lys	Ile	Lys	Asp	Ser	Tyr	Ile	Glu	Ser	Lys	Ile	
			580					585					590			
Lys	Ser	Gly	Glu	Leu	Phe	Leu	Phe	Glu	Ile	Tyr	Asn	Lys	Asp	Phe	Ser	
	595						600					605				
Pro	Tyr	Ser	Lys	Gly	Asn	Pro	Asn	Leu	His	Thr	Ile	Tyr	Trp	Lys	Ser	
610						615					620					
Ile	Phe	Asp	Lys	Glu	Asn	Leu	Ser	Asn	Val	Val	Ile	Lys	Leu	Asn	Gly	
625					630					635					640	
Gln	Ala	Glu	Ile	Phe	Phe	Arg	Pro	Ala	Ser	Leu	Lys	Arg	Asn	Glu	Val	
			645						650					655		
Val	Val	His	Arg	Ala	Lys	Glu	Asn	Ile	Leu	Asn	Lys	Asn	Pro	Leu	Asn	
			660					665					670			
Pro	Lys	Lys	Glu	Ser	Met	Phe	Glu	Tyr	Asp	Ile	Val	Lys	Asp	Lys	Arg	
	675						680					685				
Tyr	Thr	Gln	Asp	Lys	Phe	Phe	Phe	His	Cys	Pro	Ile	Thr	Leu	Asn	Phe	
690					695						700					
Lys	Ser	Gly	Asn	Val	Gly	Lys	Phe	Asn	Asp	Lys	Val	Asn	Gln	Phe	Leu	
705				710						715					720	
Lys	Asn	Asn	Pro	Asp	Val	Asn	Val	Ile	Gly	Phe	Asp	Arg	Gly	Glu	Arg	
			725					730					735			
His	Leu	Leu	Tyr	Cys	Asn	Val	Leu	Asn	Gln	Lys	Gly	Glu	Ile	Ile	Glu	
	740							745					750			
Gln	Lys	Ser	Phe	Asn	Val	Ile	Glu	Asn	Lys	Asn	Asn	Gly	Ile	Thr	Gln	
	755						760					765				
Lys	Val	Asp	Tyr	His	Asn	Leu	Leu	Asp	Arg	Lys	Glu	Lys	Glu	Arg	Asp	
770					775						780					
Ala	Ser	Arg	Lys	Ser	Trp	Ser	Thr	Ile	Glu	Asn	Ile	Lys	Glu	Leu	Lys	
785					790					795					800	
Glu	Gly	Tyr	Leu	Ser	Asn	Val	Val	His	Glu	Ile	Ser	Glu	Leu	Ile	Ile	
			805					810					815			
Lys	Tyr	Asn	Ala	Ile	Leu	Val	Leu	Glu	Asp	Leu	Asn	Phe	Glu	Phe	Lys	
		820						825					830			

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Lys Gly Arg Phe Lys Ile Glu Lys Gln Val Tyr Gln Lys Phe Glu Lys
 835 840 845
 Ala Leu Ile Asp Lys Leu Ser Tyr Met Val Phe Lys Lys Glu Glu Ser
 850 855 860
 Asn Lys Pro Gly His Ser Leu Met Ala Tyr Gln Leu Ala Ser Pro Phe
 865 870 875 880
 Glu Ser Phe Gln Lys Leu Gly Lys Gln Cys Gly Phe Ile Phe Tyr Val
 885 890 895
 Asn Ser Asn Tyr Thr Ser Lys Ile Asp Pro Val Thr Gly Phe Val Asn
 900 905 910
 Leu Leu Lys Ile Lys Tyr Glu Ser Val Asp Lys Ser Cys Lys Phe Ile
 915 920 925
 Asn Asp Lys Phe Asp Asp Ile Arg Tyr Asn Ala Asp Arg Glu Tyr Phe
 930 935 940
 Glu Phe Thr Phe Asp Asn Gly Lys Trp Thr Ala Cys Ser His Gly Lys
 945 950 955 960
 Glu Arg Tyr Arg Tyr Asn Arg Asn Asp Lys Lys Tyr Asn Cys Phe Asp
 965 970 975
 Val Thr Glu Glu Leu Lys Ser Leu Phe Asn Lys Tyr Glu Ile Asp Phe
 980 985 990
 Lys Ala Gly Thr Asp Ile Lys Lys Ser Ile Cys Gln Val Gln Asp Lys
 995 1000 1005
 Asn Phe His Ser Glu Leu Leu Phe Asn Leu Ser Leu Ile Val Gln
 1010 1015 1020
 Leu Arg His Thr Tyr Lys Asn Gly Asp Ile Glu Lys Asp Phe Ile
 1025 1030 1035
 Leu Ser Pro Ile Met Asp Lys Glu Thr Gly Lys Phe Phe Asp Ser
 1040 1045 1050
 Arg Glu Tyr Glu Asn Leu Glu Asn Ser Leu Leu Pro Thr Asn Ala
 1055 1060 1065
 Asp Ser Asn Gly Ala Tyr Asn Ile Ala Arg Lys Gly Leu Leu Thr
 1070 1075 1080
 Leu Arg Gln Ile Asp Lys Asp Gly Lys Pro Ser Asn Ile Ser Asn
 1085 1090 1095
 Lys Glu Trp Phe Asp Phe Val Gln Lys
 1100 1105

<210> SEQ ID NO 3
 <211> LENGTH: 1323
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Bacterial species

 <400> SEQUENCE: 3

Met Glu Lys Asn Leu Asn Tyr Leu Glu Arg Phe Thr Lys His Tyr Asn
 1 5 10 15
 Thr Lys Lys Thr Leu Lys Asn Lys Leu Ile Pro Tyr Gly Asn Thr Ala
 20 25 30
 Glu Asn Met Ile Lys Asn Asn Ile Ile Ser Asn Glu Lys Gln Ile Ile
 35 40 45
 Leu Ser Ala Lys Lys Gln Lys Gln Ser Ile Asp Phe Leu Gln Lys Glu
 50 55 60

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Tyr	Ile	Glu	Asn	Lys	Leu	Ser	Glu	Ile	Thr	Leu	Pro	Tyr	Leu	Asn	Asp	
65					70					75					80	
Tyr	Tyr	Asn	Glu	Phe	Ile	Lys	Asn	Lys	Lys	Glu	Arg	Asp	Thr	Asp	Val	
			85						90					95		
Ile	Asp	Asn	Ile	Glu	Ile	Ala	Met	Arg	Lys	His	Ile	Ser	Lys	Ser	Leu	
			100					105						110		
Thr	Glu	Asn	Gly	Asn	His	Lys	Lys	Tyr	Leu	Asn	Lys	Glu	Val	Phe	Asp	
			115					120					125			
Ile	Ile	Ser	Glu	Lys	Lys	Glu	Leu	Tyr	Tyr	Asp	Val	Thr	Phe	Lys	Arg	
			130				135				140					
Asn	Ala	Thr	Tyr	Leu	Ser	Asp	Tyr	Phe	Gln	Ser	Arg	Val	Asn	Leu	Tyr	
145					150					155					160	
Lys	Asp	Ser	Asn	Lys	Ser	Ser	Thr	Ile	Ala	Ser	Arg	Cys	Ile	Asn	Ile	
				165					170						175	
Asn	Leu	Pro	Ile	Phe	Ala	Lys	Asn	Ile	Val	Leu	Phe	Asn	Phe	Ile	Lys	
			180					185					190			
Asn	Lys	Ala	Asn	Ile	Ile	Phe	Asp	Asp	Leu	Lys	Glu	Ile	Thr	Asp	Asp	
			195					200					205			
Glu	Tyr	Thr	Leu	Asp	Ser	Ile	Phe	Ser	Ile	Asp	Phe	Phe	Asn	Met	Val	
			210				215						220			
Leu	Ser	Gln	Lys	Gly	Ile	Asp	Tyr	Tyr	Asn	Thr	Ile	Leu	Gly	Gly	Met	
225					230					235					240	
Thr	Lys	Glu	Asp	Gly	Lys	Lys	Ile	Lys	Gly	Ile	Asn	Glu	Tyr	Ile	Asn	
				245					250					255		
Leu	Tyr	Asn	Gln	Asn	Val	Lys	Asp	Glu	Lys	Asn	Lys	Leu	Pro	Tyr	Pro	
			260					265					270			
Lys	Lys	Leu	Lys	Lys	Gln	Leu	Leu	Ser	Asp	Ile	Asn	Ser	Tyr	Ser	Ala	
			275					280					285			
Arg	Phe	Glu	Lys	Phe	Asp	Thr	Glu	Gln	Glu	Met	Val	Lys	Ser	Ile	Lys	
			290				295				300					
Ser	Leu	Val	Glu	Asn	Asp	Leu	Phe	Gln	Gly	Glu	Leu	Phe	Asp	Lys	Lys	
305					310					315					320	
Val	Asp	Ile	Leu	Lys	Glu	Thr	Glu	Arg	Leu	Leu	Glu	Arg	Ile	Ser	Glu	
				325					330					335		
Tyr	Asp	Ser	Asn	Ala	Leu	Phe	Ile	Thr	Glu	Lys	Asn	Ile	Ser	Tyr	Ile	
				340				345						350		
Ser	Ile	Asp	Ile	Phe	Asn	Asp	Lys	Phe	Phe	Ile	Lys	Thr	Ala	Ile	Glu	
			355					360					365			
Tyr	Phe	Tyr	Glu	Asn	Asn	Ile	Cys	Pro	Asp	Tyr	Arg	Lys	Ile	Tyr	Asp	
			370				375				380					
Asn	Ala	Ser	Lys	Asn	Lys	Arg	Lys	Gln	Leu	Gly	Lys	Glu	Lys	Asn	Lys	
385					390					395					400	
Val	Ile	Lys	Gln	Lys	Ser	Phe	Ser	Ile	Ser	Phe	Leu	Gln	Asp	Ala	Ile	
				405					410					415		
Thr	Phe	Tyr	Ile	Lys	Asp	Ser	Gly	Ile	Asn	Lys	Ile	Ser	Glu	Asn	Cys	
				420				425					430			
Ile	Ile	Asn	Tyr	Phe	Lys	Lys	His	Thr	Ile	Lys	Leu	Thr	Glu	Leu	Phe	
				435				440					445			
Gly	Lys	Val	Tyr	Glu	Asp	Tyr	Asn	Val	Ile	Lys	Pro	Ile	Leu	Glu	Gln	
				450			455				460					

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His	Leu	Val	Glu	Tyr	Glu	Gly	Lys	Ser	Ile	Ser	Lys	Asp	Ser	Ile	Lys	
465					470					475					480	
Arg	Ser	Lys	Ile	Lys	Leu	Phe	Ser	Glu	Asn	Leu	Lys	Asn	Ile	Phe	Tyr	
			485						490					495		
Phe	Ile	Arg	Pro	Leu	Asn	Ile	Ile	Glu	Glu	Ala	Leu	Asn	Tyr	Asp	Thr	
			500					505					510			
Ser	Phe	Tyr	Thr	Pro	Phe	Asn	Ile	Leu	Phe	Glu	Glu	Ile	Lys	Lys	Phe	
		515					520					525				
Asn	Lys	Leu	Tyr	Asp	Lys	Ile	Arg	Asn	Phe	Ile	Thr	Lys	Lys	Pro	Phe	
	530					535					540					
Asn	Asp	Glu	Glu	Ile	Asn	Leu	Tyr	Phe	Gly	Ile	Pro	Asn	Leu	Gly	Gly	
545					550					555					560	
Gly	Phe	Ile	Asp	Ser	Gln	Thr	Asp	Lys	Ser	Asn	Asn	Gly	Thr	Gln	Tyr	
			565						570					575		
Cys	Thr	Tyr	Leu	Phe	Arg	Lys	Lys	Asn	Gln	Leu	Leu	Asn	Trp	Glu	Tyr	
			580					585					590			
Phe	Val	Gly	Ile	Ser	Lys	Asn	Lys	His	Leu	Phe	Arg	Glu	Lys	Glu	Asn	
		595					600					605				
Ile	Glu	Leu	Asn	Ser	Asp	Glu	Thr	Ser	Phe	Gln	Arg	Tyr	Ser	Phe	Tyr	
	610					615					620					
Thr	Pro	Lys	Asp	Lys	Ser	Ile	Tyr	Gly	Ser	Ser	Tyr	Phe	Ser	Ala	Asn	
625					630					635					640	
Glu	Lys	Asn	Tyr	Lys	Asp	Asp	Lys	Gln	Glu	Phe	Ile	Asn	Ile	Ile	Asn	
			645						650					655		
Asn	Ile	Val	Asn	Asn	Ser	Gly	Asn	Glu	Leu	Ala	Ile	Lys	Glu	Leu	Lys	
		660						665					670			
Lys	Tyr	Ile	Asn	Asn	Ser	Thr	Glu	Asn	Ser	Glu	Thr	Pro	Asn	Gly	Cys	
		675					680					685				
Leu	Ser	Val	Leu	Lys	Asn	Lys	Cys	Asn	Glu	Ile	Tyr	Asn	Leu	Val	Ile	
	690					695					700					
Asn	His	Asp	Asp	Phe	Lys	Glu	Lys	Asn	Glu	Asp	Ile	Ile	Asn	Lys	Leu	
705					710					715					720	
Lys	Asn	Thr	Leu	Ser	Lys	Leu	Ser	Lys	Val	Pro	Gln	Ala	Lys	Glu	Leu	
			725						730					735		
Ile	Asn	Lys	Lys	Tyr	Asn	Leu	Phe	Ser	Glu	Ile	Ile	Ser	Asp	Ile	Ser	
		740						745					750			
Glu	Ile	Cys	Leu	Thr	Ser	Thr	Gln	Arg	Tyr	Tyr	Pro	Ile	Asp	Asp	Glu	
	755						760					765				
Glu	Leu	Asn	Ser	Ala	Leu	Asn	Asp	Glu	Asn	Lys	Pro	Leu	Tyr	Phe	Phe	
	770					775					780					
Lys	Ile	Ser	Asn	Lys	Asp	Leu	Ser	Ala	Asp	Glu	Asn	Ile	Leu	Asn	Gly	
785					790					795					800	
Lys	Arg	Lys	Ser	Lys	Gly	Lys	Asp	Asn	Ile	His	Thr	Met	Ile	Leu	Arg	
			805					810						815		
Ala	Met	Met	Asp	Asp	Asn	Val	Thr	Asn	Ile	Ile	Pro	Thr	Ser	Cys	Lys	
			820					825					830			
Ile	Ser	Met	Arg	Glu	Ala	Ser	Ile	Lys	Lys	Asp	Asp	Leu	Val	Ile	His	
		835					840					845				
Lys	Ala	Asn	Glu	Pro	Ile	Lys	Leu	Lys	Asn	Ser	Leu	Ala	Asn	Lys	Lys	
	850					855					860					
Glu	Ser	Thr	Phe	Ser	Tyr	Asp	Ile	Thr	Lys	Asp	Arg	Arg	Tyr	Ser	Arg	

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865					870						875					880
Asp	Glu	Phe	Phe	Phe	Ser	Ile	Thr	Ala	Ser	Ile	Asn	Ser	Asp	Cys	Lys	
				885					890					895		
Glu	Asn	Asp	Tyr	Tyr	Phe	Asn	Gln	Lys	Val	Asn	Glu	Tyr	Leu	Lys	Asn	
			900					905					910			
Asn	Ser	Lys	Ile	Asn	Leu	Leu	Ala	Val	Asp	Leu	Gly	Glu	Thr	Asn	Ile	
		915					920					925				
Ile	Thr	Ile	Ser	Val	Ile	Asp	Gln	Lys	Gly	Asn	Ile	Ile	Leu	Gln	Lys	
	930					935				940						
Asp	Leu	Asp	Lys	Phe	Ile	Asn	Lys	Glu	Lys	Asn	Ile	Ile	Thr	Asp	Phe	
945					950					955					960	
Asn	Leu	Leu	Leu	Ser	Asn	Arg	Ser	Lys	Glu	Arg	Asp	Ile	Ala	Lys	Arg	
			965						970					975		
Asp	Trp	Gln	Glu	Gln	Gln	Gln	Ile	Lys	Asn	Leu	Lys	Glu	Gly	Met	Ile	
		980						985					990			
Ser	Cys	Ile	Ile	His	Glu	Ile	Cys	Lys	Leu	Met	Ile	Glu	His	Asn	Ala	
	995						1000					1005				
Ile	Leu	Ile	Met	Glu	Asp	Leu	Asp	Ala	Asn	Phe	Lys	Asn	Arg	Lys		
1010						1015						1020				
Lys	Arg	Ile	Glu	Lys	Ala	Ile	Tyr	Gln	Lys	Phe	Glu	Ile	Ala	Ile		
1025						1030						1035				
Leu	Glu	Lys	Leu	Asn	Asn	Leu	Val	Phe	Lys	Asp	Ile	Pro	Ile	Asn		
1040						1045						1050				
Glu	Val	Gly	Ser	Val	Thr	Lys	Pro	Leu	Gln	Leu	Ser	Asp	Lys	Phe		
1055						1060						1065				
Glu	Thr	Tyr	Glu	Lys	Val	Gly	Asn	Gln	Ser	Gly	Phe	Val	Phe	Lys		
1070						1075						1080				
Val	Ser	Pro	Phe	Tyr	Thr	Ser	Ile	Ile	Asp	Pro	Thr	Thr	Gly	Phe		
1085						1090						1095				
Ile	Asn	Leu	Phe	Lys	Lys	Asn	Phe	Glu	Ser	Val	Lys	Tyr	Ser	Ile		
1100						1105						1110				
Glu	Phe	Phe	Ser	Lys	Phe	Glu	Ser	Ile	Arg	Tyr	Asn	Thr	Lys	Glu		
1115						1120						1125				
Lys	Tyr	Phe	Glu	Phe	Ala	Phe	Asp	Tyr	Lys	Asn	Phe	Lys	Glu	Ile		
1130						1135						1140				
Lys	Tyr	Thr	Glu	Asn	Ile	Lys	Thr	Asp	Trp	Val	Ala	Cys	Thr	Thr		
1145						1150						1155				
Asn	Ile	Asp	Arg	Tyr	Glu	Tyr	Asp	Lys	Lys	Asn	Lys	Ile	Tyr	Lys		
1160						1165						1170				
Lys	Tyr	Asp	Val	Thr	Thr	Asp	Leu	Lys	Asn	Leu	Phe	Glu	Asn	Glu		
1175						1180						1185				
Glu	Ile	Tyr	Tyr	Gln	Lys	Gly	Glu	Asn	Ile	Leu	Asp	Val	Ile	Leu		
1190						1195						1200				
Lys	Lys	Asn	Asn	Arg	Glu	Phe	Phe	Glu	Lys	Leu	Thr	Asn	Leu	Leu		
1205						1210						1215				
Lys	Ile	Thr	Met	Leu	Phe	Arg	Tyr	Arg	Asn	Ser	His	Leu	Lys	Leu		
1220						1225						1230				
Asp	Tyr	Ile	Ser	Ser	Pro	Val	Lys	Asn	Ser	Asn	Gly	Glu	Phe	Phe		
1235						1240						1245				
Ser	Thr	Glu	Asn	Gly	Leu	Glu	Asn	Tyr	Pro	Ile	Asp	Ser	Asp	Thr		
1250						1255						1260				

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Asn Gly  Ala Tyr His Ile Ala  Leu Lys Gly Lys Met  Ile Leu Asp
1265                      1270                      1275

Arg Ile  Asn Ser Asn Ser Ser  Glu Lys Leu Asp Thr  Tyr Ile Ser
1280                      1285                      1290

Ile Glu  Asp Trp Leu Lys Phe  Ile Gln Lys Phe Ser  Val Asn Lys
1295                      1300                      1305

Ile Thr  Glu Thr Lys Lys Asn  Lys Lys Ile Asn Ile  Lys Tyr Val
1310                      1315                      1320

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<210> SEQ ID NO 4
<211> LENGTH: 1324
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Bacterial species

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<400> SEQUENCE: 4

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Met Lys Asn Leu Thr Glu Phe Thr Gly Leu Tyr Pro Val Ser Lys Thr
1      5      10      15

Leu Arg Phe Glu Leu Lys Pro Thr Asp Asp Phe Asn Trp Glu Thr Phe
20     25     30

Leu Glu Ser Thr Ile Phe Lys His Asp Gln Glu Arg Ala Glu Ala Tyr
35     40     45

Pro Ile Val Lys Val Ile Val Asp Gln Phe His Lys Trp Phe Ile Glu
50     55     60

Asp Ala Leu Asn Lys Ser Thr Ile Asn Trp Asn Ser Leu Tyr Asp Ala
65     70     75     80

Tyr Phe Ala Pro Lys Asn Glu Asn Ser Val Glu Asn Leu Arg Lys Glu
85     90     95

Gln Asp Lys Ile Arg Lys Glu Ile Val Asp Thr Tyr Phe Lys Lys His
100    105    110

Asp Trp Trp Lys Tyr Val Ser Lys Asp His Ser Lys Leu Phe Lys Ile
115    120    125

Glu Leu Pro Ala Leu Leu Ser Asp Asp Ala Phe Ile Tyr Glu Ile Asn
130    135    140

Asp Lys Tyr Pro Asn Tyr Thr Gln Glu Ile Leu Ile Asp Ala Leu Ala
145    150    155    160

Lys Phe Gln Asn Phe Ser Val Tyr Phe Gly Gly Tyr Phe Lys Asn Arg
165    170    175

Asp Asn Met Tyr Lys Ser Asp Ala Gln Ser Thr Ser Ile Ala Asn Arg
180    185    190

Ile Val Asn Glu Asn Phe Thr Lys Phe Ala Asp Asn Ile Lys Ile Tyr
195    200    205

Asn Arg Leu Lys Glu Asn Cys Leu Ser Glu Leu Gln Lys Val Glu Leu
210    215    220

Asp Phe Thr Asp Glu Leu Thr Gly Leu Thr Phe Asp Asp Ile Phe Ser
225    230    235    240

Pro Ser Tyr Phe Asn Lys Cys Leu Thr Gln Lys Gly Ile Glu Lys Leu
245    250    255

Asn Leu Tyr Ile Gly Gly Lys Thr Gly Lys Asn Lys Glu Asp Lys Val
260    265    270

Phe Gly Ile Asn Arg Val Gly Asn Glu Phe Leu Gln Phe Asn Lys Glu
275    280    285

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Ser	Lys	Leu	Lys	Leu	Lys	Asp	Leu	Lys	Met	Val	Lys	Leu	Tyr	Lys	Gln
290						295					300				
Ile	Leu	Ser	Asp	Arg	Glu	Gln	Pro	Ser	Phe	Leu	Pro	Glu	Gln	Phe	Arg
305					310					315					320
Asn	Glu	Asp	Glu	Leu	Ile	Lys	Ser	Ile	Glu	Asp	Phe	His	Asn	Leu	Ile
				325					330					335	
Thr	Glu	Gln	Lys	Leu	Phe	Glu	Arg	Leu	Leu	Lys	Leu	Met	Gly	Arg	Leu
			340					345					350		
Lys	Asn	Gly	Glu	Cys	Glu	Asp	Leu	Asn	Lys	Ile	His	Val	Val	Gly	Ser
		355					360					365			
Ser	Leu	Thr	Gln	Leu	Ser	Lys	Val	Leu	Tyr	Gly	Asn	Trp	Glu	Val	Leu
	370					375					380				
Gly	Thr	Ala	Leu	Arg	Asn	Lys	Phe	Gln	Thr	Asn	Lys	Thr	Lys	Lys	Asp
385					390					395					400
Lys	Leu	Glu	Ser	Glu	Lys	Asp	Ile	Gln	Glu	Trp	Met	Glu	Arg	Lys	Ser
				405					410					415	
Phe	Ser	Leu	Ala	Gln	Ile	Ile	Glu	Val	Glu	Ser	Ser	Leu	Gln	Asp	Asp
			420					425					430		
Lys	Ser	Ile	Lys	Val	Ile	Asp	Leu	Phe	Thr	Thr	Phe	Asn	Ala	Trp	Gln
		435					440					445			
Lys	Val	Asn	Glu	Lys	Pro	Gln	Leu	Val	Asp	Leu	Ile	Lys	Leu	Cys	Lys
	450					455					460				
Asp	Asp	Phe	Gln	Thr	Arg	Phe	Arg	Ala	Val	Lys	Asp	Leu	Ile	Glu	Lys
465					470					475					480
Gly	Glu	Gln	Ile	Gln	Gly	Asn	Glu	Ser	Ala	Lys	Glu	Glu	Ile	Lys	Ala
			485						490					495	
Val	Leu	Asp	Asn	Tyr	Gln	Asn	Leu	Leu	His	Val	Val	Lys	Leu	Leu	Asn
		500					505						510		
Leu	Gly	Lys	Lys	Glu	Ser	Tyr	Leu	Asp	Lys	Asp	Glu	Thr	Phe	Tyr	Asn
		515					520					525			
Glu	Tyr	Lys	Glu	Ile	Leu	Ser	Ser	Thr	Glu	Ser	Asp	Asn	Val	Cys	Leu
	530					535					540				
Glu	Asp	Ile	Ile	Pro	Leu	Tyr	Asn	Lys	Val	Arg	Ser	Phe	Leu	Thr	Arg
545					550					555					560
Lys	Leu	Gly	Asp	Glu	Gly	Lys	Met	Leu	Leu	Lys	Phe	Asp	Cys	Ser	Thr
				565					570					575	
Leu	Ala	Asp	Gly	Trp	Asp	Val	Gly	Lys	Glu	Ser	Ala	Asn	Asn	Ser	Thr
		580						585					590		
Ile	Leu	Ile	Asp	Asn	Ser	Lys	Tyr	Tyr	Leu	Ile	Ile	Thr	Asn	Pro	Glu
		595					600					605			
Asn	Lys	Pro	Asp	Leu	Ser	Thr	Ala	Ile	Thr	Ser	Asn	Thr	Asp	Asn	Val
	610					615					620				
Tyr	Lys	Lys	Ile	Val	Tyr	Arg	Gln	Ile	Ala	Asp	Pro	Thr	Lys	Asp	Leu
625					630					635					640
Pro	Asn	Leu	Met	Val	Ile	Asp	Gly	Lys	Thr	Gln	Arg	Lys	Thr	Gly	Asn
				645					650					655	
Lys	Asp	Asp	Asp	Gly	Ile	Asn	Arg	Val	Leu	Asp	Gln	Leu	Lys	Asp	Lys
			660					665				670			
Tyr	Leu	Pro	Gln	Glu	Val	Asn	Arg	Ile	Arg	Lys	Leu	Gly	Ser	Tyr	Leu
	675						680					685			

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Lys	Thr	Ser	Glu	His	Phe	Asn	Lys	Lys	Asp	Ser	Gln	Val	Tyr	Leu	Ala	690	695	700
Tyr	Tyr	Met	Gln	Arg	Leu	Ile	Glu	Tyr	Lys	Gln	Gly	Glu	Met	Glu	Phe	705	710	715
Ser	Phe	Lys	Asn	Ser	Glu	Glu	Tyr	Asp	Ser	Tyr	Ser	Asp	Phe	Leu	Asp	725	730	735
Asp	Ile	Thr	Lys	Gln	Lys	Tyr	Ser	Leu	Ser	Phe	Val	Asn	Val	Ser	Lys	740	745	750
Glu	Ile	Ile	Thr	Gln	Trp	Ile	Ser	Glu	Gly	Lys	Ile	Phe	Leu	Phe	Gln	755	760	765
Ile	Tyr	Asn	Lys	Asp	Phe	Glu	Glu	Lys	Ala	Thr	Gly	Thr	Pro	Asn	Leu	770	775	780
His	Thr	Leu	Tyr	Trp	Lys	Glu	Leu	Phe	Ser	Glu	Glu	Asn	Leu	Lys	Asp	785	790	795
Ile	Val	Tyr	Lys	Leu	Asn	Gly	Glu	Ala	Glu	Leu	Phe	Tyr	Arg	Lys	Lys	805	810	815
Met	Asp	Gly	Lys	Pro	Phe	Thr	His	Asn	Lys	Gly	Ala	Val	Leu	Val	Asn	820	825	830
Lys	Thr	Phe	Ala	Asp	Gly	Ser	Pro	Val	Glu	Pro	Glu	His	Tyr	Lys	Glu	835	840	845
Tyr	Val	Glu	Tyr	Ile	Thr	Gly	Lys	Val	Ile	Glu	Lys	Gln	Leu	Ser	Lys	850	855	860
Glu	Ala	Lys	Asp	Lys	Leu	His	Leu	Val	Lys	Thr	Asn	Lys	Ala	Lys	Leu	865	870	875
Asp	Ile	Ile	Lys	Asp	Lys	Arg	Tyr	Phe	Gln	His	Lys	Leu	Leu	Phe	His	885	890	895
Val	Pro	Ile	Thr	Ile	Asn	Phe	Lys	Ser	Glu	Gly	Val	Pro	Lys	Phe	Asn	900	905	910
Asp	Tyr	Thr	Leu	Asn	Tyr	Leu	Arg	Glu	Asn	Lys	Lys	Asp	Ile	Asn	Ile	915	920	925
Ile	Gly	Ile	Asp	Arg	Gly	Glu	Arg	Asn	Leu	Ile	Tyr	Val	Ser	Val	Ile	930	935	940
Asn	Gln	Lys	Gly	Glu	Asn	Ile	Ile	Pro	Pro	Lys	His	Phe	Asn	Ile	Val	945	950	955
Glu	Ser	Asp	Met	Phe	Gly	Met	Glu	Asp	Lys	Arg	Lys	Phe	Asn	Tyr	Leu	965	970	975
Glu	Lys	Leu	Ile	Gln	Lys	Glu	Gly	Asn	Arg	Asp	Asp	Ala	Arg	Lys	Asn	980	985	990
Trp	Ser	Lys	Ile	Glu	Thr	Ile	Lys	Asp	Leu	Lys	Thr	Gly	Tyr	Leu	Ser	995	1000	1005
Leu	Val	Val	His	Glu	Ile	Ala	Lys	Leu	Val	Val	Glu	His	His	Ala		1010	1015	1020
Ile	Val	Val	Leu	Glu	Asp	Leu	Asn	Tyr	Gly	Phe	Lys	Arg	Gly	Arg		1025	1030	1035
Phe	Asn	Val	Glu	Arg	Gln	Ile	Tyr	Gln	Asn	Phe	Glu	Lys	Met	Leu		1040	1045	1050
Ile	Glu	Lys	Leu	Asn	Leu	Leu	Val	Phe	Lys	Asn	Asn	Ser	Asn	Ser		1055	1060	1065
Pro	Asp	Tyr	Gly	Asn	Ile	Leu	Asn	Gly	Leu	Gln	Leu	Thr	Ala	Pro		1070	1075	1080
Phe	Gly	Ser	Phe	Lys	Glu	Leu	Gly	Lys	Gln	Ser	Gly	Trp	Leu	Phe				

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1085	1090	1095
Tyr Val Asn Ala Ser Tyr Thr	Ser Lys Ile Asp Pro	Gln Thr Gly
1100	1105	1110
Phe Ala Asn Leu Phe Asn Met	Lys Asp Ala Lys Lys	Asp Thr Lys
1115	1120	1125
Ser Phe Phe Glu Lys Ile Thr	Glu Ile Lys Tyr Asp	Asp Gly Met
1130	1135	1140
Phe Lys Phe Thr Phe Asp Tyr	Arg Asn Gly Phe Ser	Ile Val Gln
1145	1150	1155
Thr Asp Tyr Lys Asn Ile Trp	Thr Val Cys Thr Asn	Asp Lys Arg
1160	1165	1170
Ile Leu Val Ser Lys Asp Asn	Ile Ser Gly Lys Phe	Lys His Glu
1175	1180	1185
Tyr Val Asp Ile Thr Glu Ser	Ile Lys Asn Leu Phe	Ile Asn Asn
1190	1195	1200
Asn Ile Asn Asp Tyr His Ser	Ile Ser Lys Glu Thr	Ile Leu Ser
1205	1210	1215
Ile Lys Glu Lys Lys Phe Phe	Asp Asp Leu Phe Phe	Tyr Phe Lys
1220	1225	1230
Leu Ser Leu Gln Met Arg Asn	Ser Ile Pro Asn Ser	Asp Ile Asp
1235	1240	1245
Tyr Leu Ile Ser Pro Val Gln	Ile Lys Gly Lys Pro	Phe Phe Asp
1250	1255	1260
Ser Arg Ile Pro Asn Asn Ile	Asn Ile Val Asp Ala	Asp Ala Asn
1265	1270	1275
Gly Ala Tyr His Ile Ala Leu	Lys Gly Leu Tyr Leu	Val Ile Asn
1280	1285	1290
Asp Phe Pro Thr Glu Lys Lys	Gly Lys Ser Glu Tyr	Leu Lys Lys
1295	1300	1305
Ile Thr Asn Glu Asp Trp Phe	Glu Phe Ala Gln Arg	Arg Ser Leu
1310	1315	1320

Lys

<210> SEQ ID NO 5
 <211> LENGTH: 1230
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 5

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Glu Asp Ala Tyr Gln Val Leu Lys Pro Val Phe Asp Lys Leu His Glu
20 25 30
Asn Phe Ile Ile Gly Ser Leu Glu Asn Thr Asn Asn Lys Lys Leu Phe
35 40 45
Ser Phe Asp Lys Tyr Leu Lys Leu Lys Asn Asp Leu Leu His Val Asn
50 55 60
Lys Lys Glu Gln Glu Ser Asp Tyr Lys Lys Lys Glu Lys Glu Phe Glu
65 70 75 80
Thr Glu Gly Lys Leu Leu Arg Asn Thr Phe Ala Thr Val Trp Ile Asn
85 90 95

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Glu	Gly	Lys	Asn	Phe	Lys	Asn	Thr	Ile	Val	Gly	Gly	Glu	Asn	Asp	Arg
			100							105					
Glu	Ile	Leu	Lys	Glu	Gly	Gly	Tyr	Lys	Ile	Leu	Thr	Glu	Ala	Gly	Ile
			115				120								
Leu	Lys	Tyr	Ile	Lys	Met	Asn	Ile	Asp	Lys	Phe	Val	Glu	Leu	Lys	Leu
			130				135								
Lys	Thr	Arg	Glu	Asp	Ile	Leu	Trp	Lys	Lys	Glu	Asn	Arg	Asn	Leu	Val
			145				150								
Glu	Met	Ala	Asp	Leu	Glu	Lys	Ser	Leu	Gly	Thr	Ile	Glu	Ser	Trp	Gly
			165												
Val	Phe	Glu	Gly	Phe	Phe	Thr	Tyr	Phe	Ser	Gly	Phe	Asn	Gln	Asn	Arg
			180				185								
Glu	Asn	Tyr	Tyr	Ser	Thr	Asp	Glu	Lys	Ala	Thr	Ala	Val	Ala	Ser	Arg
			195				200								
Val	Ile	Asp	Glu	Asn	Leu	Pro	Lys	Phe	Ser	Asp	Asn	Val	Leu	Glu	Phe
			210				215								
Asn	Lys	Lys	Asn	Asp	Val	Tyr	Ile	Gly	Ile	Phe	Ser	Phe	Leu	Lys	Gly
			225				230								
Lys	Asn	Ile	Val	Leu	Lys	Gly	Lys	Ser	Gly	Asn	Gly	Glu	Glu	Gln	Asp
			245												
Leu	Leu	Pro	Ile	Thr	Glu	Lys	Ile	Phe	Glu	Ile	Glu	Tyr	Phe	Lys	Asn
			260				265								
Cys	Leu	Ser	Glu	Gly	Glu	Ile	Glu	Arg	Tyr	Asn	Ser	Asp	Ile	Gly	Asn
			275				280								
Ala	Asn	Phe	Leu	Ile	Asn	Leu	Tyr	Asn	Gln	Gln	Gln	Asp	Lys	Lys	Glu
			290				295								
Asn	Lys	Leu	Arg	Ile	Phe	Lys	Thr	Leu	Tyr	Lys	Gln	Ile	Gly	Cys	Gly
			305				310								
Ile	Lys	Gly	Asp	Phe	Ile	Gln	Leu	Ile	Lys	Thr	Asp	Asp	Glu	Leu	Lys
			325												
Lys	Ile	Phe	Glu	Asp	Leu	Lys	Ile	Thr	Gly	Asp	Asn	Phe	Phe	Lys	Asn
			340				345								
Thr	Gln	Asn	Leu	Lys	Glu	Ile	Ile	Leu	Ser	Leu	Glu	Asn	Phe	Ser	Gly
			355				360								
Ile	Tyr	Trp	Ser	Asp	Lys	Ala	Leu	Asn	Thr	Val	Ser	Gly	Lys	Tyr	Phe
			370				375								
Ala	Asn	Trp	Ala	Ser	Leu	Lys	Glu	Leu	Leu	Lys	Asn	Ala	Lys	Ile	Phe
			385				390								
Lys	Lys	Glu	Lys	Asp	Glu	Ile	Lys	Ile	Pro	Gln	Thr	Ile	Glu	Leu	Ser
			405												
Asp	Leu	Phe	Gly	Val	Leu	Asp	Ser	Asn	Glu	Leu	Ile	Phe	Lys	Glu	Ser
			420				425								
Phe	Asn	Glu	Asn	Asp	Glu	Leu	Lys	Gln	Ile	Ile	Leu	Lys	Ser	Tyr	Glu
			435				440								
Lys	Asn	Ser	Ile	Lys	Leu	Leu	Lys	Met	Ile	Phe	Val	Asp	Val	Glu	Glu
			450				455								
Asn	Gln	Lys	Ile	Phe	Gly	Asn	Leu	Lys	Asp	Gly	Leu	Pro	Ile	Asn	Asp
			465				470								
Phe	Lys	Lys	Asp	Glu	Asn	Thr	Gln	Ile	Ile	Lys	Thr	Trp	Leu	Asp	Gly
			485												

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Leu	Leu	Asn	Thr	Asn	Gln	Ile	Leu	Lys	Tyr	Phe	Lys	Val	Arg	Glu	Ser	500	505	510
Lys	Ile	Lys	Gly	Ala	Pro	Leu	Asn	Pro	Glu	Val	Ser	Glu	Arg	Leu	Asn	515	520	525
Lys	Ile	Leu	Asn	Val	Glu	Asn	Pro	Thr	Val	Ile	Tyr	Asp	Val	Val	Arg	530	535	540
Asn	Tyr	Leu	Thr	Lys	Lys	Pro	Thr	Glu	Gly	Leu	Asn	Lys	Leu	Lys	Leu	545	550	555
Asn	Phe	Asp	Asn	Ala	Val	Leu	Ala	Ala	Gly	Trp	Asp	Val	Asn	Lys	Glu	565	570	575
Ser	Glu	Arg	Gly	Cys	Leu	Ile	Leu	Lys	Asp	Gly	Asp	Asn	Lys	Lys	Tyr	580	585	590
Leu	Ala	Ile	Leu	Thr	Asn	Lys	Thr	Gln	Lys	Phe	Phe	Gly	Glu	Lys	Val	595	600	605
Lys	Tyr	Lys	Glu	Phe	Val	Gly	Asp	Glu	Asn	Trp	Gln	Lys	Met	Asp	Tyr	610	615	620
Lys	Leu	Leu	Pro	Gly	Pro	Asn	Lys	Met	Leu	Pro	Lys	Val	Leu	Leu	Pro	625	630	635
Lys	Ser	Asp	Arg	Tyr	Lys	Phe	Gly	Ala	Thr	Asp	Glu	Ile	Leu	Lys	Ile	645	650	655
Tyr	Asn	Glu	Gly	Gly	Phe	Lys	Lys	Asn	Glu	Pro	Thr	Phe	Thr	Lys	Ala	660	665	670
Lys	Leu	Ala	Lys	Ile	Val	Asp	Phe	Phe	Lys	Asp	Gly	Leu	Lys	Asn	Tyr	675	680	685
Pro	Ser	Ala	Lys	Ser	Ser	Trp	Tyr	Asn	Leu	Phe	Ala	Phe	Asp	Phe	Ser	690	695	700
Asp	Thr	Glu	Lys	Tyr	Glu	Ser	Ile	Asp	Arg	Phe	Tyr	Thr	Glu	Val	Glu	705	710	715
Lys	Gln	Gly	Tyr	Lys	Leu	Ser	Trp	Ser	Ala	Ile	Ser	Lys	Asn	Phe	Ile	725	730	735
Phe	Glu	Lys	Val	Asp	Ala	Gly	Asp	Met	Tyr	Leu	Phe	Glu	Ile	Arg	Asn	740	745	750
Lys	Asp	Asn	Asn	Leu	Lys	Asn	Gly	Lys	Ala	Lys	Thr	Gly	Ala	Lys	Asn	755	760	765
Leu	His	Thr	Ile	Tyr	Trp	Gly	Thr	Ile	Phe	Gly	Glu	Ser	Glu	Asn	Lys	770	775	780
Pro	Lys	Leu	Asn	Gly	Glu	Ala	Glu	Ile	Phe	Tyr	Arg	Pro	Val	Val	Lys	785	790	795
Asp	Leu	Ile	Lys	Asp	Lys	Asp	Lys	Asn	Gly	Asp	Ile	Ile	Lys	Ala	Ser	805	810	815
Glu	Lys	Arg	Phe	Glu	Gln	Glu	Lys	Phe	Val	Phe	His	Cys	Pro	Ile	Thr	820	825	830
Leu	Asn	Phe	Cys	Leu	Lys	Ser	Thr	Arg	Leu	Asn	Asp	Val	Ile	Asn	Gln	835	840	845
Ile	Met	Ile	Glu	Asn	Lys	Lys	Asp	Val	Cys	Phe	Ile	Gly	Ile	Asp	Arg	850	855	860
Gly	Glu	Lys	His	Leu	Ala	Tyr	Tyr	Ser	Val	Val	Asn	Gln	Lys	Gly	Glu	865	870	875
Ile	Leu	Glu	Gln	Gly	Ser	Phe	Asn	Glu	Ile	Asn	Gly	Gln	Asn	Tyr	Ala	885	890	895
Lys	Lys	Leu	Glu	Glu	Lys	Ala	Gly	His	Arg	Asp	Glu	Ala	Arg	Lys	Asn			

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900					905					910					
Trp	Lys	Thr	Ile	Gly	Thr	Ile	Lys	Glu	Leu	Lys	Asn	Gly	Tyr	Ile	Ser
	915						920					925			
Gln	Val	Val	Arg	Arg	Ile	Val	Asp	Leu	Ala	Val	Lys	Tyr	Asn	Ala	Tyr
	930					935					940				
Ile	Val	Leu	Glu	Asp	Leu	Asn	Ser	Gly	Phe	Lys	Arg	Gly	Arg	Gln	Lys
945					950					955					960
Ile	Glu	Lys	Ser	Val	Tyr	Gln	Lys	Leu	Glu	Leu	Ala	Leu	Ala	Lys	Lys
				965					970					975	
Leu	Asn	Phe	Leu	Val	Asp	Lys	Ser	Lys	Lys	Asp	Gly	Glu	Ile	Gly	Ser
			980					985					990		
Val	Gln	Lys	Ala	Leu	Gln	Leu	Thr	Pro	Pro	Ala	Thr	Asn	Phe	Ala	Asp
		995					1000						1005		
Ile	Glu	Lys	Ala	Lys	Gln	Phe	Gly	Ile	Met	Leu	Tyr	Val	Arg	Ala	
	1010					1015							1020		
Asn	Tyr	Thr	Ser	Gln	Thr	Asp	Pro	Val	Thr	Gly	Trp	Arg	Lys	Thr	
	1025					1030							1035		
Ile	Tyr	Phe	Lys	Ser	Thr	Thr	Gln	Glu	Asn	Leu	Lys	Lys	Glu	Ile	
	1040					1045							1050		
Cys	Glu	Lys	Phe	Ser	Glu	Ile	Gly	Phe	Asp	Gly	Asn	Asp	Tyr	Tyr	
	1055					1060							1065		
Phe	Glu	Tyr	Lys	Asp	Glu	Asn	Ala	Glu	Lys	Lys	Trp	Thr	Met	Tyr	
	1070					1075							1080		
Ser	Gly	Val	Ser	Gly	Lys	Ser	Leu	Asp	Arg	Phe	Arg	Gly	Lys	Lys	
	1085					1090							1095		
Asp	Thr	His	Gly	Ile	Trp	Lys	Val	Glu	Lys	Gln	Asp	Ile	Val	Glu	
	1100					1105							1110		
Leu	Leu	Lys	Lys	Ile	Phe	Gly	Gln	Gln	Thr	Ser	Val	Val	Gly	Asp	
	1115					1120							1125		
Leu	Lys	Thr	Lys	Ile	Thr	Asn	Asp	Asn	Val	Asn	Asp	Leu	Lys	Tyr	
	1130					1135							1140		
Thr	Ile	Asp	Leu	Ile	Gln	Gln	Ile	Arg	Asn	Thr	Gly	Phe	Asn	Glu	
	1145					1150							1155		
Ile	Asp	Asn	Asp	Phe	Ile	Leu	Ser	Pro	Val	Arg	Asp	Glu	Lys	Gly	
	1160					1165							1170		
Asn	His	Phe	Asp	Ser	Arg	Lys	Asp	Gly	Ala	Ile	Leu	Ser	Asn	Gly	
	1175					1180							1185		
Asp	Ala	Asn	Gly	Ala	Tyr	Asn	Ile	Ala	Arg	Lys	Gly	Val	Leu	Ala	
	1190					1195							1200		
Phe	Glu	Arg	Ile	Asn	Ala	Lys	Pro	Glu	Lys	Pro	Glu	Leu	Tyr	Ile	
	1205					1210							1215		
Ala	Asp	Val	Glu	Trp	Asp	Lys	Trp	Leu	Gln	Ser	Lys				
	1220					1225							1230		

<210> SEQ ID NO 6

<211> LENGTH: 1256

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 6

Met Asp Ser Phe Thr Gln Phe Thr Gly Leu Tyr Ser Leu Ser Lys Thr

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1	5	10	15
Leu Arg Phe	Glu Leu Lys Pro Ile Gly Lys Thr Leu Ala Tyr Ile Glu		
	20	25	30
Asn Lys Gly	Leu Leu Val Gln Asp Glu His Arg Ala Asp Ser Tyr Lys		
	35	40	45
Ile Val Lys Lys Ile Ile Asp Glu Tyr His Lys Ser Phe Ile Glu Lys			
	50	55	60
Ser Leu Asn Gly Leu Cys Leu Asp Gly Leu Glu Asp Tyr Tyr Phe Tyr			
65	70	75	80
Tyr Gln Ile Pro Lys Lys Asp Asp Asn Gln Lys Lys Ile Val Glu Asp			
	85	90	95
Ile Leu Thr Lys Leu Arg Lys Gln Ile Ala Glu Arg Phe Ser Lys Gln			
	100	105	110
Asp Ile Tyr Lys Asn Leu Phe Ala Lys Glu Leu Ile Lys Asp Asp Leu			
	115	120	125
Asn Ser Phe Val Gln Glu Val Glu Gln Lys Asp Leu Ile Lys Glu Phe			
	130	135	140
Glu Asn Phe Thr Thr Tyr Phe Thr Gly Phe His Glu Asn Arg Lys Asn			
145	150	155	160
Met Tyr Ser Ala Glu Asp Lys Ser Thr Ala Ile Ala Phe Arg Leu Ile			
	165	170	175
His Gln Asn Leu Pro Lys Phe Leu Asp Asn Met Arg Ala Phe Asn Lys			
	180	185	190
Ile Ser Val Ser Pro Leu Ala Glu Lys Phe Lys His Ile Leu Ser Asp			
	195	200	205
Ser Glu Leu Gly Pro Ile Val Gln Val Val Ala Met Glu Asp Val Phe			
	210	215	220
Asn Leu Ala Tyr Phe Asn Glu Thr Leu Thr Gln Ser Gly Ile Asp Ile			
225	230	235	240
Tyr Asn His Leu Leu Gly Gly Tyr Thr Pro Glu Glu Gly Lys Glu Lys			
	245	250	255
Ile Lys Gly Leu Asn Glu Tyr Ile Asn Leu Tyr Asn Gln Thr Val Lys			
	260	265	270
Lys Glu Glu Arg Leu Pro Lys Leu Lys Pro Leu Phe Lys Gln Ile Leu			
	275	280	285
Ser Asp Arg Ser Thr Ala Ser Phe Ile Pro Glu Gln Tyr Lys Asn Asp			
	290	295	300
Asn Glu Val Leu Glu Ser Ile Glu Lys Leu Tyr Gln Glu Ile Lys Glu			
305	310	315	320
His Val Phe His Ser Leu Lys Glu Leu Phe Val His Ile Asn Glu Tyr			
	325	330	335
Asp Leu His Lys Ile Tyr Leu Arg Asn Asp Val Ser Met Thr Asp Ile			
	340	345	350
Ser Gln Lys Met Phe Gly Asp Trp Gly Val Phe Thr Lys Ala Met Asn			
	355	360	365
Leu Tyr Phe Asp Lys Gln Tyr Lys Gly Lys Ala Lys Leu Gly Thr Glu			
	370	375	380
Lys Tyr Glu Asp Glu Gln Lys Lys Tyr Phe Ser Asn Gln Glu Ser Phe			
385	390	395	400
Ser Ile Gly Tyr Ile Asn Glu Cys Leu Leu Leu Leu Gly Ser Asn Tyr			
	405	410	415

His	Lys	Lys	Val	Glu	Asp	Tyr	Phe	Lys	Val	Ala	Gly	Lys	Thr	Glu	Glu	
			420					425					430			
Gln	Val	Gln	Met	Leu	Phe	Glu	Ile	Ile	Glu	Thr	Lys	Tyr	Gln	Asn	Ile	
			435					440					445			
Gln	Asp	Leu	Leu	Asn	Ser	Pro	Tyr	Pro	Thr	Glu	Lys	Asn	Leu	Ala	Gln	
			450					455					460			
Asp	Gln	Val	Gln	Val	Asp	Lys	Ile	Lys	Gly	Leu	Leu	Asp	Ser	Ile	Lys	
			465					470					475		480	
Asn	Leu	Gln	Trp	Phe	Ile	Lys	Pro	Leu	Leu	Gly	Lys	Gly	Asn	Glu	Ala	
			485					490					495			
Glu	Lys	Asp	Glu	Arg	Phe	Tyr	Gly	Glu	Phe	Thr	Ala	Leu	Trp	Glu	Thr	
			500					505					510			
Leu	Asp	Gln	Ile	Thr	Pro	Leu	Tyr	Asn	Lys	Val	Arg	Asn	Tyr	Met	Thr	
			515					520					525			
Arg	Lys	Pro	Tyr	Ser	Thr	Glu	Lys	Met	Lys	Leu	Asn	Phe	Asp	Asn	Ser	
			530					535					540			
Thr	Leu	Leu	Asp	Gly	Trp	Asp	Ile	Asn	Lys	Glu	Pro	Asp	Asn	Thr	Ser	
			545					550					555		560	
Val	Val	Leu	Arg	Lys	Asp	Gly	Leu	Phe	Tyr	Leu	Gly	Ile	Met	Asp	Lys	
			565					570					575			
Lys	Tyr	Asn	Lys	Thr	Phe	Lys	Gln	Glu	Phe	Ile	Glu	Ser	Asn	Glu	Pro	
			580					585					590			
Cys	Phe	Glu	Lys	Met	Glu	Tyr	Lys	Leu	Leu	Pro	Gly	Ala	Asn	Lys	Met	
			595					600					605			
Leu	Pro	Lys	Val	Phe	Phe	Ser	Asn	Ser	Arg	Ile	Glu	Glu	Phe	Asn	Pro	
			610					615					620			
Thr	Val	Asp	Leu	Leu	Glu	Asn	Tyr	Lys	Asn	Gln	Thr	His	Lys	Lys	Gly	
			625					630					635		640	
Asp	Lys	Phe	Asn	Ile	Ala	His	Cys	Arg	Asn	Leu	Ile	Asp	Phe	Phe	Lys	
			645					650					655			
Gln	Ser	Ile	Asn	Lys	His	Asp	Asp	Trp	Lys	Gln	Phe	Gly	Phe	Ala	Phe	
			660					665					670			
Ser	Asp	Thr	Lys	Asn	Tyr	Asp	Asp	Leu	Ser	Gly	Phe	Tyr	Arg	Glu	Val	
			675					680					685			
Glu	Gln	Gln	Gly	Tyr	Lys	Ile	Thr	Phe	Arg	Asn	Ile	Pro	Glu	Lys	Phe	
			690					695					700			
Ile	Asn	Gln	Met	Val	Glu	Glu	Ser	Lys	Leu	Tyr	Leu	Phe	Gln	Ile	Tyr	
			705					710					715		720	
Asn	Lys	Asp	Phe	Ser	Pro	Tyr	Ser	Lys	Gly	Thr	Pro	Asn	Met	His	Thr	
			725					730					735			
Leu	Tyr	Trp	Lys	Met	Leu	Phe	Asp	Thr	Glu	Asn	Leu	Lys	Asp	Val	Val	
			740					745					750			
Tyr	Lys	Leu	Asn	Gly	Gln	Ala	Glu	Val	Phe	Tyr	Arg	Lys	Ala	Ser	Ile	
			755					760					765			
Asn	Asp	Glu	Asn	Ile	Val	Val	His	Lys	Ala	Asn	Glu	Val	Ile	Ile	Asn	
			770					775					780			
Lys	Asn	Thr	Leu	Asn	Glu	Lys	Lys	Gln	Ser	Arg	Phe	Asp	Tyr	Asp	Ile	
			785					790					795		800	
Ile</																

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Ile	Thr	Met	Asn	Phe	Lys	Ala	Arg	Gly	Leu	Asn	Asn	Ile	Asn	Leu	Glu	820	825	830
Val	Asn	Gln	Tyr	Leu	Gln	Lys	Glu	Asn	Asp	Ile	His	Ile	Ile	Gly	Ile	835	840	845
Asp	Arg	Gly	Glu	Arg	His	Leu	Leu	Tyr	Leu	Ser	Leu	Ile	Asn	Ser	Lys	850	855	860
Gly	Asn	Ile	Ile	Glu	Gln	Tyr	Ser	Leu	Asn	Glu	Ile	Ile	Asn	Glu	Tyr	865	870	875
Asn	Gly	Asn	His	Tyr	His	Thr	Asn	Tyr	His	Asp	Leu	Leu	Asp	Lys	Arg	885	890	895
Glu	Gly	Asn	Arg	Thr	Glu	Glu	Arg	Gln	Asn	Trp	Lys	Thr	Ile	Glu	Ser	900	905	910
Ile	Lys	Glu	Leu	Lys	Glu	Gly	Tyr	Leu	Ser	Gln	Val	Val	His	Lys	Ile	915	920	925
Ser	Glu	Leu	Met	Val	Glu	Tyr	Asn	Ala	Ile	Val	Val	Leu	Glu	Asp	Leu	930	935	940
Asn	Met	Gly	Phe	Ile	Arg	Gly	Arg	Gln	Lys	Val	Glu	Lys	Ser	Val	Tyr	945	950	955
Gln	Gln	Phe	Glu	Lys	Met	Leu	Ile	Asp	Lys	Leu	Asn	Tyr	Leu	Val	Asp	965	970	975
Lys	Lys	Lys	Lys	Ser	Phe	Glu	Leu	Gly	Gly	Thr	Leu	His	Ala	Tyr	Gln	980	985	990
Leu	Thr	Asn	Lys	Phe	Glu	Ser	Phe	Gln	Lys	Met	Gly	Lys	Gln	Ser	Gly	995	1000	1005
Phe	Leu	Phe	Tyr	Ile	Pro	Ala	Trp	Asn	Thr	Ser	Lys	Met	Asp	Pro		1010	1015	1020
Val	Thr	Gly	Phe	Val	Asn	Leu	Phe	Asp	Thr	Arg	Tyr	Glu	Asn	Val		1025	1030	1035
Val	Lys	Ala	Lys	Ala	Phe	Phe	Asn	Lys	Phe	Glu	Ser	Ile	Arg	Tyr		1040	1045	1050
Asn	Lys	Asp	Lys	Asp	Tyr	Phe	Glu	Phe	Glu	Val	Lys	Lys	Tyr	Ser		1055	1060	1065
Asp	Phe	Asn	Ala	Lys	Ala	Glu	Asp	Thr	Arg	Gln	Glu	Trp	Ile	Ile		1070	1075	1080
Cys	Thr	His	Gly	Glu	Arg	Ile	Ile	Asn	Tyr	Arg	Asn	Pro	Glu	Lys		1085	1090	1095
Asn	Asn	Glu	Trp	Asp	Asp	Lys	Thr	Val	His	Pro	Thr	Thr	Glu	Leu		1100	1105	1110
Lys	Ser	Leu	Phe	Thr	Ser	Lys	Asn	Ile	Ile	Phe	Glu	Asn	Gly	Ser		1115	1120	1125
Cys	Leu	Lys	Glu	Gln	Ile	Ala	Leu	Gln	Lys	Asp	Thr	Asp	Lys	Glu		1130	1135	1140
Phe	Phe	Glu	Gly	Leu	Leu	Lys	Gln	Phe	Lys	Asn	Thr	Leu	Gln	Met		1145	1150	1155
Arg	Asn	Ser	Lys	Thr	Lys	Ser	Glu	Ile	Asp	Tyr	Leu	Phe	Ser	Pro		1160	1165	1170
Val	Ser	Asn	Glu	Asn	Gly	Val	Phe	Phe	Asp	Ser	Arg	Asp	Tyr	Val		1175	1180	1185
Asp	Ile	Asp	Asn	Arg	Asp	Arg	Lys	Phe	Cys	Val	Ser	Thr	Gly	Lys		1190	1195	1200
Pro	Thr	Leu	Pro	Val	Asn	Ala	Asp	Ala	Asn	Gly	Ala	Tyr	Asn	Ile				

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1205	1210	1215
Ala Arg Lys Gly Leu Trp Ile Val Glu Gln Ile Lys Asn Pro Asn		
1220	1225	1230
Thr Asp Leu Lys Lys Leu Lys Leu Ala Met Thr Asn Lys Glu Trp		
1235	1240	1245
Leu Gln Phe Val Gln Asn Lys Gly		
1250	1255	

<210> SEQ ID NO 7
 <211> LENGTH: 1274
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 7

Met Asp Asn Ala Phe Ser Asp Phe Thr Gln Lys Tyr Thr Leu Ser Lys		
1	5	10
Thr Leu Arg Phe Glu Leu Arg Pro Val Gly Asn Thr Glu Lys Met Leu		
20	25	30
Glu Asp Glu Lys Val Phe Glu Lys Asp Lys Leu Ile Gln Glu Lys Tyr		
35	40	45
Ile Lys Thr Lys Pro Tyr Phe Asp Leu Leu His Arg Glu Phe Val Glu		
50	55	60
Glu Ala Leu Lys Asp Val Asp Ile Ser Gly Leu His Asn Tyr Phe Glu		
65	70	75
Thr Tyr Gln Lys Trp Ala Lys Asp Lys Lys Lys Tyr Gln Lys Glu Leu		
85	90	95
Gln Asn Lys Glu Gln Ile Leu Arg Lys Glu Ile Leu Val Phe Leu Asp		
100	105	110
Ser Thr Ala Lys Tyr Trp Ala Glu Lys Lys Tyr Ser Glu Leu Arg Ile		
115	120	125
Lys Lys Lys Asp Ile Glu Ile Phe Phe Glu Glu Asp Val Phe Thr Ile		
130	135	140
Leu Lys Lys Arg Tyr Gly Glu Asp Ser Glu Ala Gln Ile Ile Asp Glu		
145	150	155
Val Ser Gly Glu Thr Val Ser Ile Phe Asp Ser Trp Lys Gly Phe Thr		
165	170	175
Gly Tyr Phe Lys Lys Phe Gln Glu Thr Arg Lys Asn Leu Tyr Arg Asp		
180	185	190
Asp Gly Thr Ala Thr Ala His Ala Thr Arg Ile Ile Asp Gln Asn Leu		
195	200	205
Lys Arg Phe Cys Asp Asn Leu Glu Ile Ile Lys Arg Ile Ala Gly Ile		
210	215	220
Ile Glu Phe Ser Glu Val Glu Gly Asn Phe Lys His Ser Met Gly Asp		
225	230	235
Val Phe Ser Leu Ser Phe Tyr Asn Lys Cys Leu Leu Gln Asp Gly Ile		
245	250	255
Asn Phe Tyr Asn Arg Ile Leu Gly Gly Glu Val Leu Gln Asp Gly Thr		
260	265	270
Lys Leu Lys Gly Ile Asn Glu Leu Ile Asn Lys Tyr Arg Gln Asp Asn		
275	280	285
Lys Gly Val Lys Ile Pro Phe Leu Lys Leu Leu Asp Lys Gln Ile Leu		

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290	295	300
Ser Glu Lys Glu Glu Phe Leu Asp Gly Ile Glu Asp Asp Lys Glu Leu 305 310 315 320		
Leu Ala Val Leu Lys Lys Phe Tyr Glu Val Ala Glu Lys Lys Thr Ser 325 330 335		
Ile Leu Lys Ser Leu Ile Gln Asp Phe Ala Gln Asn Asn Arg Gln Tyr 340 345 350		
Asn Leu Glu Glu Val Tyr Ile Ser Lys Glu Ala Phe Asn Thr Ile Ser 355 360 365		
Arg Lys Trp Thr His Glu Thr Ser Lys Phe Glu Glu Trp Leu Tyr Asn 370 375 380		
Val Met Lys Pro Asn Lys Pro Thr Gly Leu Lys Tyr Asp Lys Lys Glu 385 390 395 400		
Glu Ser Tyr Lys Phe Pro Asp Phe Ile Pro Leu Ser Tyr Ile Gln Thr 405 410 415		
Ala Leu Glu Gln Ala Asp Ile Asp Gly Asp Phe Trp Lys Glu His Tyr 420 425 430		
Ser Glu Asn Ser Lys Ala Asn Asp Gly Cys Leu Met Gly Asp Glu Ser 435 440 445		
Ile Trp Glu Gln Phe Ile Lys Ile Phe Glu Tyr Glu Phe Gln Ser Leu 450 455 460		
Phe Glu Lys Glu Ile Ile Asp Arg Glu Thr Gly Gln Pro Lys Lys Asn 465 470 475 480		
Gly Tyr Asn Tyr Val Lys Asp Asp Phe Lys Gly Leu Leu Asn Gly Glu 485 490 495		
Asn Phe Ser Val Glu Ile Ile Lys Asp Phe Ala Asp Thr Val Leu Ser 500 505 510		
Ile Tyr Gln Met Ala Lys Tyr Phe Ala Ile Glu Lys Lys Arg Lys Trp 515 520 525		
Leu Asp Glu Tyr Asp Thr Gly Asp Phe Tyr Glu Asn Pro Glu Phe Gly 530 535 540		
Tyr Lys Leu Phe Tyr Asp Asp Ala Tyr Lys Glu Ile Val Gln Thr Tyr 545 550 555 560		
Asn Asn Leu Arg Asn Tyr Leu Thr Lys Lys Ser Tyr Ser Glu Glu Lys 565 570 575		
Trp Lys Leu Asn Phe Glu Asn Pro Thr Leu Ala Asp Gly Trp Asp Lys 580 585 590		
Asn Lys Glu Pro Asp Asn Ser Ala Val Ile Leu Arg Lys Asp Gly Arg 595 600 605		
Tyr Tyr Leu Gly Leu Met Lys Lys Gly Cys Asn Lys Ile Phe Asp Asp 610 615 620		
Arg Asn Lys Val Glu Phe Ser Gly Gly Val Asp Lys Asp Lys Tyr Glu 625 630 635 640		
Lys Ile Val Tyr Lys Phe Phe Pro Asp Gln Ala Lys Met Phe Pro Lys 645 650 655		
Val Cys Phe Ser Ala Lys Gly Leu Asp Phe Phe Gln Pro Ser Glu Glu 660 665 670		
Ile Leu Asn Ile Tyr Lys Asn Ser Glu Phe Lys Lys Gly Asp Thr Phe 675 680 685		
Ser Val Gln Ser Met Gln Lys Leu Ile Asp Phe Tyr Lys Asp Cys Leu 690 695 700		

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Thr Lys Tyr Glu Gly Trp Ile Ala Tyr Glu Phe Lys His Leu Lys Ser
 705 710 715 720
 Thr Asp Leu Tyr Arg Asn Asn Ile Ser Glu Phe Phe Ser Asp Val Ala
 725 730 735
 Glu Asp Gly Tyr Lys Ile Thr Phe Gln Asp Ile Ser Asp Asn Tyr Ile
 740 745 750
 Asp Lys Lys Asn Gln Ser Glu Glu Leu Tyr Leu Phe Glu Ile His Asn
 755 760 765
 Lys Asp Trp Asn Leu Lys Asp Glu Val Lys Lys Thr Gly Ser Lys Asn
 770 775 780
 Leu His Thr Leu Tyr Phe Glu Ala Leu Phe Ser His Glu Asn Ile Gln
 785 790 795 800
 Asn Asn Phe Pro Ile Lys Leu Asn Gly Gln Ala Glu Val Phe Tyr Arg
 805 810 815
 Pro Lys Thr Asp Glu Glu Lys Leu Val Lys Lys Lys Asp Lys Lys Gly
 820 825 830
 Arg Glu Val Ile Asp His Lys Arg Tyr Ala Glu Asn Lys Ile Phe Phe
 835 840 845
 His Val Pro Leu Thr Leu Asn Arg Gly Lys Gly Asp Ala Tyr Gln Phe
 850 855 860
 Asn Ala Lys Ile Asn Asn Phe Leu Ala Asn Asn Ser Asp Ile Asn Val
 865 870 875 880
 Ile Gly Val Asp Arg Gly Glu Lys His Leu Ala Tyr Tyr Ser Val Ile
 885 890 895
 Asn Gln Lys Gly Glu Thr Leu Asp Ser Gly Ser Leu Asn Val Val Asn
 900 905 910
 Lys Ile Asn Tyr Gly Glu Lys Leu Gln Glu Lys Ala Ser Asn Arg Lys
 915 920 925
 Gln Ser Ile Arg Asp Trp Lys Ala Val Glu Gly Ile Lys Asn Leu Lys
 930 935 940
 Lys Gly Tyr Ile Ser Gln Val Val Arg Lys Leu Ala Asp Leu Ala Ile
 945 950 955 960
 Glu His Asn Ala Ile Ile Ile Phe Glu Asp Leu Asn Met Arg Phe Lys
 965 970 975
 Gln Ile Arg Gly Gly Ile Glu Lys Ser Val Tyr Gln Gln Leu Glu Gly
 980 985 990
 Ala Leu Ile Glu Lys Leu Ser Phe Leu Val Asn Lys Gly Glu Lys Asp
 995 1000 1005
 Pro Lys Gln Ala Gly Asn Leu Leu Lys Ala Tyr Gln Leu Ala Ala
 1010 1015 1020
 Pro Phe Thr Thr Phe Lys Asp Met Gly Lys Gln Thr Gly Ile Ile
 1025 1030 1035
 Phe Tyr Thr Gln Ala Ser Tyr Thr Ser Lys Ile Asp Pro Leu Thr
 1040 1045 1050
 Gly Trp Arg Pro Asn Leu Tyr Leu Lys Tyr Thr Asn Ala Glu Lys
 1055 1060 1065
 Thr Lys Glu Asp Ile Gly Asn Phe Ser Asn Ile Glu Phe Lys Asn
 1070 1075 1080
 Gly Ile Phe Glu Phe Thr Tyr Asp Leu Arg Asn Phe Gln Lys Gln
 1085 1090 1095

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Lys	Glu	Tyr	Pro	Lys	Lys	Thr	Glu	Trp	Thr	Leu	Cys	Ser	Cys	Val
1100						1105					1110			
Glu	Arg	Phe	Arg	Trp	Asn	Arg	Val	Leu	Asn	Gln	Asn	Lys	Gly	Gly
1115						1120					1125			
Tyr	Asp	His	Tyr	Glu	Asp	Ile	Thr	His	Asn	Phe	Arg	Asp	Leu	Phe
1130						1135					1140			
Glu	Lys	Tyr	Asp	Ile	Asn	Phe	Met	Ser	Ala	Asp	Ile	Lys	Gly	Gln
1145						1150					1155			
Ile	Asp	Thr	Leu	Asp	Ala	Lys	Gly	Asn	Glu	Asn	Phe	Phe	Lys	Asp
1160						1165					1170			
Phe	Ile	Phe	Phe	Phe	Asn	Leu	Ile	Cys	Gln	Ile	Arg	Asn	Thr	Gln
1175						1180					1185			
Gln	Asp	Lys	Asp	Gly	Asp	Glu	Asn	Asp	Phe	Ile	Leu	Ser	Pro	Ile
1190						1195					1200			
Lys	Pro	Phe	Phe	Asp	Ser	Arg	Asp	Ser	Lys	Lys	Phe	Gly	Glu	Asn
1205						1210					1215			
Leu	Pro	Asn	Asn	Gly	Asp	Asp	Asn	Gly	Ala	Tyr	Asn	Ile	Ser	Arg
1220						1225					1230			
Lys	Gly	Ile	Ile	Ile	Leu	Asn	Lys	Ile	Ser	Glu	Phe	Phe	Asp	Glu
1235						1240					1245			
Asn	Gly	Gly	Cys	Glu	Lys	Met	Lys	Trp	Gly	Asp	Leu	Tyr	Ile	Ser
1250						1255					1260			
His	Lys	Asp	Trp	Asp	Asp	Phe	Ala	Arg	Gln	Ile				
1265						1270								

<210> SEQ ID NO 8
 <211> LENGTH: 1335
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 8

Met	Met	Gln	Ile	Met	Lys	Asn	Phe	Asp	Lys	Phe	Thr	Asn	Leu	Tyr	Ser
1				5					10					15	
Val	Ser	Lys	Thr	Leu	Arg	Phe	Glu	Leu	Arg	Pro	Glu	Pro	Lys	Thr	Leu
			20					25					30		
Glu	Tyr	Met	Arg	Ser	Asn	Leu	Arg	Phe	Asp	Lys	Asn	Leu	Gln	Thr	Phe
		35				40					45				
Leu	Ala	Asp	Gln	Glu	Ile	Glu	Asp	Ala	Tyr	Gln	Ala	Leu	Lys	Pro	Ile
	50					55					60				
Phe	Asp	Ser	Leu	His	Glu	Arg	Phe	Ile	Thr	Glu	Ser	Leu	Glu	Ser	Gly
65				70					75					80	
Ser	Ala	Gln	Lys	Ile	Asp	Phe	Ser	Lys	Tyr	Leu	Glu	Lys	Tyr	Arg	Asn
			85					90						95	
Lys	Arg	Asp	Leu	Gly	Ile	Lys	Ala	Leu	Glu	Gly	Thr	Glu	Lys	Leu	Leu
		100						105					110		
Arg	Asn	Asn	Phe	Ala	Glu	Ile	Tyr	Lys	Ala	Thr	Ala	Lys	Ser	Trp	Lys
		115					120					125			
Glu	Asn	Ala	Gly	Lys	Asp	Gly	Lys	Gly	Lys	Glu	Val	Phe	Lys	Lys	Glu
	130					135					140				
Gly	Phe	Asn	Ile	Leu	Thr	Glu	Lys	Gly	Ile	Leu	Glu	Tyr	Ile	Glu	Lys
145				150						155					160

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Asn	Ile	Asp	Ser	Phe	Ser	Ala	Ile	Lys	Ser	Pro	Glu	Glu	Ile	Arg	Gly	165	170	175
Ala	Leu	Gly	Ala	Phe	Asp	Gly	Phe	Phe	Thr	Tyr	Phe	Thr	Gly	Phe	Asn	180	185	190
Gln	Asn	Arg	Glu	Asn	Tyr	Tyr	Glu	Thr	Lys	Lys	Glu	Ala	Ser	Thr	Ala	195	200	205
Val	Ala	Thr	Arg	Ile	Val	His	Glu	Asn	Leu	Pro	Lys	Phe	Cys	Asp	Asn	210	215	220
Ile	Leu	Ile	Phe	Asp	Glu	Arg	Ala	Glu	Asp	Tyr	Ile	Gly	Ala	Tyr	Lys	225	230	235
Ala	Leu	Gln	Lys	Met	Gly	Arg	Ala	Leu	Val	Asn	Lys	Glu	Gly	Gly	Glu	245	250	255
Leu	Pro	Ser	Ile	Ser	Gly	Asp	Leu	Phe	Lys	Ile	Thr	Phe	Phe	Asn	Lys	260	265	270
Cys	Phe	Ser	Gln	Lys	Gln	Ile	Glu	Glu	Tyr	Asn	Thr	Ala	Ile	Gly	Asn	275	280	285
Ala	Asn	Ser	Leu	Val	Asn	Leu	Phe	Asn	Gln	Ala	Lys	Arg	Asp	Glu	Asp	290	295	300
Gly	Tyr	Lys	Lys	Leu	Ala	Leu	Phe	Lys	Thr	Leu	Tyr	Lys	Gln	Ile	Gly	305	310	315
Cys	Asp	Lys	Lys	Asp	Ser	Leu	Phe	Phe	Ala	Val	Thr	His	Asp	Arg	Arg	325	330	335
Ala	Asp	Ala	Glu	Lys	Ala	Arg	Glu	Asn	Gly	Gln	Glu	Ala	Phe	Ser	Val	340	345	350
Glu	Glu	Val	Leu	Val	Leu	Ala	Lys	His	Ala	Gly	Glu	Lys	Tyr	Phe	Asn	355	360	365
Lys	Gly	Asn	Asp	Asp	Gly	Glu	Val	Asn	Thr	Thr	Gln	Glu	Phe	Ile	Ser	370	375	380
Tyr	Ile	Lys	Asp	Arg	Ser	Asp	Tyr	Gln	Gly	Ile	Tyr	Trp	Ser	Lys	Ala	385	390	395
Ala	Leu	Asn	Thr	Ile	Ser	Asn	Lys	Tyr	Phe	Asp	Asn	Trp	Tyr	Glu	Leu	405	410	415
Ile	Asp	Gln	Leu	Lys	Glu	Ala	Lys	Val	Phe	Thr	Lys	Thr	Gly	Ser	Gly	420	425	430
Ser	Glu	Asp	Asn	Val	Lys	Ile	Pro	Asp	Ala	Ile	Glu	Leu	Glu	Gly	Phe	435	440	445
Phe	Gln	Val	Leu	Asn	Lys	Ile	Gln	Asp	Trp	Lys	Thr	Val	Phe	Phe	Lys	450	455	460
Lys	Ser	Ile	Thr	Ala	Asp	Pro	Gln	Lys	Leu	Gly	Ile	Ile	Glu	Ser	Ser	465	470	475
Glu	Thr	Ala	Ser	Ala	Ala	Leu	Leu	Ser	Leu	Ile	Phe	Asp	Asp	Val	Ala	485	490	495
Lys	His	Thr	Lys	Leu	Phe	Ile	Asp	Gln	Ser	Glu	Asp	Ile	Leu	Lys	Val	500	505	510
Glu	Asn	Phe	Val	Lys	Pro	Glu	Asn	Lys	Glu	Asp	Ile	Lys	Arg	Trp	Leu	515	520	525
Asp	His	Ser	Leu	Ala	Ile	Asn	Gln	Met	Leu	Lys	Tyr	Phe	Leu	Val	Lys	530	535	540
Glu	Ser	Arg	Thr	Lys	Gly	Ala	Pro	Ile	Asp	Pro	Thr	Leu	Thr	Lys	Ala	545	550	555
Leu	Asp	Thr	Leu	Leu	Arg	Ser	Gln	Asp	Ala	Glu	Trp	Phe	Lys	Trp	Tyr			

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565							570					575				
Asp	Val	Leu	Arg	Asn	Tyr	Leu	Thr	Lys	Lys	Pro	Gln	Asp	Gly	Thr	Lys	
			580					585					590			
Glu	Asn	Lys	Leu	Lys	Leu	Ser	Phe	Glu	Asn	Gly	Thr	Leu	Ala	Asn	Gly	
		595					600					605				
Trp	Asp	Val	Asn	Lys	Glu	Pro	Asp	Asn	Phe	Cys	Val	Ile	Leu	Gln	Asn	
	610					615					620					
Pro	Glu	Gly	Lys	Lys	Phe	Leu	Ala	Ile	Ile	Ala	Arg	Gln	Glu	Gly	Gln	
625					630					635					640	
Lys	Gly	Phe	Asn	Gln	Val	Phe	Ala	Lys	Lys	His	Asp	Asn	Pro	Leu	Tyr	
			645						650					655		
Lys	Val	Asp	Glu	Gly	Gly	Val	Phe	Trp	Ser	Lys	Met	Glu	Tyr	Lys	Leu	
		660						665					670			
Leu	Pro	Gly	Pro	Asn	Lys	Met	Leu	Pro	Lys	Cys	Leu	Met	Pro	Lys	Ser	
		675					680					685				
Asn	Arg	Glu	Lys	Tyr	Gly	Ala	Thr	Glu	Glu	Val	Leu	Lys	Ile	Tyr	Asn	
	690					695					700					
Gln	Gly	Ser	Phe	Lys	Lys	Thr	Glu	Ser	Asn	Phe	Ser	Lys	Lys	Asp	Leu	
705					710					715					720	
Ser	Arg	Leu	Ile	Asn	Phe	Tyr	Lys	Ser	Ala	Leu	Gln	Gln	Tyr	Glu	Asp	
			725						730					735		
Trp	Arg	Cys	Phe	Asn	Phe	Ser	Phe	Arg	Ala	Thr	Asp	Ser	Tyr	Glu	Asp	
		740						745					750			
Ile	Gly	Gln	Phe	Tyr	Arg	Asp	Val	Glu	Ser	Gln	Gly	Tyr	Lys	Leu	Asp	
	755					760						765				
Phe	Gln	Ser	Ile	Asn	Thr	Asp	Val	Leu	Asp	Glu	Leu	Val	Glu	Glu	Gly	
	770					775					780					
Lys	Ile	Tyr	Leu	Phe	Glu	Ile	Lys	Asn	Gln	Asp	Ser	Asn	Gln	Gly	Lys	
785					790					795					800	
Ser	Ser	Ile	His	Arg	Asp	Asn	Leu	His	Thr	Met	Tyr	Trp	Asn	Ala	Leu	
			805						810					815		
Phe	Gln	Glu	Val	Leu	Asn	Arg	Pro	Lys	Leu	Asn	Gly	Gly	Ala	Glu	Leu	
		820						825					830			
Phe	Tyr	Arg	Lys	Ala	Leu	Ser	Pro	Glu	Lys	Ile	Lys	Glu	Leu	Gly	Ser	
	835						840					845				
Val	Asp	Lys	Asn	Gly	Lys	Arg	Ile	Ile	Arg	Asn	Tyr	Arg	Phe	Ser	Lys	
	850					855					860					
Glu	Lys	Phe	Ile	Phe	His	Ile	Pro	Ile	Thr	Leu	Asn	Phe	Cys	Leu	Ser	
865					870					875					880	
Asp	Thr	Arg	Val	Asn	Asp	Thr	Val	Asn	Gln	Glu	Leu	Ser	Arg	Thr	Ser	
			885						890					895		
Ser	Ser	His	Phe	Leu	Gly	Ile	Asp	Arg	Gly	Glu	Lys	His	Leu	Ala	Tyr	
		900						905					910			
Tyr	Phe	Leu	Val	Asp	Gln	Asn	Gly	Lys	Ile	Val	Leu	Asp	Glu	Tyr	Gly	
	915						920					925				
Lys	Ala	Val	Gln	Gly	Thr	Leu	Asn	Ile	Pro	Phe	Leu	Asp	Asn	Asn	Gly	
	930					935					940					
Asn	Val	Arg	Lys	Ile	Lys	Ala	Lys	Arg	Arg	Ser	Leu	Asp	Glu	Asn	Gly	
945					950					955					960	
Lys	Glu	Lys	Ile	Glu	Glu	Val	Trp	Cys	Lys	Asp	Tyr	Asn	Glu	Leu	Leu	
			965					970						975		

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Glu	Ala	Arg	Ala	Gly	Asp	Arg	Ala	Tyr	Ala	Arg	Lys	Asn	Trp	Gln	Thr
	980							985					990		
Ile	Gly	Asn	Ile	Lys	Glu	Leu	Lys	Glu	Gly	Tyr	Ile	Ser	Gln	Val	Val
	995						1000					1005			
Arg	Lys	Ile	Val	Asp	Leu	Ala	Ile	Glu	Tyr	Glu	Ala	Phe	Ile	Val	
	1010					1015						1020			
Leu	Glu	Asp	Leu	Asn	Val	Gly	Phe	Lys	Arg	Gly	Arg	Gln	Lys	Ile	
	1025					1030						1035			
Glu	Lys	Ser	Val	Tyr	Gln	Lys	Leu	Glu	Leu	Ala	Leu	Ala	Lys	Lys	
	1040					1045						1050			
Leu	Asn	Phe	Val	Val	Asp	Lys	Ser	Ala	Lys	Ile	Gly	Gly	Leu	Lys	
	1055					1060						1065			
Ser	Val	Thr	Asn	Ala	Leu	Gln	Leu	Ala	Pro	Pro	Val	Ser	Asn	Phe	
	1070					1075						1080			
Gly	Asp	Ile	Glu	Gly	Arg	Lys	Gln	Phe	Gly	Ile	Met	Leu	Tyr	Thr	
	1085					1090						1095			
Arg	Ala	Asn	Tyr	Thr	Ser	Gln	Thr	Asp	Pro	Ala	Thr	Gly	Trp	Arg	
	1100					1105						1110			
Lys	Ser	Ile	Tyr	Leu	Lys	Arg	Gly	Ser	Glu	Glu	Ser	Ile	Arg	Lys	
	1115					1120						1125			
Gln	Ile	Ile	Asp	Ser	Phe	Glu	Glu	Ile	Gly	Phe	Asp	Gly	Glu	Asp	
	1130					1135						1140			
Tyr	Phe	Phe	Thr	Tyr	Thr	Asp	Ser	Val	Ala	Gly	Arg	Thr	Trp	Ile	
	1145					1150						1155			
Leu	Tyr	Ser	Gly	Lys	Asn	Gly	Gly	Ser	Leu	Asp	Arg	Phe	Tyr	Gly	
	1160					1165						1170			
Lys	Arg	Asp	Asn	Asp	Lys	Asn	Gln	Trp	Val	Ser	Met	Arg	Gln	Asp	
	1175					1180						1185			
Val	Ser	Lys	Gln	Leu	Asp	Gly	Ile	Leu	Ala	Asn	Phe	Glu	Lys	Asp	
	1190					1195						1200			
Arg	Ser	Ile	Leu	Ala	Gln	Ile	Ile	Asp	Gly	Glu	Val	Asp	Leu	Ile	
	1205					1210						1215			
Lys	Val	Glu	Gln	Lys	Tyr	Thr	Ala	Trp	Glu	Ser	Phe	Arg	Ser	Thr	
	1220					1225						1230			
Ile	Asp	Leu	Ile	Gln	Gln	Ile	Arg	Asn	Thr	Gly	Thr	Ser	Glu	Arg	
	1235					1240						1245			
Asp	Gly	Asp	Phe	Ile	Leu	Ser	Pro	Val	Arg	Asp	Glu	Arg	Gly	Ile	
	1250					1255						1260			
His	Phe	Asp	Ser	Arg	Asp	Thr	Arg	Glu	Gly	Met	Pro	Thr	Ser	Gly	
	1265					1270						1275			
Asp	Ala	Asn	Gly	Ala	Tyr	Asn	Ile	Ala	Arg	Lys	Gly	Thr	Ile	Met	
	1280					1285						1290			
Gly	Glu	His	Ile	Lys	Arg	Glu	Tyr	Ser	Arg	Met	Phe	Ile	Ser	Asp	
	1295					1300						1305			
Glu	Glu	Trp	Asp	Ala	Trp	Leu	Ala	Gly	Lys	Gln	Val	Trp	Glu	Lys	
	1310					1315						1320			
Trp	Leu	Lys	Asp	Asn	Glu	Lys	Ile	Leu	Lys	Lys	Lys				
	1325					1330						1335			

<210> SEQ ID NO 9

<211> LENGTH: 1274

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 9

Met Lys Asn Ser Leu Glu Asp Phe Thr Asn Leu Tyr Ser Leu Gln Lys
 1             5             10             15

Thr Leu Arg Phe Glu Leu Lys Pro Ile Gly Asn Thr Gln Ser Met Leu
 20             25             30

Glu Glu Asp Gly Val Phe Asp Thr Asp Glu Lys Arg Lys Ile Ala Tyr
 35             40             45

Ser Lys Thr Lys Pro Tyr Ile Asp Arg Leu His Arg Glu Phe Ile Glu
 50             55             60

Glu Ser Leu Ser Asp Ala Gln Ile Ser Lys Leu Asp Glu Tyr Phe Lys
 65             70             75             80

Ala Tyr Val Asp Tyr Lys Lys Asp Lys Lys Asp Thr Lys Arg Phe Asn
 85             90             95

Arg Ile Lys Gln Phe Lys Ser Val Leu Arg Lys Glu Val Val Asp His
100            105            110

Phe Asn Lys Gln Gly Lys Glu Trp Thr Thr Val Lys Phe Ala His Leu
115            120            125

Lys Ile Lys Lys Lys Asp Leu Glu Val Leu Phe Glu Lys Gln Leu Pro
130            135            140

Asn Ile Leu Lys Glu Glu Tyr Gly Thr Glu Lys Glu Thr Gln Ile Ile
145            150            155            160

Asp Glu Asp Ser Gly Glu Val Thr Ser Ile Phe Asp Met Trp Asn Gly
165            170            175

Phe Met Gly Tyr Phe Thr Lys Phe Phe Glu Thr Arg Lys Asn Phe Tyr
180            185            190

Lys Ser Asp Gly Thr Ser Thr Ala Ile Ala Thr Arg Ile Ile Asp Gln
195            200            205

Asn Leu Asp Arg Phe Ile Glu Asn Ile Leu Ile Tyr Asp Ser Ile Lys
210            215            220

Pro Lys Ile Asp Thr Ser Glu Val Arg Glu Phe Phe Asn Leu Glu Ser
225            230            235            240

Asp Thr Ile Phe Ser Met Glu Phe Tyr Asn Asn Cys Leu Leu Gln Ala
245            250            255

Gly Ile Asp Gln Tyr Asn Asn Phe Leu Gly Gly Lys Thr Leu Glu Asn
260            265            270

Gly Arg Lys Ile Arg Gly Ile Asn Glu Leu Ile Asn Lys Tyr Arg Gln
275            280            285

Glu Asn Pro Glu Asp Lys Ile Pro Phe Leu Lys Lys Leu Asp Lys Gln
290            295            300

Ile His Ser Glu Lys Glu Lys Phe Ile Gln Gln Ile Glu Thr Leu Glu
305            310            315            320

Asp Leu Lys Glu Glu Leu Gln Lys Phe Tyr Asn Ser Ser Asn Glu Lys
325            330            335

Ile Lys Ile Leu Asp Asn Leu Leu Ser Arg Ile Glu Glu Phe Lys Pro
340            345            350

Glu Gly Ile Phe Ile Ser Lys Gln Ala Phe Asn Thr Ile Ser Arg Arg
355            360            365

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Trp	Thr	Asp	Gln	Ser	Glu	Ala	Phe	Glu	Thr	Ser	Leu	Phe	Glu	Ser	Leu
370						375					380				
Lys	Glu	Glu	Lys	Pro	Ile	Thr	Gly	Thr	Ala	Lys	Lys	Lys	Asp	Asp	Gly
385					390					395					400
Tyr	Asn	Phe	Pro	Glu	Phe	Ile	Ser	Leu	Gln	Ser	Ile	Arg	Asn	Thr	Leu
			405					410						415	
Lys	Lys	Val	Gln	Gly	Glu	Glu	Arg	Phe	Trp	Lys	Glu	Arg	Tyr	Tyr	Arg
			420					425					430		
Asp	Asn	Ser	Glu	Ser	Gly	Ile	Leu	Ala	Gly	Asn	Glu	Glu	Ile	Trp	Thr
		435					440					445			
Gln	Phe	Leu	Met	Ile	Phe	Lys	Ser	Glu	Phe	Asn	Ser	Lys	Phe	Glu	Arg
450						455					460				
Asn	Asp	Pro	Glu	Asp	Asn	Gly	Thr	Ile	Gly	Tyr	Asn	Leu	Phe	Lys	Glu
465					470					475					480
Asp	Leu	Glu	Lys	Leu	Leu	Lys	Asp	Leu	Lys	Ile	Thr	Lys	Asp	Thr	Lys
			485					490						495	
Ser	Ile	Ile	Lys	Arg	Phe	Ala	Asp	Glu	Ala	Leu	His	Ile	Tyr	Gln	Val
			500					505					510		
Gly	Lys	Tyr	Phe	Ala	Leu	Glu	Lys	Asp	Arg	Val	Trp	Ile	Ser	Ser	Tyr
		515					520					525			
Asp	Asp	Leu	Leu	Asp	Thr	Phe	Tyr	Thr	Asp	Pro	Asn	Thr	Gly	Tyr	Leu
530						535					540				
Ser	Phe	Tyr	Glu	Gly	Ala	Tyr	Glu	Gln	Ile	Val	Gln	Pro	Tyr	Asn	Met
545					550					555					560
Ile	Arg	Asn	Tyr	Leu	Thr	Arg	Lys	Pro	Tyr	Ser	Asp	Glu	Lys	Trp	Lys
			565					570						575	
Leu	Asn	Phe	Glu	Asn	Pro	Thr	Leu	Ala	Asn	Gly	Trp	Asp	Lys	Asn	Lys
			580					585					590		
Glu	Thr	Asp	Asn	Ser	Ser	Ile	Met	Leu	Arg	Lys	Glu	Gly	Ala	Tyr	Tyr
		595					600					605			
Leu	Gly	Ile	Met	Lys	Lys	Gly	Lys	Asn	Lys	Leu	Phe	Glu	Glu	Arg	Asn
610					615						620				
Arg	Gln	Leu	Phe	Glu	Pro	Lys	Asn	Gly	Glu	Asp	Thr	Tyr	Glu	Lys	Leu
625					630					635					640
Ser	Tyr	Lys	Leu	Phe	Pro	Asp	Pro	Ala	Lys	Met	Ile	Pro	Lys	Val	Cys
			645					650						655	
Phe	Ser	Asn	Lys	Asn	Ile	Gln	Met	Phe	Ser	Pro	Ser	Thr	Glu	Ile	Met
		660						665					670		
Asn	Ile	Tyr	Asn	Gly	Glu	Thr	Phe	Lys	Lys	Asn	Ser	Asp	Asp	Phe	Ser
		675					680					685			
Val	Ser	Ser	Met	Gln	Lys	Leu	Ile	Ala	Phe	Tyr	Thr	Lys	Cys	Leu	Ser
690					695						700				
Gln	Tyr	Glu	Gly	Trp	Lys	Tyr	Tyr	Asp	Phe	Lys	Tyr	Ile	Lys	Ser	Pro
705					710					715					720
Asp	Gln	Tyr	Lys	Asp	Asn	Ile	Gly	Glu	Phe	Tyr	Asn	Asp	Val	Ala	Lys
			725					730						735	
Ser	Gly	Tyr	Arg	Val	Trp	Phe	Glu	Asn	Ile	Ser	Gln	Ser	Tyr	Val	Asp
			740					745					750		
Ser	Lys	Asn	Thr	Met	Gly	Glu	Leu	Tyr	Leu	Phe	Lys	Ile	His	Asn	Lys
		755					760					765			
Asp	Trp	Asn	Gln	Lys	Asp	Lys	Lys	Thr	Lys	Val	Gly	Ser	Lys	Asn	Leu

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770	775	780
His Thr His Tyr Phe Glu Glu Leu Phe Ser Gln Asp Asn Ile Glu Asn		
785	790	795 800
Asn Phe Pro Leu Lys Leu Asn Gly Glu Ala Glu Val Phe Tyr Arg Pro		
	805	810 815
Lys Thr Asn Pro Glu Lys Leu Gly Thr Lys Lys Asp Ser Lys Gly Arg		
	820	825 830
Glu Val Ile Asp Arg Lys Arg Tyr Ala Ser Asp Lys Val Leu Phe His		
	835	840 845
Val Pro Ile Thr Leu Asn Arg Thr Pro Val Thr Thr Thr Lys Leu Asn		
	850	855 860
Lys Glu Ile Asn Gly Phe Leu Ala Asn Asn Pro Ser Ile Asn Ile Ile		
865	870	875 880
Gly Val Asp Arg Gly Glu Lys His Leu Val Tyr Tyr Ser Val Val Asn		
	885	890 895
Gln Arg Gly Lys Met Leu Glu Ser Gly Ser Phe Asn Thr Ile Asn Gly		
	900	905 910
Val Asp Tyr His Gly Lys Leu Glu Glu Arg Ala Asp Arg Arg Glu Gln		
	915	920 925
Ala Arg Arg Asp Trp Gln Asp Val Glu Gly Ile Lys Asn Leu Lys Lys		
	930	935 940
Gly Tyr Ile Ser Leu Val Val Arg Glu Leu Ala Asn Leu Ser Ile Lys		
945	950	955 960
Tyr Asn Ala Ile Ile Val Met Glu Asp Leu Asn Met Arg Phe Lys Gln		
	965	970 975
Ile Arg Gly Gly Ile Glu Lys Ser Ala Tyr Gln Gln Leu Glu Lys Ala		
	980	985 990
Leu Ile Glu Lys Leu Asn Tyr Leu Val Asn Lys Thr Glu Thr Asp Pro		
	995	1000 1005
Gln Lys Thr Gly His Ile Leu Lys Ala Tyr Gln Leu Thr Ser Pro		
1010	1015	1020
Ile Lys Ser Phe Lys Glu Met Gly Lys Gln Thr Gly Ile Ile Phe		
1025	1030	1035
Tyr Thr Gln Ala Ser Tyr Thr Ser Val Thr Asp Pro Ile Thr Gly		
1040	1045	1050
Trp Arg Pro Asn Leu Tyr Leu Lys Tyr Ser Ser Ala Ser Lys Ala		
1055	1060	1065
Lys Ser Asp Ile Leu Lys Phe Ser Lys Ile Ser Tyr Asn Thr Asn		
1070	1075	1080
Asn Asn Arg Phe Glu Phe Thr Tyr Asp Leu Arg Asn Phe Val Asn		
1085	1090	1095
Met Lys Ala Tyr Pro Gln Lys Thr Ala Trp Thr Ile Cys Ser Asn		
1100	1105	1110
Val Glu Arg Phe Arg Trp Asp Arg Lys Gly Asn Lys Asn Asn Gly		
1115	1120	1125
Glu Tyr Ile Gln Tyr Lys Asp Leu Thr Glu Asn Phe Lys Thr Phe		
1130	1135	1140
Phe Glu Glu Val Ser Ile Asn Tyr Lys Gly Asp Ile Leu Phe Gln		
1145	1150	1155
Ile Lys Asn Leu Ser Glu Lys Gly Asn Glu Lys Phe Phe Arg Asp		
1160	1165	1170

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Leu Ile  Phe Tyr Ile Ser Leu  Ile Ser Gln Ile Arg  Asn Thr Gln
1175                                1180                                1185

Lys Asp  Lys Lys Gly Asp Glu  Asn Asp Phe Ile Leu  Ser Pro Val
1190                                1195                                1200

Glu Pro  Phe Phe Asp Ser Arg  Lys Ser Ser Thr Phe  Gly Glu Asn
1205                                1210                                1215

Leu Pro  Leu Asn Gly Asp Ala  Asn Gly Ala Tyr Asn  Ile Ala Arg
1220                                1225                                1230

Lys Gly  Ile Ile Met Leu Asn  Lys Ile Ser Lys Gly  Ser Lys Asn
1235                                1240                                1245

Lys Val  Lys Glu Asp Ile Gly  Trp Gly Asp Leu Tyr  Ile Pro His
1250                                1255                                1260

Thr Glu  Trp Asp Asp Phe Ala  Thr Gly Ser Ile
1265                                1270

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<210> SEQ ID NO 10
<211> LENGTH: 1456
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Bacterial species

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<400> SEQUENCE: 10

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Met Ser Thr Lys Arg Ser Phe Ser Asp Phe Thr Asn Leu Tyr Ser Val
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Ser Lys Thr Leu Arg Phe Glu Leu Lys Pro Ile Gly Lys Thr Leu Glu
20     25     30

Asn Met Arg Glu Arg Ile Tyr Asn Asp Lys Lys Asp Tyr Asp Ser Ala
35     40     45

Leu Gln Thr Phe Leu His Asp Gln Ala Ile Glu Asp Ala Tyr Lys Thr
50     55     60

Leu Lys Pro Ile Leu Asp Ser Leu His Glu Glu Phe Ile Asn Thr Ser
65     70     75     80

Leu Asn Ser Ser Lys Ala Lys Asn Ile Asp Leu Ser Glu Tyr Leu Asn
85     90     95

Ala Tyr Arg Glu Arg Gly Asn Asp Thr Lys Thr Gly Glu Glu Ser Lys
100    105    110

Leu Ser Gly Ile Glu Lys Ser Leu Arg Lys Ala Ile Gly Glu Thr Tyr
115    120    125

Leu Thr Trp Ala Lys Ser Phe Thr Glu Gln Ala Lys Asn Leu Ile Gly
130    135    140

Ile Ile Glu Asp Ile Trp Asp Thr Glu Glu Glu Trp Asp Glu Glu Lys
145    150    155    160

Lys Thr Lys Trp Leu Phe Lys Lys Lys Asn Phe Glu Leu Leu Thr Glu
165    170    175

Ser Gly Ile Leu Val Phe Ile Glu Lys Lys Leu Asp Thr Met Asn Ile
180    185    190

Ser Glu Gln Glu Lys Thr Asp Ile Lys Lys Ala Leu Glu Glu Phe Lys
195    200    205

Gly Phe Phe Thr Tyr Phe Ser Gly Tyr Asn Gln Asn Arg Lys Asn Tyr
210    215    220

Tyr Glu Thr Lys Ala Glu Lys Lys Thr Ala Ile Ala Thr Arg Ile Val
225    230    235    240

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His	Glu	Asn	Leu	Pro	Lys	Phe	Cys	Asp	Asn	Val	Ile	Leu	Phe	His	Gly
				245					250					255	
Tyr	Gln	Lys	Ile	Leu	Lys	Asp	Gly	Ser	Lys	Arg	Glu	Tyr	Lys	Lys	Lys
			260					265					270		
Glu	Glu	Tyr	Leu	Gly	Met	Tyr	Ala	Phe	Leu	Lys	Leu	Arg	Asn	Ile	Glu
		275					280					285			
Thr	Cys	Ile	Lys	Asp	Ala	Glu	Ser	Gly	Glu	Met	Ile	Glu	Leu	Tyr	Ala
	290					295					300				
Ile	Thr	Glu	Asp	Ile	Phe	Asp	Ile	Ser	Phe	Phe	Ser	Ser	Cys	Leu	Ala
305					310					315					320
Gln	Arg	Glu	Ile	Asp	Glu	His	Asn	Arg	Ile	Ile	Gly	Gly	Ile	Asp	Lys
			325						330					335	
Tyr	Asn	Arg	Ile	Ile	Gly	His	Tyr	Asn	Ala	Leu	Ile	Asn	Leu	Tyr	Asn
		340						345					350		
Gln	Ala	Arg	Lys	Lys	Asp	Glu	Lys	Phe	Thr	Lys	Leu	Ser	Pro	Phe	Lys
		355						360				365			
Glu	Leu	Tyr	Lys	Gln	Ile	Trp	Cys	Gly	Asn	Lys	Lys	Trp	Ser	Trp	Ile
	370					375					380				
Lys	Ala	Ile	Thr	His	Asp	Thr	Asp	Glu	Gln	Ile	Leu	Ala	Asp	Thr	Asn
385					390					395					400
His	Thr	Gly	Glu	Ala	Ile	Ser	Val	Glu	Arg	Ile	Leu	Ser	Leu	Ala	Ser
			405						410					415	
Lys	Ala	Gly	Lys	Lys	Tyr	Phe	Gln	Pro	Trp	Lys	Ser	Thr	Asp	Asp	Gly
			420					425					430		
Ile	Lys	Thr	Val	Pro	Asp	Phe	Leu	Asp	Trp	Leu	Arg	Gly	Gln	Thr	Asp
		435					440					445			
Trp	Asn	Gly	Ile	Tyr	Trp	Ser	Lys	Ala	Ala	Ile	Asn	Ser	Ile	Ser	Asn
	450					455					460				
Val	Tyr	Phe	Pro	Asn	Trp	Gly	Ser	Ile	Lys	Glu	Thr	Met	Lys	Gly	Asp
465					470					475					480
Lys	Thr	Leu	Val	Ser	Tyr	Asp	Lys	Lys	Arg	Glu	Glu	Gln	Ile	Lys	Ile
			485						490					495	
Asn	Glu	Ala	Val	Glu	Leu	Ser	Gly	Leu	Phe	Asp	Ile	Leu	Asp	Ser	Thr
			500					505					510		
Asp	Gly	Asp	Trp	Lys	Gln	Glu	Trp	Val	Leu	Phe	Lys	Ala	Ser	Leu	Thr
		515					520					525			
Lys	Leu	Leu	Asp	Ala	Ser	Ala	Glu	Asn	Ala	Glu	Glu	Asn	Ser	Lys	Arg
	530					535					540				
Ala	Arg	Arg	Lys	Asp	Ile	Ile	Asp	Arg	Ser	Ser	Ser	Pro	Ser	Gln	Ala
545					550					555					560
Leu	Leu	Ala	Leu	Ile	Thr	Asp	Phe	Ile	Glu	Glu	Asn	Met	Lys	His	Phe
			565						570					575	
Leu	Asp	Gln	Ser	His	Thr	Ile	Leu	Arg	Leu	Thr	Glu	Tyr	Ser	Ser	Pro
			580					585					590		
Lys	Ser	Lys	Glu	Ala	Ile	Lys	Ser	Trp	Met	Asp	Leu	Ala	Leu	Ser	Val
		595					600					605			
Ser	Gln	Thr	Ile	Arg	Tyr	Phe	Arg	Val	Lys	Glu	Ser	Lys	Thr	Lys	Gly
	610					615					620				
Asp	Thr	Leu	Asn	Ala	Glu	Leu	Val	Gly	Ile	Leu	Thr	Asn	Leu	Leu	Asp
625					630					635					640

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Ala	Glu	Asp	Ala	Thr	Trp	Phe	Glu	Trp	Tyr	Asp	Leu	Leu	Arg	Asn	Tyr	645	650	655
Leu	Thr	Lys	Lys	Pro	Gln	Asp	Asp	Ala	Lys	Glu	Asn	Lys	Leu	Lys	Leu	660	665	670
Asn	Phe	Ala	Asn	Ser	Thr	Leu	Ala	Ala	Gly	Trp	Asp	Val	Asn	Lys	Glu	675	680	685
Thr	Asp	Asn	Thr	Cys	Val	Ile	Leu	Gln	Asn	Pro	Glu	Trp	Lys	Thr	Tyr	690	695	700
Leu	Ala	Val	Met	Asn	Lys	Asn	Lys	Lys	Asn	Val	Phe	Gln	Lys	Glu	Trp	705	710	715
Asn	Glu	Trp	Arg	Trp	Lys	Lys	Lys	Thr	Thr	Lys	Leu	Asn	Pro	Leu	Tyr	725	730	735
Glu	Ile	Asp	Trp	Gly	Glu	Ser	Trp	Lys	Lys	Met	Glu	Tyr	Asp	Phe	Trp	740	745	750
Ser	Asp	Val	Ser	Lys	Met	Ile	Pro	Lys	Cys	Ser	Thr	Gln	Leu	Lys	Lys	755	760	765
Val	Ile	Lys	His	Phe	Lys	Glu	Ser	Asp	Glu	Asp	Phe	Ile	Phe	Pro	Ser	770	775	780
Gly	Tyr	Lys	Val	Thr	Ser	Gly	Glu	Arg	Phe	Ile	Glu	Glu	Cys	Arg	Ile	785	790	795
Thr	Lys	Glu	Gln	Phe	Glu	Leu	Asn	Asn	Lys	Val	Tyr	Lys	Arg	Asp	Gly	805	810	815
Asp	Arg	Ile	Ile	Ser	Ala	Phe	Arg	Tyr	Glu	Leu	Ser	Glu	Thr	Glu	Glu	820	825	830
Lys	Thr	Tyr	Ile	Lys	Ser	Phe	Gln	Lys	Gly	Tyr	Leu	Asp	Met	Leu	Leu	835	840	845
Lys	Ser	Asn	Asn	Leu	Pro	Glu	Thr	Glu	Gln	Glu	Ile	Tyr	Arg	Lys	Lys	850	855	860
Tyr	Glu	Asp	Ser	Leu	Ser	Lys	Trp	Ile	Asn	Phe	Cys	Lys	Tyr	Phe	Ile	865	870	875
Trp	Lys	Tyr	Pro	Lys	Thr	Ser	Leu	Phe	Glu	Tyr	Gln	Phe	Asp	Glu	Thr	885	890	895
Asp	His	Tyr	Lys	Ser	Val	Asp	Lys	Phe	Asn	Leu	Asp	Val	Asp	Ile	Trp	900	905	910
Ser	Tyr	Lys	Leu	Lys	Val	Asp	Thr	Lys	Ile	Asn	Lys	Thr	Ile	Leu	Asp	915	920	925
Thr	Leu	Val	Glu	Asn	Gly	Asp	Ile	Tyr	Leu	Phe	Glu	Ile	Lys	Asn	Gln	930	935	940
Asp	Ser	Asn	Ile	Gly	Lys	Trp	Glu	Asn	His	Lys	Asn	Asn	Leu	His	Thr	945	950	955
Thr	Tyr	Trp	Lys	Ser	Ile	Phe	Glu	Ser	Val	Gln	Asn	Arg	Pro	Lys	Leu	965	970	975
Asn	Gly	Glu	Ala	Glu	Ile	Phe	Tyr	Met	Lys	Pro	Leu	Ser	Pro	Glu	Lys	980	985	990
Leu	Gln	Lys	Lys	Ile	Asp	Lys	Lys	Gly	Lys	Glu	Ile	Ile	Asp	Gly	Tyr	995	1000	1005
Arg	Phe	Ser	Arg	Glu	Arg	Phe	Ile	Phe	His	Cys	Pro	Ile	Thr	Leu		1010	1015	1020
Asn	Phe	Cys	Leu	Gly	Asn	Glu	Lys	Ile	Asn	Asn	Ile	Ile	Asn	Phe		1025	1030	1035
Glu	Leu	Ser	Pro	Lys	Ser	Asp	Ile	Tyr	Phe	Leu	Gly	Leu	Asp	Arg				

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1040	1045	1050
Gly Glu Lys His Leu Val	Tyr Tyr Ser Ile Val	Asp Gln Asn Gly
1055	1060	1065
Lys Met Ile Asp Gln Trp	Ser Phe Asn Glu Ile	Lys Trp Lys Asp
1070	1075	1080
Tyr His Ala Leu Leu Thr	Lys Arg Glu Trp Asp	Arg Met Glu Ser
1085	1090	1095
Arg Lys Asn Trp Gln Thr	Ile Ser Asn Ile Ala	Lys Leu Lys Glu
1100	1105	1110
Trp Tyr Ile Ser Leu Val	Ile His Glu Ile Ile	Glu Lys Leu Lys
1115	1120	1125
Leu Asn Pro Trp Phe Ile	Val Leu Glu Asp Leu	Asn Thr Gly Phe
1130	1135	1140
Lys Arg Gly Arg Gln Lys	Ile Glu Lys Ser Ile	Tyr Gln Lys Phe
1145	1150	1155
Glu Leu Ala Leu Ala Lys	Lys Leu Asn Phe Val	Val Asp Lys Ser
1160	1165	1170
Ala Lys Leu Gly Glu Val	Gly Ser Val Thr Asn	Ala Leu Gln Leu
1175	1180	1185
Thr Pro Pro Val Ser Asn	Tyr Gly Asp Ile Glu	Asn Arg Lys Gln
1190	1195	1200
Val Gly Ile Met Leu Tyr	Thr Arg Ala Asn Tyr	Thr Ser Gln Thr
1205	1210	1215
Asp Pro Ala Thr Gly Trp	Arg Lys Thr Ile Tyr	Leu Lys Thr Gly
1220	1225	1230
Ser Glu Glu Asn Ile Lys	Glu Gln Ile Val Thr	Gln Phe Ser Asp
1235	1240	1245
Ile Gly Phe Asp Gly Lys	Asp Tyr Tyr Phe Glu	Tyr Thr Asp Lys
1250	1255	1260
Ile Gly Lys Thr Trp Ile	Leu Tyr Ser Gly Lys	Asn Gly Lys Ser
1265	1270	1275
Leu Thr Arg Phe Arg Gly	Val Arg Gly Lys Glu	Lys Asn Glu Trp
1280	1285	1290
Asn Ile Lys Glu Ile Asn	Val Arg Asn Met Leu	Asp Gly Ile Phe
1295	1300	1305
Ala Asn Phe Asp Lys Asp	Arg Ser Phe Leu Ser	Gln Ile Leu Asp
1310	1315	1320
Glu Trp Val Glu Ile Lys	Lys Ile Asp Glu His	Thr Ala Trp Glu
1325	1330	1335
Ser Leu Arg Phe Ala Ile	Asp Leu Ile Gln Gln	Ile Arg Asn Ser
1340	1345	1350
Gly Asp Lys Thr Gln Trp	Glu Asp Asp Asn Phe	Leu Phe Ser Pro
1355	1360	1365
Val Arg Asp Ala Gln Gly	Asn His Phe Asp Thr	Arg Glu Gln Lys
1370	1375	1380
Glu Gly Leu Pro Lys Asp	Ala Asp Ala Asn Gly	Ala Tyr Asn Ile
1385	1390	1395
Ala Arg Lys Trp Ile Ile	Met Asn Glu His Ile	Arg Ile Asn Glu
1400	1405	1410
Asp Thr Lys Asp Leu Asp	Leu Phe Val Ser Asp	Glu Glu Trp Asp
1415	1420	1425

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Met Trp Leu Thr Asp Arg Glu Lys Trp Lys Glu Met Leu Pro Ile
1430 1435 1440

Phe Ala Ser Arg Lys Ala Met Glu Lys Arg Arg Gly Lys
1445 1450 1455

<210> SEQ ID NO 11
<211> LENGTH: 1317
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 11

Met Ser Gln Asn Asn Thr Phe Glu Lys Phe Thr Asn Gln Tyr Ser Leu
1 5 10 15

Ser Lys Thr Leu Arg Phe Glu Leu Arg Pro Val Gly Asn Thr Glu Gln
20 25 30

Met Leu Glu Asp Glu Asn Val Phe Lys Lys Asp Glu Ile Ile Arg Lys
35 40 45

Lys Tyr Glu Gln Thr Lys Pro Phe Ile Asp Lys Leu His Lys Glu Val
50 55 60

Ile Lys Asp Ser Leu His Gly Lys Lys Ile Glu Gly Leu Asp Asp Tyr
65 70 75 80

Phe Lys Lys Phe Glu Ile Tyr Ser Lys Asn Lys Lys Asp Ser Lys Ile
85 90 95

Lys Lys Glu Phe Thr Asp Lys Glu Ser Glu Leu Arg Lys Gln Leu Asn
100 105 110

Ser His Phe Lys Ala Glu Ser Leu Phe Ser Glu Lys Val Phe Ser Leu
115 120 125

Leu Lys Glu Lys Tyr Gly Thr Glu Asp Glu Ser Phe Val Lys Asp Glu
130 135 140

Asn Gly Asn Phe Val Leu Asp Thr Val Gly Glu Lys Ile Ser Ile Phe
145 150 155 160

Asp Glu Trp Lys Gly Phe Thr Gly Tyr Phe Thr Lys Phe Gln Lys Thr
165 170 175

Arg Glu Asn Phe Tyr Lys Asp Asp Gly Thr Ser Thr Ala Ile Val Thr
180 185 190

Arg Thr Ile Asp Glu Asn Leu Tyr Arg Phe Cys Glu Asn Ile Lys His
195 200 205

Phe Glu Ser Ile Lys Asn Arg Val Asn Phe Ser Glu Ile Glu Lys Asn
210 215 220

Phe Asn Phe Lys Leu Glu Asn Leu Phe Lys Ala Asp Phe Tyr Asn Ser
225 230 235 240

Cys Leu Leu Gln Asp Gly Ile Asp Lys Tyr Asn Asp Ile Leu Gly Gly
245 250 255

Lys Thr Leu Glu Ser Gly Glu Lys Leu Lys Gly Leu Asn Glu Ile Ile
260 265 270

Asn Lys Tyr Arg Gln Asp Asn Lys Val Glu Lys Ile Gly Phe Phe Lys
275 280 285

Met Leu Asp Lys Gln Ile Leu Gly Asp Lys Glu Lys Pro Ser Phe Ile
290 295 300

Glu Ser Ile Ala Asp Asp Asn Glu Leu Leu Leu Lys Leu Lys Glu Phe
305 310 315 320

Tyr	Thr	Asn	Ala	Glu	Glu	Lys	Thr	Glu	Val	Leu	Lys	Lys	Leu	Phe	Ser
				325						330		335			
Asp	Phe	Ser	Lys	Asn	Asn	Asp	Ser	Tyr	Asp	Leu	Ser	Lys	Ile	Tyr	Ile
				340						345		350			
Asn	Lys	Val	Gly	Ile	Asn	Thr	Ile	Leu	Leu	Lys	Trp	Phe	Asp	Val	Ala
				355						360		365			
Gly	Arg	Ser	Asp	Phe	Glu	Lys	Asn	Ile	Ser	Thr	Gln	Thr	Lys	Lys	Glu
				370						375		380			
Lys	Ile	Val	Thr	Phe	Asp	Lys	Asp	Ser	Asn	Ser	Tyr	Lys	Phe	Pro	Glu
				385						390		395			
Phe	Leu	Ala	Phe	Ser	His	Ile	Lys	Glu	Ala	Leu	Ser	Asn	Gly	Thr	Tyr
				405						410		415			
Glu	Val	Lys	Glu	Ile	Trp	Lys	Glu	Arg	Tyr	Tyr	Gln	Ser	Glu	Asn	Lys
				420						425		430			
Glu	Lys	Ser	Glu	Lys	Ala	Pro	Leu	Lys	Lys	Asp	Ser	Ala	Ile	Ser	His
				435						440		445			
Trp	Glu	Glu	Phe	Leu	Gln	Ile	Phe	Ser	Tyr	Glu	Phe	Asp	Leu	Leu	Phe
				450						455		460			
Val	Gly	Ala	Glu	Ser	Gln	Ala	Gly	Tyr	Asn	Ser	Asn	Lys	Asn	Leu	Phe
				465						470		475			
Glu	Ser	Leu	Ile	Lys	Lys	Asn	Glu	Lys	Gly	Phe	Ser	Ile	Ser	Pro	Glu
				485						490		495			
Glu	Lys	Leu	Val	Ile	Lys	Asn	Phe	Val	Asp	Asn	Thr	Leu	Trp	Ile	Tyr
				500						505		510			
Gln	Met	Ala	Lys	Tyr	Phe	Ala	Ile	Glu	Lys	Lys	Arg	Lys	Trp	Leu	Glu
				515						520		525			
Ser	Glu	Tyr	Pro	Thr	Asp	Ser	Ser	Phe	Tyr	Asp	Ser	Glu	Glu	Phe	Gly
				530						535		540			
Phe	Lys	Asn	Lys	Phe	Tyr	Asp	Asp	Ala	Tyr	Asp	Lys	Ile	Val	Lys	Leu
				545						550		555			
Arg	Met	Leu	Leu	Gln	Ser	His	Leu	Thr	Lys	Lys	Pro	Phe	Ser	Thr	Asp
				565						570		575			
Lys	Trp	Lys	Leu	Asn	Phe	Glu	Asn	Pro	Thr	Leu	Ala	Lys	Gly	Trp	Asp
				580						585		590			
Lys	Asn	Lys	Glu	Ser	Asp	Asn	Ser	Ala	Val	Leu	Leu	Arg	Lys	Glu	Gly
				595						600		605			
Arg	Tyr	Tyr	Leu	Ala	Val	Met	Lys	Lys	Gly	Asn	Asn	Lys	Ile	Phe	Asp
				610						615		620			
Asp	Lys	Asn	Lys	Ser	Asn	Phe	Leu	Glu	Asn	Ile	Glu	Gly	Gly	Lys	Tyr
				625						630		635			
Glu	Lys	Met	Val	Tyr	Lys	Gln	Met	Ser	Asp	Pro	Ser	Lys	Asp	Ile	Gln
				645						650		655			
Asn	Leu	Met	Val	Ile	Asp	Asp	Lys	Thr	Val	Arg	Lys	Val	Gly	Lys	Lys
				660						665		670			
Asp	Pro	Leu	Asp	Gly	Val	Asn	Arg	Arg	Leu	Glu	Glu	Leu	Lys	Lys	Glu
				675						680		685			
Tyr	Leu	Pro	Arg	Asp	Ile	Asn	Thr	Ile	Arg	Glu	Gln	Lys	Ala	Tyr	Leu
				690						695		700			
Lys	Ser	Ser	Asp	Asn	Phe	Asn	Leu	Gly	Asp	Ala	Asn	Leu	Phe	Ile	Asn
				705						710		715			
												720			

Tyr	Tyr	Lys	Asp	Arg	Leu	Val	Glu	Tyr	His	Lys	Asp	Ile	Phe	Val	Phe
				725					730					735	
Ser	Phe	Arg	Asp	Arg	Tyr	Ser	Asp	Phe	His	Asp	Phe	Ser	Lys	His	Val
			740					745					750		
Ala	Glu	Gln	Thr	Tyr	Ser	Leu	Ser	Phe	Glu	Asp	Ile	Ser	Glu	Phe	Tyr
		755					760					765			
Ile	Gln	Glu	Lys	Asn	Asn	Asn	Gly	Glu	Leu	Phe	Leu	Phe	Glu	Ile	His
	770					775					780				
Asn	Lys	Asp	Trp	Asn	Leu	Glu	Lys	Lys	Gly	Gly	Asp	Arg	Lys	Ser	Gly
785					790					795					800
Ala	Lys	Asn	Leu	His	Thr	Val	Tyr	Phe	Glu	Ser	Leu	Phe	Ser	Lys	Glu
				805					810					815	
Asn	Glu	Asn	Asn	Asn	Phe	Ser	Ile	Lys	Leu	Asn	Gly	Glu	Ala	Glu	Leu
			820					825					830		
Phe	Tyr	Arg	Pro	Lys	Thr	Asp	Glu	Gln	Lys	Leu	Gly	Asn	Lys	Asn	Asp
	835						840					845			
Leu	Lys	Gly	Lys	Ile	Val	Leu	Asn	Lys	Lys	Arg	Tyr	Ala	Glu	Asn	Lys
	850					855					860				
Thr	Phe	Ile	His	Ile	Pro	Ile	Thr	Leu	Asn	Arg	Val	Ala	Ser	Glu	Ser
865					870					875					880
Lys	Tyr	Phe	Asn	Gln	Lys	Leu	Asn	Asp	Phe	Leu	Val	Gly	Asn	Pro	Asp
			885						890					895	
Ile	Asn	Ile	Ile	Gly	Ile	Asp	Arg	Gly	Glu	Lys	His	Leu	Ile	Tyr	Tyr
			900					905					910		
Ala	Gly	Ile	Asn	Gln	Ala	Gly	Glu	Phe	Leu	Lys	Asp	Glu	Lys	Gly	Asn
	915						920					925			
Leu	Val	Leu	Gly	Ser	Leu	Asn	Thr	Ile	Asn	Asp	Val	Asn	Tyr	Ala	Gln
	930					935					940				
Lys	Leu	Glu	Glu	Arg	Ala	Lys	Gly	Arg	Val	Lys	Ala	Lys	Gln	Asp	Trp
945					950					955					960
Gln	Glu	Ile	Glu	Asn	Ile	Lys	Asp	Leu	Lys	Arg	Gly	Tyr	Ile	Ser	Leu
				965					970					975	
Val	Val	Arg	Glu	Leu	Ala	Asp	Leu	Ile	Ile	Lys	His	Asn	Ala	Ile	Ile
			980					985					990		
Val	Phe	Glu	Asp	Leu	Asn	Met	Arg	Phe	Lys	Gln	Ile	Arg	Gly	Gly	Ile
	995						1000					1005			
Glu	Lys	Ser	Val	Tyr	Gln	Gln	Leu	Glu	Lys	Ala	Leu	Ile	Asp	Lys	
	1010					1015					1020				
Leu	Asn	Phe	Leu	Val	Asn	Lys	Gly	Glu	Lys	Asp	Pro	Thr	Lys	Ala	
	1025					1030					1035				
Gly	His	Leu	Leu	Arg	Ala	Phe	Gln	Leu	Thr	Ala	Pro	Ile	Ser	Ala	
	1040					1045					1050				
Tyr	Lys	Asp	Met	Gly	Lys	Gln	Thr	Gly	Val	Ile	Phe	Tyr	Thr	Gln	
	1055					1060					1065				
Ala	Ser														

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1115	1120	1125
Arg Asn Ile Leu Tyr Thr	Asn Val Ser Lys Gln Asp	Lys Phe Asn
1130	1135	1140
Leu Asn Thr Lys Asp Ala Val	Arg Cys Lys Trp Phe	Arg Lys Thr
1145	1150	1155
Leu Ser Glu Asn Glu Leu Asn	Lys Gly Glu Gln Lys	Leu Asn Ile
1160	1165	1170
Gln Thr Glu Thr Gly Val Asn	Ile Glu Tyr Lys Ile	Ser Asp Cys
1175	1180	1185
Leu Ile Gly Leu Phe Glu Lys	Tyr Gly Leu Asp Tyr	Gln Asn Asn
1190	1195	1200
Leu Gln Glu Glu Ile Lys Asn	Ser Gly Asp Ser Leu	Pro Val Lys
1205	1210	1215
Phe Tyr Asp Lys Leu Ser Phe	Tyr Leu His Leu Leu	Thr Asn Thr
1220	1225	1230
Arg Ser Ser Val Ser Gly Thr	Asp Ile Asp His Ile	Asn Cys Pro
1235	1240	1245
Asn Cys Gly Phe Cys Ser Lys	Asn Gly Phe Lys Gly	Gly Glu Phe
1250	1255	1260
Asn Gly Asp Ala Asn Gly Ala	Tyr Asn Ile Ala Arg	Lys Gly Ile
1265	1270	1275
Ile Ile Leu Asp Lys Leu Lys	Asn Tyr Lys Thr Glu	Asn Ser Asn
1280	1285	1290
Leu Glu Lys Met Thr Trp Gly	Asp Leu Phe Ile Asp	Ile Asp Glu
1295	1300	1305
Trp Asp Lys Phe Thr Gln Asn	Lys Thr	
1310	1315	

<210> SEQ ID NO 12

<211> LENGTH: 1385

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 12

Met Glu Thr Lys Asn Lys Ser Ile Trp Gly Asp Phe Thr Asn Lys Tyr
1 5 10 15
Ser Leu Ser Lys Thr Leu Arg Phe Glu Leu Val Pro Val Gly Lys Thr
20 25 30
Arg Glu Asn Ile Gln Lys His Asn Pro Glu Phe Val Gln Asp Gln Lys
35 40 45
Ile Glu Glu Ala Tyr Gln Ile Leu Lys Ser Val Phe Asp Lys Ile His
50 55 60
Glu Asp Phe Ile Thr Lys Ser Leu Glu Ser Asp Glu Ala Lys Ser Ile
65 70 75 80
Asn Phe Ser Glu Tyr Phe Asp Leu Tyr Lys Lys Trp Asn Glu Leu Lys
85 90 95
Lys Lys Lys Thr Asn Glu Lys Asn Ile Glu Ile Lys Lys Glu Ile Gln
100 105 110
Asn Glu Ile Glu Lys Ile Tyr Lys Asp Asn Gly Gly Ser Lys Gly Glu
115 120 125
Ile Gln Lys Ile Glu Asp Glu Leu Arg Lys Arg Phe Glu Glu Ile Phe

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130	135	140
Lys Ile Gln Gly Lys Ile Phe Lys Glu Lys Ala Cys Glu Leu Asn Ile		
145	150	155 160
Lys Glu Gly Gln Glu Lys Asp Asp Asp Glu Glu Lys Asp Asp Asn Lys		
	165	170 175
Lys Gly Phe Arg Lys Leu Leu Lys Ala Lys Phe Leu Tyr Asp Tyr Leu		
	180	185 190
Cys Asn Leu Ile Glu Ser Lys Asn Ile Ile Tyr Lys Asp Phe Phe Glu		
	195	200 205
Asn Ile Lys Asn Lys Glu Gly Glu Ser Ile Ser Lys Glu Lys Thr Lys		
	210	215 220
Asp Ala Leu Ile Arg Phe Lys Gly Phe Thr Thr Tyr Phe Gly Gly Phe		
	225	230 235 240
Glu Leu Asn Arg Leu Asn Tyr Tyr Thr Thr Lys Glu Glu Lys Ser Thr		
	245	250 255
Ala Val Ala Thr Arg Ile Val Asn Gln Asn Leu Pro Lys Phe Cys Asp		
	260	265 270
Asn Val Ile Leu Phe Glu Ile Lys Lys Ser Glu Tyr Leu Lys Ile Asp		
	275	280 285
Glu Phe Leu Lys Asn Lys Asn Ile Ser Leu Ile Ser Lys Asn Gln Asn		
	290	295 300
Gly Gly Glu Val Glu Leu His Lys Ile Asn Lys Asn Phe Phe Glu Met		
	305	310 315 320
Met Phe Phe Ser Lys Cys Leu Ser Gln Lys Glu Ile Gln Lys Tyr Asn		
	325	330 335
Leu Glu Ile Gly Asn Ala Asn Asn Leu Ile Asn Arg Tyr Asn Gln Gln		
	340	345 350
Gln Ser Asp Lys Ser Gln Lys Leu Lys Leu Phe Lys Thr Leu His Lys		
	355	360 365
Gln Ile Gly Cys Gly Asp Arg Gly Gly Phe Ile Pro Ser Ile Lys Gly		
	370	375 380
Glu Glu Asp Leu Arg Glu Arg Leu Gln Glu Ile Lys Asn Asn Ser Ile		
	385	390 395 400
Glu Tyr Phe Glu Asn Ile Asn Asp Phe Ile Glu Tyr Leu Lys Asn His		
	405	410 415
Glu Asn Tyr Glu Asn Val Tyr Trp Ser Asp Lys Ala Ile Asn Thr Ile		
	420	425 430
Ser Ser Lys Tyr Phe Ser Asp Trp Leu Asn Leu Lys Lys Glu Ile Trp		
	435	440 445
Gly Lys Arg Asp Arg Lys Gly Asn Leu Lys Asp Glu Glu Thr Lys Ile		
	450	455 460
Pro Arg Ala Val Gln Leu Lys Asp Leu Leu Glu Asn Leu Asp Lys Ile		
	465	470 475 480
Thr Asp Trp Lys Leu Glu Gly Arg Leu Phe Lys Leu Ser Leu Phe Glu		
	485	490 495
Asn Gly Arg Lys Ala Lys Lys Leu Gln Gln Glu Asp Leu Asn Lys Phe		
	500	505 510
Asn Lys Asn Lys Ile Glu Asn Glu Leu Glu Ile Glu Lys Leu Gln Ile		
	515	520 525
Ile Glu Gln Asn Pro Ser Pro Phe Gln Ala Leu Leu Asn Met Ile Phe		
	530	535 540

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Ala	Asp	Ile	Lys	Ser	Lys	Glu	Ser	Ala	Phe	Leu	Glu	Ser	Arg	Ile	Phe
545					550					555					560
Glu	Ile	Ser	Asp	Phe	Val	His	Asn	Glu	Asp	Lys	Gln	Ile	Ile	Lys	Gln
				565					570					575	
Trp	Leu	Asp	Ser	Ile	Leu	Ala	Ile	Asn	Gln	Ile	Ile	Lys	Tyr	Trp	Arg
			580					585					590		
Val	Lys	Asp	Thr	Phe	Gly	Thr	Glu	Gly	Thr	Leu	Asp	Glu	Lys	Leu	Lys
	595						600					605			
Asn	Ile	Ile	Tyr	Ser	Glu	Lys	Asn	Pro	Thr	Arg	Phe	Tyr	Asp	Ile	Ile
610					615					620					
Arg	Asn	Tyr	Leu	Thr	Lys	Lys	Pro	Gln	Asp	Glu	Leu	Asn	Lys	Leu	Lys
625					630					635					640
Leu	Asn	Phe	Glu	Asn	Ser	Thr	Leu	Ala	Gln	Gly	Leu	Asp	Val	Asn	Lys
			645						650					655	
Glu	Lys	Asp	Asn	Phe	Cys	Ile	Ile	Leu	Arg	Asp	Asp	Lys	Gln	Asn	Gln
			660					665					670		
Tyr	Leu	Gly	Ile	Leu	Asn	Ser	Lys	Asn	Lys	Asn	Ile	Phe	Glu	Ile	Asp
	675						680					685			
Gln	Asn	Glu	Asp	Ile	Tyr	Gln	Asp	Asp	Gly	Leu	Gly	Trp	Ser	Lys	Met
690						695					700				
Met	Tyr	Lys	Leu	Ile	Pro	Gly	Ala	Ser	Lys	Thr	Leu	Pro	Lys	Ile	Phe
705					710					715					720
Phe	Ser	Lys	Arg	Trp	Thr	Glu	Asn	Asn	Pro	Thr	Pro	Asp	Glu	Ile	Ser
			725						730					735	
Lys	Ile	Lys	Lys	Gly	Glu	Thr	Phe	Lys	Lys	Gly	Asp	Asn	Phe	Ile	Lys
			740					745					750		
Arg	Asp	Leu	His	Glu	Leu	Ile	Asn	Phe	Tyr	Lys	Ala	Asn	Leu	Glu	Lys
		755					760					765			
Tyr	Pro	Ser	Val	Asn	Glu	Ser	Trp	Ala	Lys	Leu	Phe	Ile	Phe	Asn	Phe
	770					775					780				
Ser	Asp	Thr	Lys	Thr	Tyr	Glu	Ser	Ile	Asp	Gln	Phe	Tyr	Asn	Glu	Val
785					790					795					800
Asp	Lys	Gln	Gly	Tyr	Lys	Val	Ser	Phe	Ile	Ser	Ile	Asn	Lys	Asn	Thr
			805						810					815	
Leu	Asp	Asn	Phe	Ile	Asp	Lys	Glu	Lys	Leu	Tyr	Leu	Phe	Gln	Ile	Lys
			820					825					830		
Asn	Lys	Asp	Asn	Asn	Leu	Asp	Lys	Gly	Glu	Lys	Lys	Gln	Ser	Asn	Lys
			835				840					845			
Asn	Leu	His	Ser	Ile	Tyr	Trp	Glu	Ala	Ile	Phe	Gly	Lys	Ala	Leu	Asn
	850					855					860				
Lys	Pro	Lys	Leu	Asn	Gly	Gly	Ala	Glu	Ile	Phe	Tyr	Arg	Pro	Ala	Leu
865					870					875					880
Ser	Glu	Lys	Lys	Ile	Ser	Glu	Leu	Lys	Ile	Lys	Asp	Lys	Asn	Gly	Lys
			885						890					895	
Asn	Ile	Ile	Ile	Ile	Lys	Asn	Tyr	Arg	Tyr	Ser	Lys	Asp	Lys	Phe	Ile
			900					905						910	
Phe	His	Cys	Pro	Ile	Thr	Leu	Asn	Phe	Ser	Ala	Lys	Ser	Ser	Lys	Leu
		915						920				925			
Asn	Asp	Glu	Ile	Asn	Asp	His	Ile	Lys	Asn	Lys	Lys	Glu	Phe	Cys	Phe
	930					935						940			

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Met	Gly	Ile	Asp	Arg	Gly	Glu	Lys	His	Leu	Ala	Tyr	Tyr	Ser	Leu	Val	945	950	955	960
Asn	Gln	Asn	Gly	Lys	Ile	Leu	Asp	Lys	Gly	Gln	Gly	Thr	Leu	Asn	Leu	965	970	975	
Pro	Phe	Val	Asp	Lys	Asp	Gly	Asn	Lys	Arg	Cys	Ile	Lys	Thr	Glu	Lys	980	985	990	
Tyr	Phe	Glu	Glu	Asp	Lys	Lys	Glu	Asn	Glu	Lys	Trp	Lys	Pro	Arg	Ile	995	1000	1005	
Ile	Asp	Cys	Pro	Asp	Tyr	Asn	Cys	Leu	Leu	Asp	Ala	Arg	Ala	Ser		1010	1015	1020	
Asn	Arg	Asp	Leu	Ala	Arg	Lys	Asn	Trp	Gln	Thr	Ile	Gly	Thr	Ile		1025	1030	1035	
Lys	Glu	Leu	Lys	Glu	Gly	Tyr	Ile	Ser	Gln	Val	Val	Arg	Lys	Ile		1040	1045	1050	
Val	Asp	Leu	Ala	Ile	Glu	Asn	Asn	Ala	Phe	Ile	Val	Leu	Glu	Asn		1055	1060	1065	
Leu	Asn	Ile	Gly	Phe	Lys	Arg	Gly	Arg	Gln	Lys	Ile	Glu	Lys	Gln		1070	1075	1080	
Val	Tyr	Gln	Lys	Leu	Glu	Leu	Ala	Leu	Ala	Arg	Lys	Leu	Asn	Phe		1085	1090	1095	
Leu	Val	Asp	Lys	Lys	Ala	Ile	Ile	Gly	Glu	Val	Gly	Ser	Val	Thr		1100	1105	1110	
Lys	Ala	Leu	Gln	Leu	Thr	Pro	Pro	Val	Asn	Asn	Phe	Gly	Asp	Ile		1115	1120	1125	
Gly	Gly	Lys	Ser	Gln	Phe	Gly	Ile	Met	Phe	Tyr	Thr	Lys	Ala	Asp		1130	1135	1140	
Tyr	Thr	Ser	Gln	Thr	Asp	Pro	Val	Thr	Gly	Trp	Arg	Lys	Ser	Ile		1145	1150	1155	
Tyr	Leu	Lys	Arg	Gly	Pro	Glu	Asp	Tyr	Ile	Lys	Asp	Gln	Ile	Leu		1160	1165	1170	
Gly	Asn	Lys	Asn	Lys	Asn	Ile	Glu	Pro	Ala	Phe	Glu	Asp	Ile	Cys		1175	1180	1185	
Phe	Asp	Gly	Gln	Asp	Tyr	Cys	Phe	Thr	Tyr	Ile	Asn	Lys	Asn	Thr		1190	1195	1200	
Gly	Lys	Lys	Trp	Thr	Leu	Tyr	Ser	Ser	Lys	Asn	Gly	Lys	Ser	Leu		1205	1210	1215	
Asp	Arg	Tyr	His	Arg	Glu	Leu	Val	Tyr	Glu	Asn	Ser	Asp	Lys	Lys		1220	1225	1230	
Trp	Leu	Pro	Lys	Lys	Gln	Asp	Val	Leu	Glu	Met	Leu	Asn	Asn	Leu		1235	1240	1245	
Phe	Glu	Gly	Phe	Asp	Lys	Lys	Lys	Ser	Leu	Leu	Lys	Gln	Leu	Glu		1250	1255	1260	
Thr	Lys	Asn	Pro	Asn	Lys	Thr	Gly	Glu	His	Pro	Ala	Trp	Glu	Ser		1265	1270	1275	
Leu	Arg	Phe	Thr	Ile	Asp	Leu	Ile	Gln	Gln	Ile	Arg	Asn	Thr	Gly		1280	1285	1290	
Ile	Lys	Glu	Arg	Asp	Glu	Asp	Phe	Ile	Leu	Ser	Pro	Val	Arg	Asp		1295	1300	1305	
Lys	Lys	Gly	Asp	His	Phe	Asp	Ser	Arg	Glu	Ala	Ser	Pro	Asp	Leu		1310	1315	1320	
Pro	Asn	Ser	Gly	Asp	Ala	Asn	Gly	Ala	Tyr	Asn	Ile	Ala	Arg	Lys					

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1325	1330	1335
Gly Ile Ile Met Ala Lys His	Ile Glu Lys Gly Tyr	Phe Leu Tyr
1340	1345	1350
Ile Ser Asp Glu Glu Trp Asp	Ala Trp Leu Ala Gly	Glu Glu Cys
1355	1360	1365
Trp Asn Arg Trp Ala Glu Lys	Asn Thr Lys Ser Leu	Leu Lys Asn
1370	1375	1380
Asn Tyr		
1385		

<210> SEQ ID NO 13
 <211> LENGTH: 1385
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 13

Met Ser Thr Lys Thr Ile Phe Ser Asp Phe Thr Asn Leu Tyr Glu Leu
1 5 10 15
Ser Lys Thr Leu Arg Phe Glu Leu Lys Pro Val Gly Glu Thr Glu Asn
20 25 30
Leu Leu Asn Glu Asn Gln Val Phe Leu Thr Asp Lys Ile Arg Gln Lys
35 40 45
Lys Tyr Glu Glu Ile Lys Pro Phe Leu Asp Glu Phe His Leu Asp Phe
50 55 60
Ile His Phe Cys Leu Ser Asp Leu His Leu Asp Tyr Thr Glu Tyr Lys
65 70 75 80
Lys Ser Leu Asp Asn Tyr Gln Lys Asp Lys Lys Asn Lys Asp Leu Glu
85 90 95
Lys Lys Lys Glu Asn Glu Glu Lys Lys Leu Arg Glu Gln Ile Val Gly
100 105 110
Lys Phe Asp Ser Lys Val Glu Asp Phe Leu Lys Thr Phe Gly Lys Val
115 120 125
Glu Lys Ile Lys Gly Lys Lys Asp Asn Glu Lys Phe Lys Val Ser Leu
130 135 140
Gly Lys Asp Trp Glu Ile Glu Phe Ser Lys Asn Asn Tyr Glu Phe Leu
145 150 155 160
Phe Glu Ile Gly Ile Phe Asp Leu Met Lys Lys Lys Phe Glu Gly Asn
165 170 175
Gly Asp Ile Tyr Val Ala Asp Lys Glu Thr Gly Glu Ile Tyr Gln Asp
180 185 190
Glu Lys Thr Gly Lys Asp Ile Thr Ile Phe Asp Asp Trp Asn Gly Trp
195 200 205
Leu Gly Tyr Leu Thr Lys Phe Phe Glu Thr Arg Lys Asn Leu Tyr Lys
210 215 220
Ser Asp Gly Thr Ser Thr Ala Ile Ala Thr Arg Ile Ile Asn Glu Asn
225 230 235 240
Leu Lys Lys Tyr Cys Glu Asn Leu Asp Ile Tyr Asn Lys Leu Ser Gln
245 250 255
Ile Glu Asn Leu Lys Asn Lys Phe Gln Asn Leu Glu Ala Asp Phe Gly
260 265 270
Ile Lys Leu Glu Lys Phe Phe Ser Leu Glu Asn Tyr Asn Ser Cys Ile

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275	280	285
Leu Gln Asn Gly Ile Glu Asn Tyr Asn Asp Ile Arg Gly Gly Lys Leu		
290	295	300
Glu Lys Asn Asn Asn Lys Ile Pro Gly Ile Asn Glu Tyr Ile Asn Lys		
305	310	315
Tyr Arg Gln Asp Ser Gly Glu Lys Leu Pro Phe Leu Gln Lys Leu Asp		
325	330	335
Lys Gln Ile Leu Ala Gly Gly Lys Glu Asn Phe Ile Glu Gln Ile Glu		
340	345	350
Asn Glu Pro Ser Phe Glu Lys Cys Leu Lys Asn Phe Tyr Asn Asn Ser		
355	360	365
Ile Lys Lys Val Asp Ile Leu Thr Gln Ile Phe Gln Asp Leu Ser Thr		
370	375	380
Tyr Thr Asn Glu Asp Tyr Lys Thr Ile Tyr Phe Ser Lys Glu Ala Phe		
385	390	395
Asn Thr Leu Ser His Lys Phe Thr Asp Gln Val Leu Asn Phe Glu Lys		
405	410	415
Leu Val Phe Glu Glu Leu Leu Leu Asn Lys Leu Val Glu Lys Lys Asp		
420	425	430
Phe Asp Lys Lys Glu Glu Lys Tyr Lys Phe Pro Asp Phe Ile Pro Leu		
435	440	445
Phe Tyr Val Lys Lys Gly Leu Glu Asn Tyr His Thr Lys Asn Leu Phe		
450	455	460
Tyr Lys Ser Arg Tyr Tyr Glu Asn Glu Ile Ile Glu Glu Asp Asn Asp		
465	470	475
Asn Ile Trp Gln Lys Phe Cys Thr Ile Leu Asn Tyr Glu Phe Gln Ser		
485	490	495
Leu Leu Ser Asn Thr Ile Ile Asn Gln Asn Gly Glu Glu Ile Glu Val		
500	505	510
Gly Phe Thr Ile Ser Lys Asn Lys Leu Glu Lys Ile Leu Asp Asn Phe		
515	520	525
Ser Leu Gly Glu Asn Asn Asn Gly Ile Ile Lys Asp Phe Ala Asp Ile		
530	535	540
Ser Lys Thr Ile Tyr Gln Met Gly Lys Tyr Phe Ala Leu Glu Lys Lys		
545	550	555
Arg Glu Trp Asn Asn Asn Phe Asp Leu Asn Asp Asp Phe Tyr Lys Thr		
565	570	575
Glu Tyr Ser Gln Glu Asn Glu Lys Tyr Gly Tyr Leu Glu Phe Tyr Asn		
580	585	590
Glu Ala Tyr Glu Gln Ile Ile Val Pro Tyr Asn Leu Met Arg Asn Phe		
595	600	605
Ile Ala Lys Lys Pro Trp Glu Asp Asn Lys Lys Trp Lys Leu Asn Phe		
610	615	620
Glu Asn Ser Ser Leu Leu Lys Gly Trp Asp Lys Glu Phe Glu Ser Tyr		
625	630	635
Gly Ser Tyr Ile Phe Glu Lys Ala Gly Leu Tyr Tyr Leu Gly Ile Ile		
645	650	655
Asn Gly Thr Lys Leu Asn Lys Asn Glu Ile Glu Lys Leu Tyr Asn Tyr		
660	665	670
Asn Ala Asn Asn Gly Ala Lys Arg Phe Val Tyr Asp Phe Gln Lys Pro		
675	680	685

Asp	Asn	Lys	Asn	Thr	Pro	Arg	Leu	Phe	Ile	Arg	Ser	Lys	Gly	Asp	Asn	
690						695					700					
Phe	Ala	Pro	Ser	Val	Lys	Glu	Leu	Asn	Leu	Pro	Ile	Asn	Asn	Ile	Ile	
705					710					715					720	
Glu	Ile	Tyr	Asp	Lys	Glu	Leu	Tyr	Lys	Lys	Asp	Lys	Glu	Lys	Pro	Asn	
				725					730					735		
Lys	His	Lys	Glu	Ser	Leu	Met	Lys	Leu	Ile	Asp	Tyr	Phe	Lys	Leu	Gly	
			740					745					750			
Phe	Arg	Lys	His	Ile	Ser	Tyr	Lys	His	Phe	Asn	Phe	Val	Trp	Lys	Glu	
	755						760					765				
Ser	Asn	Lys	Tyr	Asp	Asn	Ile	Ala	Asp	Phe	Tyr	Arg	Asp	Val	Glu	Lys	
770						775					780					
Ser	Cys	Tyr	Lys	Pro	Tyr	Trp	Glu	Glu	Asp	Ile	Asn	Phe	Asp	Glu	Leu	
785					790					795					800	
Lys	Asn	Leu	Thr	Lys	Glu	Lys	Arg	Met	Tyr	Leu	Phe	Gln	Ile	Tyr	Asn	
				805					810					815		
Lys	Asn	Phe	Glu	Leu	Asp	Glu	Ser	Ile	Ser	Thr	Asp	Asp	Tyr	Thr	Phe	
			820					825					830			
Lys	Gly	Asn	Gly	Lys	Asp	Ser	Val	His	Thr	Met	Tyr	Phe	Lys	Gly	Leu	
	835						840					845				
Phe	Ser	Lys	Asp	Asn	Leu	Glu	Asn	Lys	Asn	Gly	Val	Asn	Leu	Lys	Leu	
850						855					860					
Ser	Gly	Gly	Gly	Glu	Leu	Phe	Phe	Arg	Pro	Lys	Ser	Ile	Glu	Lys	Lys	
865					870					875					880	
Ile	Asp	Lys	Asn	Arg	Lys	Ser	Lys	Arg	Glu	Ile	Ile	Glu	Asn	Lys	Arg	
				885					890					895		
Tyr	Ser	Lys	Asp	Lys	Ile	Leu	Leu	His	Phe	Pro	Ile	Gln	Val	Asn	Phe	
			900					905					910			
Lys	Glu	Asn	Lys	Thr	Ser	Asn	Phe	Asn	Asn	Tyr	Ile	Asn	Asn	Phe	Leu	
	915						920					925				
Ala	Asn	Asn	Pro	Asp	Ile	Asn	Ile	Ile	Gly	Ile	Asp	Arg	Gly	Glu	Lys	
930						935					940					
His	Leu	Ala	Tyr	Tyr	Ser	Val	Ile	Asn	Gln	Lys	Gln	Glu	Ile	Ile	Glu	
945					950					955					960	
Ser	Gly	Ser	Leu	Asn	Tyr	Ile	Tyr	Gln	Lys	Asp	Lys	Asp	Gly	Lys	Ile	
				965					970					975		
Ile	Gln	Lys	Ser	Glu	Lys	Lys	Ile	Gln	Glu	Val	Arg	Asn	Asp	Glu	Gly	
			980					985					990			
Lys	Ile	Ile	Asp	Tyr	Glu	Leu	Val	Glu	Thr	Gly	Lys	Leu	Val	Asp	Tyr	
	995						1000						1005			
Glu	Asp	Tyr	Gly	Ile	Leu	Leu	Asp	Tyr	Lys	Glu	Lys	Lys	Arg	Arg		
1010						1015					1020					
Leu	Gln	Arg	Gln	Ser	Tr											

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Gln	Leu	Glu	Lys	Ala	Leu	Ile	Asp	Lys	Leu	Asn	Phe	Leu	Val	Asn
1085						1090					1095			
Lys	Gly	Glu	Lys	Asp	Ser	Glu	Gln	Ala	Gly	Asn	Leu	Leu	Lys	Ala
1100						1105					1110			
Phe	Gln	Leu	Thr	Ala	Pro	Ile	Gly	Thr	Phe	Lys	Asp	Met	Gly	Lys
1115						1120					1125			
Gln	Thr	Gly	Ile	Ile	Phe	Tyr	Thr	Gln	Ala	Arg	Tyr	Thr	Ser	Lys
1130						1135					1140			
Ile	Asp	Pro	Leu	Thr	Gly	Trp	Arg	Pro	Asn	Leu	Tyr	Ile	Lys	Lys
1145						1150					1155			
Gln	Ser	Ala	Glu	Leu	Asn	Lys	Glu	Ser	Ile	Leu	Lys	Phe	Asp	Ser
1160						1165					1170			
Ile	Ile	Trp	Asn	Lys	Glu	Lys	Glu	Tyr	Phe	Glu	Ile	Thr	Tyr	Asp
1175						1180					1185			
Leu	Glu	Lys	Phe	Gln	Ser	Glu	Ser	Thr	Lys	Asn	Leu	Lys	Glu	Lys
1190						1195					1200			
Lys	Glu	Glu	Lys	Leu	Glu	Arg	Thr	Lys	Trp	Thr	Leu	Ser	Thr	Arg
1205						1210					1215			
Val	Glu	Arg	Phe	Lys	Trp	Asn	Lys	Asn	Leu	Asn	Asn	Asn	Lys	Gly
1220						1225					1230			
Gly	Tyr	Glu	His	Phe	Glu	Asn	Leu	Asn	Ile	His	Phe	Lys	Glu	Leu
1235						1240					1245			
Phe	Glu	Lys	Tyr	Gly	Leu	Asp	Ile	Ser	Gly	Asp	Ile	Leu	Lys	Gln
1250						1255					1260			
Ile	His	Asn	Leu	Glu	Thr	Lys	Gly	Asn	Glu	Ala	Phe	Phe	Ser	His
1265						1270					1275			
Phe	Leu	Asp	Leu	Phe	Lys	Leu	Val	Cys	Gln	Ile	Arg	Asn	Thr	Asn
1280						1285					1290			
Gln	Asp	Lys	Lys	Gly	Asn	Glu	Asn	Asp	Phe	Ile	Tyr	Ser	Pro	Val
1295						1300					1305			
Phe	Pro	Phe	Phe	Asp	Ser	Arg	Lys	Gln	Asn	Thr	Val	Gly	Val	Lys
1310						1315					1320			
Asn	Gly	Asp	Asp	Asn	Gly	Ala	Phe	Asn	Ile	Ala	Arg	Lys	Gly	Ile
1325						1330					1335			
Ile	Ile	Leu	Glu	Arg	Ile	Gly	Lys	Trp	Lys	Lys	Glu	Asn	Asp	Met
1340						1345					1350			
Lys	Ile	Gln	Lys	Gly	Glu	Lys	Glu	Met	Tyr	Pro	Asp	Leu	Phe	Ile
1355						1360					1365			
Ser	Asn	Ile	Gly	Trp	Asp	Asn	Phe	Thr	Gln	Asn	His	Asn	Ile	Arg
1370						1375					1380			
Asp	Asn													
1385														

<210> SEQ ID NO 14

<211> LENGTH: 1310

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 14

Met Ser Gln Asn Asn Ile Lys Glu Lys Ser Ile Phe Asp Glu Phe Thr
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Asn	Lys	Tyr	Ser	Leu	Gln	Lys	Thr	Leu	Arg	Phe	Glu	Leu	Arg	Pro	Val
			20					25					30		
Leu	Asn	Thr	Glu	Gln	Met	Leu	Thr	Asp	Ser	Gly	Ile	Ile	Lys	Leu	Asp
		35					40					45			
Glu	Lys	Arg	Lys	Leu	Asn	Tyr	Glu	Lys	Thr	Lys	Pro	Phe	Leu	Asn	Arg
	50					55					60				
Leu	His	Gln	Glu	Phe	Val	Thr	Glu	Ser	Leu	Asn	Gly	Val	Arg	Leu	Lys
65					70					75					80
Ser	Leu	Asp	Gly	Tyr	Ala	Val	Leu	Tyr	Ala	Asn	Trp	Lys	Lys	Ser	Ile
			85						90					95	
Asp	Lys	Lys	Glu	Lys	Asp	Ala	Ala	Tyr	Lys	Val	Leu	Glu	Lys	Lys	Glu
			100					105					110		
Leu	Glu	Ile	Arg	Gln	Glu	Ile	Val	Val	Leu	Phe	Asp	Glu	Lys	Ala	Val
		115					120					125			
Glu	Trp	Ile	Gly	Lys	Leu	Pro	Ala	Asp	Val	Lys	Lys	Pro	Lys	Lys	Pro
	130					135						140			
Asn	Tyr	Glu	Phe	Leu	Phe	Glu	Pro	Ala	Ile	Phe	Ser	Ile	Leu	Lys	Lys
145					150					155					160
Lys	Tyr	Ser	Asp	Glu	Val	Gly	Thr	Thr	Ile	Asp	Glu	Glu	Ser	Ile	Phe
			165						170					175	
Asp	Ser	Trp	Asp	Lys	Trp	Thr	Ala	Tyr	Phe	Gly	Lys	Phe	Phe	Glu	Thr
		180						185					190		
Arg	Lys	Asn	Phe	Tyr	Lys	Ser	Asp	Gly	Lys	Ala	Thr	Ala	Val	Ala	Thr
		195					200					205			
Arg	Ile	Val	Asn	Glu	Asn	Leu	Arg	Arg	Phe	Cys	Asp	Asp	Val	Ser	Thr
	210					215					220				
Phe	Glu	Asn	Ile	Gln	Ser	Lys	Ile	Asp	Leu	Ser	Pro	Leu	Glu	Lys	Glu
225					230					235					240
Phe	Asp	Val	Ser	Leu	Lys	Lys	Val	Phe	Asp	Ile	Gln	His	Tyr	Asn	Gln
			245						250					255	
Cys	Leu	Asn	Gln	Ser	Gly	Ile	Asp	Ala	Phe	Asn	Thr	Leu	Leu	Gly	Gly
		260						265					270		
Glu	Val	His	Glu	Asn	Gly	Glu	Lys	Ile	Lys	Gly	Ile	Asn	Glu	Tyr	Ile
		275					280					285			
Asn	Glu	His	Arg	Gln	Lys	Thr	Gly	Glu	Lys	Leu	Thr	Arg	Leu	Lys	Lys
	290					295					300				
Leu	Asp	Lys	Gln	Ile	Gly	Ser	Asp	Lys	Glu	Asn	Phe	Ile	Asp	Leu	Ile
305					310					315					320
Glu	Thr	Asp	Glu	Gln	Leu	Lys	Thr	Thr	Leu	Val	Thr	Phe	Ile	Ala	Asn
			325						330					335	
Ala	Lys	Glu	Lys	Val	Asp	Leu	Leu	Asp	Lys	Ser	Val	Ser	Tyr	Leu	Thr
			340					345					350		
Lys	Asp	Thr	Asp	Val	Lys	Leu	Ser	Gly	Ile	Phe	Phe	Arg	Lys	Glu	Ala
		355					360					365			
Ile	Asn	Thr	Ile	Thr	Arg	Arg	Trp	Phe	Val	Ser	His	Glu	Lys	Ile	Ser
	370					375					380				
Asp	Ala	Leu	Val	Ser	Ala	Phe	Ser	Asp	Lys	Asn	Val	Lys	Phe	Asp	Gln
385					390					395					400
Lys	Arg	Glu	Glu	Tyr	Lys	Phe	Pro	Asp	Phe	Ile	Ser	Trp	His	Val	Ile
			405						410					415	
Gln	Asn	Ala	Val	Glu	Lys	Leu	Ala	Ser	Asp	Gly	Glu	Glu	Ile	Trp	Lys

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420							425							430						
Lys	Tyr	Tyr	Leu	Glu	Glu	Glu	Lys	Leu	Ser	Leu	Leu	Asp	Lys	Thr	Pro					
435							440							445						
Trp	Gln	Gln	Phe	Leu	Thr	Val	Phe	Glu	Cys	Glu	Tyr	Asn	Asn	Leu	Lys					
450							455							460						
Ser	Lys	Gly	His	Glu	Ser	Glu	Gly	Arg	Ser	Phe	Thr	Glu	Leu	Val	Gln					
465							470							475						
Asp	Ile	Glu	Ser	Leu	Leu	Lys	Thr	Asp	Thr	Leu	Asp	Arg	Asn	Asp	His					
485							490							495						
Val	Thr	Glu	Ile	Ile	Lys	Ser	Phe	Ser	Asp	Arg	Val	Leu	Asn	Ile	Tyr					
500							505							510						
Arg	Phe	Ala	Lys	Tyr	Phe	Ala	Leu	Asp	Lys	Ser	Cys	Gln	Trp	Asn	Pro					
515							520							525						
Asp	Gly	Leu	Asp	Thr	Asp	Asp	Phe	Tyr	Val	Ala	Tyr	Glu	Gln	Phe	Tyr					
530							535							540						
Ser	Asp	Gly	Tyr	Glu	Lys	Ile	Val	Lys	Val	Tyr	Asp	Lys	Val	Arg	Asn					
545							550							555						
Tyr	Met	Thr	Lys	Lys	Pro	Phe	Asn	Gln	Asp	Lys	Trp	Lys	Leu	Asn	Phe					
565							570							575						
Glu	Asn	Pro	Thr	Leu	Ala	Asn	Gly	Trp	Asp	Lys	Asn	Lys	Glu	Thr	Asp					
580							585							590						
Asn	Thr	Ala	Ile	Ile	Leu	Arg	Arg	Ala	Gly	Arg	Tyr	Tyr	Leu	Ala	Val					
595							600							605						
Met	Glu	Arg	Gly	His	Asn	Thr	Leu	Phe	Lys	Lys	Ile	Pro	Met	Ser	Ser					
610							615							620						
Ser	Gly	Tyr	Gln	Lys	Met	Thr	Tyr	Lys	Leu	Phe	Pro	Asp	Pro	Ser	Lys					
625							630							635						
Met	Met	Pro	Lys	Val	Cys	Phe	Ser	Lys	Lys	Gly	Leu	Glu	Phe	Phe	Lys					
645							650							655						
Pro	Ser	Ala	Glu	Ile	Met	Arg	Ile	Tyr	Lys	Asn	Gly	Glu	Phe	Lys	Lys					
660							665							670						
Gly	Asp	Thr	Phe	Ser	Leu	Ser	Ser	Met	His	Val	Leu	Ile	Asp	Phe	Tyr					
675							680							685						
Lys	Asn	Ala	Leu	Lys	Thr	Tyr	Asp	Gly	Trp	Thr	Met	Tyr	Asp	Phe	Ser					
690							695							700						
Asn	Leu	Lys	Lys	Thr	Ser	Glu	Tyr	Thr	Glu	Asn	Ile	Gly	Glu	Phe	Tyr					
705							710							715						
Arg	Asp	Val	Ala	Glu	Ser	Gly	Tyr	Gln	Ile	Asn	Phe	Asp	Tyr	Ile	Ala					
725							730							735						
Glu	Gln	Tyr	Ile	Glu	Asp	Ala	Asn	Lys	Glu	Gly	Lys	Leu	Tyr	Leu	Phe					
740							745							750						
Glu	Ile	His	Asn	Lys	Asp	Trp	Asn	Leu	Lys	Asp	Gly	Ala	Ile	Lys	Thr					
755							760							765						
Gly	Ser	Lys	Asn	Ala	His	Thr	Leu	Tyr	Phe	Glu	Gln	Val	Phe	Ser	Asp					
770							775							780						
Glu	Asn	Ala	Gln	Asn	Asn	Phe	Val	Val	Lys	Leu	Asn	Gly	Glu	Ala	Glu					
785							790							795						
Leu	Phe	Phe	Arg	Pro	Ala	Thr	Ser	Thr	Glu	Lys	Leu	Gly	Asn	His	Tyr					
805							810							815						
Asp	Ser	Lys	Gly	Asn	Val	Val	Thr	Lys	Asn	Lys	Arg	Tyr	Ala	His	Asp					
820							825							830						

Lys	Met	Phe	His	Val	Pro	Val	Thr	Leu	Asn	Arg	Thr	Ala	Pro	Asp	
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Ala	Arg	Lys	Phe	Asn	Gln	Ser	Val	Asn	Val	Phe	Leu	Ala	Asn	Asn	Pro
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Asp	Thr	Asn	Ile	Ile	Gly	Ile	Asp	Arg	Gly	Glu	Lys	His	Leu	Ala	Tyr
865				870						875		880			
Leu	Ser	Val	Ile	Asn	Gln	Lys	Gly	Asp	Ile	Leu	Lys	Ile	Lys	Ser	Leu
		885						890				895			
Asn	Lys	Ile	Glu	Val	Lys	Asp	Lys	Asp	Gly	Asn	Val	Ile	Lys	Glu	Asp
		900				905						910			
Asp	Tyr	Ala	Lys	Leu	Leu	Glu	Asp	Arg	Ala	Lys	Asn	Arg	Glu	Ser	Ala
915						920						925			
Arg	Arg	Asp	Trp	Lys	Ser	Val	Glu	Gln	Ile	Lys	Asp	Leu	Lys	Lys	Gly
930						935				940					
Tyr	Ile	Ser	Asn	Val	Val	Arg	Glu	Ile	Ala	Asp	Leu	Val	Ile	Lys	Tyr
945				950						955		960			
Asn	Ala	Ile	Val	Val	Phe	Glu	Asp	Leu	Asn	Met	Arg	Phe	Lys	Gln	Val
		965						970				975			
Arg	Gly	Gly	Ile	Glu	Lys	Ser	Val	Tyr	Gln	Gln	Leu	Glu	Lys	Ala	Leu
		980						985				990			
Ile	Asp	Lys	Leu	Asn	Phe	Leu	Val	Asp	Lys	Asn	Glu	Leu	Asp	Pro	Gln
995						1000						1005			
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Glu	Thr	Phe	Lys	Asp	Met	Gly	Lys	Gln	Thr	Gly	Val	Leu	Phe	Tyr	
1025						1030				1035					
Thr	Gln	Ala	Glu	Tyr	Thr	Ser	Gln	Thr	Asp	Pro	Val	Thr	Gly	Phe	
1040						1045				1050					
Arg	Lys	Asn	Val	Tyr	Leu	Ser	Asn	Ser	Ala	Thr	Val	Glu	Lys	Ile	
1055						1060				1065					
Lys	Ala	Phe	Val	Glu	Met	Phe	Asp	Val	Ile	Gly	Trp	Asp	Asp	Lys	
1070						1075				1080					
Leu	Lys	Ser	Tyr	Tyr	Phe	Lys	Tyr	Asn	Pro	Val	Asn	Phe	Val	Glu	
1085						1090				1095					
Thr	Lys	Phe	Lys	Glu	Asn	Thr	Phe	Ser	Lys	Asp	Trp	Val	Ile	Tyr	
1100						1105				1110					
Ala	Asn	Val	Pro	Arg	Ile	Lys	Arg	Glu	Arg	Lys	Asn	Gly	Tyr	Trp	
1115						1120				1125					
Glu	Ala	Thr	Val	Val	Asn	Pro	Asn	Glu	Glu	Phe	Leu	Lys	Leu	Phe	
1130						1135				1140					
Lys	Glu	Trp	Asp	Phe	Asp	Asn	Ile	Tyr	Val	Glu	Asp	Ile	Lys	Glu	
1145						1150				1155					
Gln	Ile	Phe	Gln	Met	Phe	Glu	Glu	Gly	Arg	Leu	Asp	Gly	Thr	Lys	
1160						1165				1170					
Glu	Phe	Asp	Gly	Lys	Asn	Arg	Asn	Phe	Trp	His	Ser	Phe	Ile	Phe	
1175						1180				1185					
Leu	Phe	Asn	Leu	Met	Leu	Gln	Val	Arg	Asn	Ser	Thr	Ala	Thr	Gln	
1190						11									

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Val	Asp	Phe	Ile	Ala	Ser	Pro	Val	Phe	Pro	Phe	Phe	Thr	Thr	Asp
1220						1225						1230		
Gly	Gly	Asp	Phe	Thr	Glu	Gly	Cys	Val	Asn	Leu	Ala	Lys	Leu	Glu
1235						1240					1245			
Asp	Lys	Phe	Val	Gly	Ser	Asn	Ala	Asp	Lys	Glu	Arg	Phe	Lys	Lys
1250						1255					1260			
Glu	Phe	Asn	Gly	Asp	Ala	Asn	Gly	Ala	Tyr	Asn	Ile	Ala	Arg	Lys
1265						1270					1275			
Gly	Ile	Ile	Met	Leu	Asn	Asn	Ile	Lys	Asn	Asn	Pro	Glu	Lys	Pro
1280						1285					1290			
Asp	Leu	Phe	Val	Ser	Lys	Lys	Asp	Trp	Asp	Lys	Phe	Ala	Gln	Ala
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Asn	Gln													
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<210> SEQ ID NO 15
 <211> LENGTH: 1275
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 15

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		20					25					30			
Pro	Ile	Gly	Lys	Thr	Gln	Lys	Ile	Leu	Glu	Asp	Asn	Lys	Val	Phe	Glu
		35				40					45				
Asn	Asp	Lys	Lys	Arg	Ala	Lys	Ser	Tyr	Glu	Glu	Ala	Lys	Lys	Tyr	Phe
50						55					60				
Asn	Lys	Leu	His	Arg	Glu	Phe	Ile	Asp	Glu	Ser	Leu	Lys	Asn	Ile	Thr
65				70					75					80	
Leu	Ser	Asn	Asn	Leu	Ile	Glu	Lys	Phe	Glu	Lys	Lys	Tyr	Leu	Thr	Trp
		85					90						95		
Lys	Asn	Ser	Lys	Asn	Lys	Asp	Asn	Ser	Thr	Glu	Leu	Lys	Lys	Ser	Ala
		100					105						110		
Lys	Arg	Leu	Arg	Ile	Val	Ile	Leu	Glu	Ser	Phe	Asn	Lys	Lys	Ala	Asn
		115				120					125				
Glu	Trp	Asn	Ser	Glu	Tyr	Ser	Asn	Gln	Val	Lys	Asn	Glu	Lys	Lys	Lys
130						135					140				
Lys	Lys	Ile	Gln	Glu	Ile	Thr	Gly	Ile	Asp	Leu	Phe	Phe	Lys	Val	Glu
145				150					155					160	
Val	Phe	Asp	Phe	Leu	Ile	His	Lys	Tyr	Pro	Glu	Val	Gln	Ile	Asn	Gly
		165					170						175		
Glu	Ser	Ile	Phe	Ser	Pro	Phe	Asn	Lys	Phe	Ser	Gly	Tyr	Phe	Lys	Lys
		180					185						190		
Phe	His	Glu	Thr	Arg	Lys	Asn	Phe	Tyr	Lys	Asp	Asp	Gly	Thr	Ser	Thr
		195				200						205			
Ala	Ile	Pro	Thr	Arg	Ile	Ile	Asp	Val	Asn	Leu	Glu	Lys	Phe	Leu	Glu
210					215						220				
Asn	Lys	Asp	Ile	Tyr	Tyr	Thr	Lys	Tyr	Phe	Gln	Lys	Tyr	Asn	Ser	Ile
225					230					235				240	

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Phe	Asn	Lys	Glu	Glu	Thr	Asp	Ile	Phe	Lys	Leu	Glu	Ser	Phe	Lys	Asn	
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Cys	Leu	Thr	Gln	Ser	Gln	Ile	Asp	Lys	Tyr	Asn	Glu	Ser	Ile	Ala	Thr	
			260					265					270			
Leu	Lys	Ser	Lys	Ile	Asn	Asn	Leu	Arg	Gln	Asn	Asn	Pro	Glu	Val	Asn	
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Lys	His	Asp	Leu	Pro	Phe	Phe	Lys	Glu	Leu	Phe	Arg	Gln	Ile	Leu	Gly	
	290					295					300					
Gln	Pro	Ile	Lys	Lys	Glu	Thr	Glu	Gln	Asp	Asn	Phe	Ile	Glu	Ile	Leu	
305					310					315					320	
Thr	Asn	Asp	Glu	Val	Phe	Pro	Val	Leu	Gln	Lys	Asn	Ile	Asp	Glu	Asn	
			325					330						335		
Glu	Leu	Tyr	Ile	Pro	Lys	Ala	Asp	Thr	Leu	Phe	Lys	Glu	Phe	Leu	Lys	
		340						345					350			
Ser	Gln	Ile	Gln	Glu	Thr	Asn	Glu	Tyr	Asn	Ile	Asn	Glu	Ile	Tyr	Val	
	355						360					365				
Ala	Ser	Arg	Phe	Ile	Asn	Ser	Ile	Ser	Asn	Asn	Trp	Phe	Ala	Glu	Trp	
	370					375					380					
Asp	Thr	Ile	Ile	Asn	Leu	Leu	Arg	Thr	Glu	Leu	Lys	Ile	Lys	Gln	Asn	
385					390					395					400	
Gln	Lys	Lys	Leu	Pro	Asp	Phe	Ile	Ser	Ile	Ala	Ser	Leu	Lys	Arg	Val	
			405						410					415		
Leu	Gln	Lys	Ser	Gln	Asp	Glu	Ile	Asp	Ala	Lys	Asp	Leu	Phe	Arg	Asn	
			420					425					430			
Asn	Tyr	Glu	Asn	Leu	Phe	Glu	Ser	Thr	Thr	Asp	Phe	Tyr	Lys	Ile	Phe	
	435						440					445				
Leu	Lys	Ile	Trp	Glu	Leu	Glu	Phe	Asn	Asp	Asn	Ile	Lys	Lys	Tyr	Asn	
	450					455					460					
Leu	Glu	Thr	Glu	Asn	Ile	Arg	Lys	Ile	Ile	Ile	Glu	Asp	Lys	Lys	Tyr	
465					470					475					480	
Leu	Pro	Asn	Lys	Lys	Ser	Ile	Leu	Lys	Asn	Gly	Glu	Thr	Gly	Ile	Ile	
			485						490					495		
His	Asn	Glu	Lys	Ile	Leu	Asp	Tyr	Ala	Gln	Ser	Ala	Leu	Asn	Ile	Tyr	
		500						505					510			
Gln	Met	Met	Lys	Tyr	Phe	Ser	Leu	Glu	Lys	Gly	Lys	Glu	Arg	Glu	Trp	
		515					520						525			
Asn	Pro	Asp	Gly	Leu	Asn	Glu	Asp	Thr	Thr	Gly	Gly	Phe	Tyr	Asp	Asp	
	530					535					540					
Phe	Asn	Lys	Tyr	Tyr	Gln	Asn	Val	Asn	Thr	Trp	Lys	Tyr	Phe	Asn	Glu	
545					550					555					560	
Phe	Arg	Asn	Tyr	Leu	Thr	Lys	Lys	Pro	Tyr	Lys	Thr	Asp	Lys	Ile	Lys	
			565						570					575		
Leu	Tyr	Phe	Gly	His	Lys	Ser	Leu	Leu	Gly	Gly	Phe	Thr	Glu	Ser	Lys	
		580						585					590			
Thr	Glu	Lys	Ser	Asn	Asn	Gly	Thr	Gln	Tyr	Gly	Ala	Tyr	Leu	Leu	Arg	
		595					600					605				
Lys	Lys	His	Gly	Leu	Gly	Gly	Phe	Asp	Tyr	Tyr	Leu	Gly	Ile	Ser	Thr	
	610				615						620					
Asp	Pro	His	Leu	Met	Ser	Tyr	Phe	Asp	Pro	Ile	Asp	Asp	Ser	Gly	Asp	
625					630					635					640	
Ser	Glu	Tyr	Glu	Arg	Leu	Asn	Tyr	Tyr	Gln	Val	Leu	Thr	Arg	Thr	Ile	

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645								650				655			
Tyr	Gly	Pro	Ser	Tyr	Glu	Gly	Asp	Tyr	Glu	Leu	Asp	Lys	Lys	Asn	Leu
			660					665					670		
Ser	Glu	Ile	Glu	Ile	Ile	Lys	Lys	Ile	Lys	Arg	Ser	Leu	Ser	Tyr	Tyr
		675					680					685			
Thr	Ser	Arg	Val	Lys	Lys	Ile	Gln	Asp	Ile	Ile	Asn	Asn	Asn	Tyr	Glu
	690					695					700				
Ser	Val	Arg	Asp	Ile	His	Lys	Asp	Ile	Thr	Asp	Val	Leu	Lys	Glu	Phe
705					710					715					720
Gly	Thr	Ile	Phe	Asp	Tyr	Lys	Val	Ile	Thr	Asn	Ser	Gln	Ile	Gln	Lys
			725						730					735	
Ala	Phe	Asn	Cys	Asp	Lys	Gly	Phe	Tyr	Leu	Phe	Glu	Ile	Tyr	Ser	Lys
			740					745					750		
Asp	Phe	Ser	Lys	Glu	Lys	Gly	Asp	Lys	Ser	Lys	Asn	Ser	Lys	Asp	Asn
		755					760					765			
Leu	His	Thr	Thr	Tyr	Phe	Lys	Ser	Leu	Met	Asp	Arg	Lys	Gln	Ser	Thr
	770					775					780				
Phe	Asp	Leu	Gly	Ser	Gly	Glu	Ile	Phe	Phe	Arg	Glu	Lys	Ser	Val	Gln
785						790				795					800
Ser	Glu	Ile	Asp	Ser	Met	Arg	Lys	Thr	Lys	Asn	Lys	Ile	Thr	Arg	Phe
			805						810					815	
Lys	Arg	Tyr	Thr	Lys	Asn	Leu	Ile	Gln	Phe	Asn	Leu	Ser	Ile	Thr	Leu
			820					825					830		
Asn	Asn	Asn	Cys	Thr	Glu	Val	Pro	Gln	Asn	Lys	Asn	Ala	Arg	Lys	Ala
		835					840					845			
Phe	Ile	Asn	Asn	Phe	Asn	Ile	Glu	Leu	Ser	Lys	Lys	Leu	Leu	Thr	Asn
	850					855					860				
Asn	Ser	Asp	Ile	Asn	Ile	Ile	Gly	Ile	Asp	Arg	Gly	Glu	Lys	His	Leu
865					870					875					880
Ala	Tyr	Tyr	Ser	Val	Ile	Asp	Gln	Gln	Ser	Asn	Ile	Leu	Glu	Thr	Gly
			885						890					895	
Ser	Phe	Asn	Lys	Ile	Gln	Glu	Arg	Lys	Asp	Arg	Glu	Pro	Thr	Asp	Tyr
		900						905					910		
Gln	Gln	Lys	Leu	Asp	Lys	Ile	Gln	Lys	Asp	Arg	Asp	Trp	Gln	Arg	Lys
		915					920					925			
Ser	Trp	Gln	Glu	Ile	Ser	Asn	Ile	Lys	Asp	Leu	Lys	Lys	Gly	Tyr	Ile
	930					935					940				
Ser	Gln	Val	Val	Tyr	Glu	Ile	Ser	Lys	Leu	Val	Lys	Lys	Tyr	Asn	Ala
945					950					955					960
Ile	Ile	Val	Phe	Glu	Asp	Leu	Asn	Ile	Gly	Phe	Lys	Arg	Gly	Arg	Phe
			965						970					975	
Ala	Ile	Glu	Lys	Gln	Val	Tyr	Gln	Asn	Leu	Glu	Leu	Ser	Leu	Ala	Lys
			980					985					990		
Lys	Leu	Asn	Tyr	Leu	Val	Phe	Lys	Asp	Ala	Asn	Glu	Gly	Glu	Ser	Gly
		995					1000						1005		
His	Tyr	Leu	Lys	Ala	Tyr	Gln	Leu	Thr	Ser	Pro	Val	Asn	Asn	Phe	
	1010					1015						1020			
Gln	Asp	Ile	Gly	Lys	Gln	Cys	Gly	Ile	Ile	Phe	Tyr	Ile	Pro	Ala	
	1025					1030						1035			
Ser	Tyr	Thr	Ser	Ala	Ile	Cys	Pro	Ser	Cys	Gly	Phe	His	Lys	Asn	
	1040					1045						1050			

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Ile Pro	Thr Ser	Ile Lys	Lys	Leu Ala	Lys Asn	Lys	Glu Phe	Val
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Glu Lys	Phe Val	Ile Thr	Tyr	Glu Leu	Lys Lys	Asp	Arg Phe	Tyr
1070			1075			1080		
Phe Gly	Tyr Lys	Ile Asn	Asp	Phe Tyr	Asn Ser	Asn	Leu Gln	Asp
1085			1090			1095		
Asn Val	Ile Phe	Tyr Ser	Asn	Val Glu	Arg Leu	Arg	Tyr Lys	Arg
1100			1105			1110		
Asn Lys	Asp Asn	Arg Ser	Gly	Glu Val	Gln Glu	Arg	Leu Pro	Asn
1115			1120			1125		
Glu Glu	Leu Lys	Lys Leu	Phe	Glu Gln	Asn His	Ile	Asn Tyr	Lys
1130			1135			1140		
Asp Asn	Pro Gln	Ile Ser	Gly	Gln Ile	Lys Asn	Gln	Lys Leu	Asp
1145			1150			1155		
Asn Glu	Lys Phe	Tyr Lys	Pro	Leu Ile	Tyr Glu	Ile	Ser Leu	Ile
1160			1165			1170		
Leu Gln	Leu Arg	Asn Ser	Lys	Thr Val	Lys Ser	Glu	Asp Gly	Thr
1175			1180			1185		
Ile Asn	Thr Asn	Ile Asn	Arg	Asp Phe	Ile Ser	Cys	Pro Ala	Cys
1190			1195			1200		
Tyr Phe	His Ser	Glu Asn	Asn	Leu Met	Asn Leu	Pro	Asn Lys	Tyr
1205			1210			1215		
Lys Gly	Gly Lys	Lys Phe	Glu	Phe Asn	Gly Asp	Ala	Asn Gly	Ala
1220			1225			1230		
Tyr Asn	Ile Ala	Arg Lys	Gly	Ile Leu	Leu Leu	Asn	Lys Leu	Asn
1235			1240			1245		
Asn Ile	Lys Asp	Ile Glu	Lys	Ile Glu	Tyr Asn	Asp	Leu Asn	Ile
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<210> SEQ ID NO 16

<211> LENGTH: 3888

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 16

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atgaagaatt acttagagca taaaaatcta attcgccaaa aggatgttga agaaaaagtt	300
aaaactaaat caatagatgt gatcaataaa attgaaaagg attggcgaaa gcgggtttct	360
gatagtttca ctaaacatcc gcaatacaag aaaatgtttg ataaaacctt atttgccgat	420
gaaagcccg tttataaatt agccgaaaat gattttcaga gaagtcagat taaaatcttt	480
gaaaaattca cgggttattt taatggtttc catgaaaacc ggaaaaacct atatgttgcc	540
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aacttatato	gccagaaaat	taatttagcc	cgcaatgact	ttcctaacct	tgcccgtta	900
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gatataaata	agtggttgga	ggatattaat	attagttccg	gacaaatctg	gatttataaa	1080
agccatctat	caggcttato	ggttatgctc	tttggtgaaa	gcggctgggg	attaattcct	1140
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tcacagcaag	aatatttttc	atttgctgaa	attggcaatg	ccattaataa	ttattctttt	1260
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<210> SEQ ID NO 17

<211> LENGTH: 3324

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

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aagtttgaaa agatagttaa aaaatatgga gatatatcta atgtattatc tacttataaa	240
gagtttttcg ctctgactg ttttaaaaat aaattaaatc agtctcaaat tgatcattac	300
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atgatgttga atgcggttga taaagaatac tcaaggatta aagagaatga tgtttttatc	540
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aaaaatgatt cctctcttac tgatatctct aataaattat gggaagattg ggcctttgtt	660
aaaaatgcta tacagaaata ttctaagag atattatgcc ttagtgacaa gaaaatagaa	720
gacatgctaa aaatgagtea ttactctatt tcttttgttc agaattctgt ttattactat	780
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<210> SEQ ID NO 18

<211> LENGTH: 3972

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 18

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gatttaaaag aaattacaga tgatgaatac actttggatt caattttttc aatcgatttc	660
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gttattaaac aaaaatcttt ttctatttct tttttacaag atgctataac cttttatatt	1260
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aaaaaacat tcaatgatga agaaattaat ttatatctcg gtattccaaa tttagggggt	1680
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<210> SEQ ID NO 19

<211> LENGTH: 3975

<212> TYPE: DNA

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<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 19

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ataacaaca atattaatga ttaccactcg atttcaaaag aaactatctt atcaatcaaa	3660
gaaaagaaat tctttgatga ttgttctttt tattttaagc ttagtcttca aatgcgaaat	3720
tcaataccaa attcggacat tgattatttg atttcaccag ttcagataaa gggaagcct	3780
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<210> SEQ ID NO 20

<211> LENGTH: 3693

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 20

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acagagggaa agttattgcg taatactttt gcaacggttt ggataaatga aggcaaaaat	300
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gaattaaaat taaagaccag agaggatatt ttatggaaaa aagagaatag aaacttggtg	480
gagatggctg atttgaaaa gtctttggga acaattgaat catggggtgt ttttgaagga	540
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aggtataatt cggatatagg aaatgcaaac tttttaatta atttatacaa tcagcaacaa	900
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ggcattgato gtggcgaaaa acaccttgct tattattcgg ttgttaatca aaagggtgaa	2640
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gaaaaggctg ggcatagaga tgaagctaga aaaaactgga aaacaattgg tacaattaaa	2760
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catggtattt ggaaagtga aaaacaggat attgttgaat tgttaaagaa aatttttggT	3360
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ttaaaatata caattgattt aattcagcaa attagaaata ccgatttaa tgaaatagac	3480
aatgatttta ttttatcgcc ggtaagagat gaaaagggga atcattttga tagtcggaaa	3540
gatggtgcaa tttgtctaa tggtagtgca aatggtgcat ataatatagc tcgtaaagga	3600
gtgttggtt ttgagaggat taatgcgaaa ccggaaaaac cggagctgta tattgcagat	3660
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<210> SEQ ID NO 21

<211> LENGTH: 3771

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 21

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gaacaccgtg cagatagtta caaatcgtt aaaaaaatta ttgacgaata ccataaatca	180
tttattgaaa aatctttaaa tgggttggtt ttggatggat tggaggatta ttatttttat	240
tatcaaatcc caaaaaaaga tgataatcag aaaaaaatag ttgaagatat actgacaaag	300
ctacgtaaac aaatagccga acgattttca aaacaggata tatataaaaa tctttttgca	360
aaggaaactg ttaaagacga tttaaattct tttgtacaag aggtagaaca aaaggattta	420
attaagaat ttgaaaattt tactacctat ttacaggat ttacagagaa ccgaaaaaat	480
atgtattcgg cagaagataa atcaacggca atagctttcc gtcttattca tcaaaactta	540
cctaagtctc ttgacaatat gcgagcattt aataaaataa gtgtttccacc gcttgccgaa	600

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tataaccatt tacttggtag atatactccc gaagaaggga aagaaaaaat taaaggacta	780
aacgaatata tcaatctata taatcaaacc gttaaaaagg aagaacgttt acctaaactt	840
aaaccattgt ttaaacagat ttttaagcgac cgttctacag catcgtttat tcccgaacaa	900
tataaaaacg ataatgaggt attagaaagt attgaaaaat tataccaaga aataaaggaa	960
catgtttttc attcattgaa agagctattt gtacacataa atgagtatga ttgcataaa	1020
atctatttac ggaatgatgt aagtatgacc gatatttcgc aaaaaatgtt tgggtgattgg	1080
ggggtgttta caaaagccat gaatctatac ttgataaac agtacaaggg gaaagcaaa	1140
cttggtagcg aaaaatacga agatgaacaa aagaagtatt ttagcaatca agagagtttt	1200
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cgtttttacg gtgaatttac tgcactttgg gaaacacttg accaaattac acctctttat	1560
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aaagggttgt	ggatttgtga	acaaataaaa	aatccaaata	cagacctgaa	aaaactgaaa	3720
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<210> SEQ ID NO 22

<211> LENGTH: 3825

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 22

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ttagccgtac	ttaaaaagtt	ttacgaagtg	gctgagaaga	agacatctat	cctaaagtcg	1020
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gaaagttata aattcccga ttttattccc ctgtcttata tccaaacgc cttggagcaa	1260
gccgatattg atggggattt ttggaaggag cattattctg agaactcaaa agcaaatgat	1320
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tttaaaaatg ggatatttga gtttacttat gatctaagaa acttccaaaa gcagaaggag	3300
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gttctaatac	aaaacaaagg	cggatatgac	cattatgaag	atataactca	taattttcga	3420
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aataacggag	atgataatgg	agcttataac	at tttcccgta	aaggaattat	tatcttaaat	3720
aagatttcag	aattctttga	tgaaaatggt	ggatgtgaga	aaatgaaatg	gggagattta	3780
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<210> SEQ ID NO 23

<211> LENGTH: 4008

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 23

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tttgataaaa	atttacagac	ctttctggct	gaccaagaga	tcgaggatgc	atatcaagct	180
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<210> SEQ ID NO 24

<211> LENGTH: 3825

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

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<210> SEQ ID NO 25

<211> LENGTH: 4371

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

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<223> OTHER INFORMATION: Bacterial species

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<211> LENGTH: 3954
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 26

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gaaaaaatgg ttataagca gatgtctgat ccgtcaaaag atatacagaa tttgatgggt      1980
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<210> SEQ ID NO 27

<211> LENGTH: 4158

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 27

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<210> SEQ ID NO 28

<211> LENGTH: 4158

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 28

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<210> SEQ ID NO 29

<211> LENGTH: 3933

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 29

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gcaaataatc ctgatacaaa catcatcggg attgatcggg gagagaagca ttggcctac	2640
ttgtccgtca ttaaccagaa aggagatata ttgaaaataa aatctttaaa caagattgag	2700
gttaaggata aggacggtaa tgtgataaag gaagatgatt atgcaaagct attggaagac	2760
cgtgctaaga atcgtgagag cgcacggcgc gactggaaga gtgtagaaca gattaagac	2820
cttaagaaag gttatatctc aaatgttgta cgagagattg cgcacctcgt cattaaatac	2880
aatgcgattg tagtatttga agatttgaat atgcgcttca agcaagtgcg cgggtggcatt	2940
gaaaagagcg tgtatcaaca acttgaaaag gccttgattg ataaacttaa cttcttggtt	3000
gataaaaatg aacttgatcc ccagaaggcg gggcacatat tgcattgcgt ccaactcact	3060
gcaccatttg agacgtttaa agatatggga aagcaaaactg gcgtactctt ctacacccaa	3120
gctgaatata cgtcgcaaac cgatccggta acaggttttc gaaaaaatgt ctatctgagc	3180
aattcggcca ccgttgagaa gataaaagca tttgtggaaa tgtttgatgt aatcggctgg	3240
gacgataaac taaaaagtta ctacttcaaa tataacccgg ttaattttgt tgaaactaag	3300
tttaaggaaa atacgtttctc aaaagatttg gtcatttatg ccaatgtgcc gcggattaag	3360
cgcgaacgca aaaatgggta ttgggaggca accgtgggta atccaaacga agaattcttg	3420
aaacttttta aggaatggga ttttgataac atctatgttg aagacataaa agaacagatc	3480
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aatttttggc acagctttat ctctctattc aatcttatgt tgcaagtaag aaattctacc	3600
gcaacccaat acaagaagga tgaagatgga aatatcattg agactgttga ggggtgttgat	3660
ttcattgctt caccagtttt tccattcttc accaccgatg gtggtgattt caccgaagga	3720
tgcgtaatc tagcaaaact tgaagataaa tttgtcggta gtaatgctga caaagagcgg	3780
ttcaagaaag aatttaacgg agacgcaaac ggtgcataca acattgctcg aaaggaatt	3840
attatgttga acaatattaa aaataatccg gagaaaccag acctgttcgt gtctaaaaaa	3900
gactgggaca aatttgctca agcaaatcaa taa	3933

<210> SEQ ID NO 30

<211> LENGTH: 3828

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 30

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tactgcttgt caaaaacct gaggtttgag ctgaaaccaa tcggcaaaac gcaaaagata	120
cttgaagata ataaggtttt cgaaaatgat aaaaagagag ctaaaagtta tgaggaagca	180
aaaaaatatt ttaataaatt acatagagaa tttattgacg aatcactcaa aaatattaca	240
ttatctaata atctaattga aaaatttgaa aaaaaatc ttacctggaa aaatagtaaa	300
aataaagata atagtactga actaaaaaag tctgctaaaa gattgagaat agtaattctt	360
gagagtttca ataaaaaagc taatgaatgg aactctgaat attcaaacca agttaagaac	420
gaaaagaaaa aaaagaaaat acaagaaata acaggaatag atttgttttt caaagttgaa	480

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gtttttgatt	ttcttataca	taaatacccc	gaggtgcaga	taaatggtga	gagtattttt	540
agtccattca	acaaatttag	tggttacttt	aaaaaatctc	acgaacaag	aaaaacttt	600
tataaagacg	atggaacctc	cacagcaatt	cccacaagaa	taattgatgt	taatttagaa	660
aaattttttag	aaaataagga	catctactat	acaaaatact	tccaaaaata	caattctatt	720
tttaataaag	aagaacaga	tattttcaaa	ctggaatcgt	ttaaaaattg	tttaacacaa	780
tctcagattg	ataaatataa	tgaatcaata	gccacattaa	aatcaaaaaa	caacaactta	840
cgtcaaaata	atcctgaggt	taataaacat	gatcttccat	tttttaaga	attattcagg	900
caaattctag	gtcaaccaat	taaaaaggag	acagaacagg	ataacttcat	agagatacta	960
acaaatgacg	aggtttttcc	tgttttacaa	aaaaatattg	atgaaaacga	gttatatatt	1020
cctaaagctg	atactctctt	taaagaattt	ttaaagagcc	aaattcaaga	gacaaatgaa	1080
tataatatta	acgaaatata	tgttgcaagt	cgttttataa	actcaatc	caacaactgg	1140
tttgcagaat	gggataccat	tattaattta	ttacgtactg	aactaaaaat	ttaacagaat	1200
cagaaaaaac	ttccagattt	catctcgatt	gcttcattaa	aaagagtgtt	acaaaaatcc	1260
caagacgaga	tagatgctaa	agatttat	agaaacaact	atgagaatct	tttcgaatct	1320
acaactgact	tctataaaat	atttcttaaa	atttggaat	tagaatttaa	tgataatatt	1380
aaaaaatata	atttgagac	agagaatata	agaaaaataa	taatagaaga	taaaaagtat	1440
cttccaaata	aaaagagtat	attaaagaat	ggtgagacag	gtattattca	taatgaaaaa	1500
atattagatt	atgcacaatc	tgcattaaat	atttatcaaa	tgatgaaata	tttttctttg	1560
gaaaaaggaa	aagaagaga	gtggaatccg	gacggtctga	atgaagatac	aacaggagga	1620
ttttatgatg	atttcaataa	atactaccaa	aatgtaaca	cgtggaata	ttttaatgaa	1680
ttcagaaatt	atttaacaaa	aaaaccatat	aagacagata	agattaaatt	atactttggt	1740
cacaagagtt	tattaggtgg	ttttactgaa	agcaaacag	agaaaagcaa	taacggcaca	1800
caatatgggg	cttatttatt	aagaaagaag	catggacttg	gaggctttga	ttattatctt	1860
ggaattagta	cagaccccca	tttaatgagt	tattttgatc	caattgatga	ttcaggagat	1920
agcgaatatg	aaaggttaaa	ttattatcaa	gtattaacaa	gaacaatcta	cggtccttct	1980
tatgaaggcg	attatgaatt	ggacaaaaaa	aatttatcag	aaatagagat	aataaaaaaa	2040
ataaaaagat	cattatctta	ttacacttca	agggttaaaa	aatccagga	cataatcaat	2100
aataattatg	agtctgttag	agatattcat	aaagatataa	ctgatgtact	aaaagaattt	2160
ggaacaatct	tgattataa	agtaataaca	aacagtcaaa	tacaaaaggc	atttaattgc	2220
gataaaggat	tttatctctt	tgagatatat	tcgaaggatt	tttcaaaaga	gaaaggatg	2280
aaaagtaaga	atagcaaaga	taatttcat	acaacgtatt	ttaaatcatt	aatggataga	2340
aagcagtcta	catttgattt	aggtagtgga	gaaatatttt	tcagagaaaa	atctgttcag	2400
tctgaaattg	attctatgag	aaaaactaag	aataaaaataa	ctagggtcaa	acgatacaca	2460
aaaaatttaa	ttcagttcaa	tttatcaatt	accttaata	ataactgtac	agaagttcct	2520
cagaataaaa	atgcaagaaa	agcatttatt	aataatttta	atattgaact	aagtaagaaa	2580
ctgttaacaa	ataattcaga	cataaatatt	attggtattg	accgtggaga	aaaacatttg	2640
gcctattact	ctgttataga	ccagcaaagc	aacattcttg	aaacaggtc	ttttaataaa	2700
attcaagaga	gaaaagacag	agaacctacc	gattaccaac	aaaaattaga	taaaattcaa	2760

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aaggacagag actggcaaag aaaatcatgg caggaaatat caaatatcaa agatttaaaa 2820
aaaggctata tttctcaagt tgtctatgaa ataagtaagt tagttaaaaa atataatgca 2880
atcatagtat ttgaagacct gaatatagga tttaagcgtg gccgttttgc aatagaaaaa 2940
caagtatacc aaaatttgga gctttcatta gcgaaaaaat taaattattt agttttttaa 3000
gatgcaaatg aaggagaatc aggacattat ctaaaagcat atcaacttac atcccctgtc 3060
aacaattttc aagatattgg taaacaatgt ggtataatat tttatattcc tgccagctac 3120
acatcagcaa tttgtccttc atgtggattc cataaaaaa taccacatc aattaaaaaa 3180
cttgcaaaga ataagaatt ttagaaaaa tttgttataa cttatgaact taaaaagat 3240
cgtttttatt ttgggtacaa aataaatgat ttttacaatt ctaatttgca agataatgtc 3300
attttctact caaatgtaga aagattaaga tataaaagaa ataaggataa ccgaagtggc 3360
gaagtacaag aacgattgcc taatgaagaa ttaaaaaaac tctttgagca aaatcacatt 3420
aattataaag ataatcctca aatatctggt caaattaaaa atcaaaaact agataatgaa 3480
aaattttata aacctcttat atatgaaatc tctttaattt tacaattaag gaatagtaaa 3540
acagtaaaaa gtgaagatgg aacaattaac acaaatataa atcgggattt catttcttgt 3600
ccagcttgtt attttcattc agagaataat ttaatgaatc ttctaataa gtataaagga 3660
ggaaaaaaat ttgaatttaa cggagatgca aatggagcat ataatttgc acgaaaaggc 3720
attttgcttt taaacaaatt aaataatatc aaagatatag aaaaaattga atataatgac 3780
ctcaatatat cacaagaaga ctgggataac tttgtcaaaa acccctaa 3828

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<210> SEQ ID NO 31
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Stem-Loop Consensus Sequence
<220> FEATURE:
<221> NAME/KEY: variation
<222> LOCATION: (1)..(1)
<223> OTHER INFORMATION: n is a,c,g,t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1)..(1)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: variation
<222> LOCATION: (11)..(11)
<223> OTHER INFORMATION: n is a,c,g,t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (11)..(11)
<223> OTHER INFORMATION: n is a, c, g, or t

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<400> SEQUENCE: 31

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natttctact nttgtagat 19

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<210> SEQ ID NO 32
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: BMC01-15 Stem Loop Sequence

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<400> SEQUENCE: 32

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gatttctact attgtagat 19

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<210> SEQ ID NO 33
<211> LENGTH: 3888
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 33

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cgctccggttg atgccggttg taatgttatt accgatctga ccattttcga agaaaccatc      120
aaaaacgatac agaaacgcta tgaagcatatc ctggcaatta aaccgctggg tgatgaaacc      180
cataaacactt ttattcagac cgttctgagt ggtctgacag atctgattaa atccgatgag      240
atgaagaact  acctggaaca caaaaatctg atccgtcaga aagatgtgga agagaaagt      300
aaaacccaaaa gcctgcagct gatcaacaaa atcgaaaaag attggcgtaa acgtgtgagc      360
gatagcttta  ccaaacatcc gcagtacaaa aagatgttcg ataaaaacct gtttgccgat      420
gaaagtccgc  tgtataaact ggccgaaaat gattttcagc gtagccagat taaaatcttc      480
gagaaattta  ccggctactt caacggcttt catgagaatc gtaaaaatct gtatgtggcg      540
gaaaaacagg  gcaccgcaat tgcaaatcgt gtgattaatg aaaacctgcc gaaatttacc      600
gagaacgcca  ataaactgaa acgtgccttt gaaaaaatc ccgaatttct gtccaaaatc      660
agcgaggata  aaagctttca ggcaactgctg attaaaaacc agctgagcct ggaaaaactg      720
ctgcagccgc  tgacctttaa tctgctgatt tctcagaccg gtatcgatag ctataatgaa      780
gtgttaggtg  gttacacacc ggaaaatagc gaaccgatta aaggcctgaa tcaactgatt      840
aatctgtacc  gccagaaaat taacctggca cgtaacgatt ttccgaatct ggcaaccgctg      900
tacaaacagc  tgctgagcga tctgtaaaac aatagcgttg tgtataaaac tctggaaaat      960
gtggcagatg  tttatagcag tgtgtttgaa ctgtgtcaga acctgctgag caaacagagc      1020
gatattaata  aatggatcga ggatatcaac atcagcagcg gtcagatttg gatctataaa      1080
agccatctga  gcggtctgtc agttatgctg tttggtgaaa gtggttgggg tctgattccg      1140
cgtattctga  atattagcga agatgatgaa gaagagatca tcaaatccaa aagcaaaaag      1200
agccagcaag  agtatttcag ctttgcgga  attggtaacg ccatcaacaa ctatagcttt      1260
gaggatgtga  acattaaagc actggcaaaa cagggtctgt gtctgtggca gaaacaagg      1320
aatgaacgct  tgatcaaatt cggcaaaactg ttagccaga tgcagaatga actgcagagc      1380
ccgaaagaaa  aatgggatag caccgaaaaa gagaaaatca aagaactgct ggatacaggc      1440
ctggaatttg  ttcattggct gaaagttatt agcaaccagc cggaagataa agatgaagtc      1500
ttttatgccg  aatggcaggc actgaccgat acctggcgtg gtctgccgaa actgtatgat      1560
cgtgttcgta  attttgccac caaaaaagat tacagccaga acaagctgaa gatcaacttt      1620
gataaaggca  cctgctgtaa tggttgggat accaataaag aaaccgataa tctgggtatc      1680
ctgctggaaa  ataaaggcca gtattatctg ggcatcatga aagatagcag catctttgat      1740
tatcagtggg  acattgacaa ttttcagaac ccgaatagca aacagtcagt ggccaaaaaa      1800
aacctgcatg  aagcaattgt tagcgataac acccaggatt gttggagcaa aattgtgtac      1860
aaactgttac  cgggtccgaa taaaatgctg ccgaaagtgt tttttagtga caagcgccag      1920
aaatactttg  gtgcggatga aaaagtgatc gacattaacg aaaatggccg tcataaaaaa      1980

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ggcgacaact ttaacattag cgattgccac tatctgatcg acttctataa aaccgccatc 2040
aataaacatc cggaatggtc ccagttcaac ttcaaattha gcgcgaccaa aagctacgaa 2100
gatatcagcc agttttatca cgaagttcag aatcagggtt atcgcatcga gtttgatcat 2160
atccgc aaaag actatatcca gaaaatggtg agcgaaggta aactgttccct gtttaaaatc 2220
cacagcaaaag atttttagcag ctatgcaaaa ggtcgtccga atatgcatac catctatttg 2280
cgtgccattt ttaaccggga aaatctggcc aatgttgttg ttaaaactgaa tggatgaagcc 2340
gaattctttt atcgcaaaaag cagtaaagat cgcattatca gccatccgca aggtctggaa 2400
gttagcaata aaaaccggag caatccgaaa aaaaccagtc gttttgcata cgacctgatc 2460
aaagataaac gtttcacgca ggacaagttt tttttccatg ttccgattac gctgaatttt 2520
cgtgaagggtg aaggttatcg ctttaaccag agcgttattc gtgagctgaa aaaatactat 2580
cagaccgata aagccaacct gcatactatt ggtattgatc gtggtgaacg tcatctgctg 2640
tattattgctg ttattaatgt tgccagcggc aaaatcgttg aacagggtag cttcaatcag 2700
atcagcacca attatacacc ggaacaaatt accgatgatg gcgaaatcat taaaggcgaa 2760
accgtgaata aaaccaccga ttatcataac ctgcttaaca ccaagaagg tgatcgccag 2820
aaagcacgta aaaattggca gaccattgaa aacatcaaag agctgaaagc cggttatctg 2880
agcaatgtga ttcataaaat cagccagctg atggtgaaat ataacgcctt tgttgttctg 2940
gaagaactga aatatggttt taaacgcggt cgcttcaagg ttgagaaaca ggtttatcag 3000
aaatttgaga aagccctgat cgacaagctt aactatctgg tgtttaaaga tcgtgccctt 3060
gccgaagttg gtggtgttct gaatgcactg cagctggcac ctccggttgc aagctatatt 3120
gatattggta aacaggcagg ctctcgtgtt tatgttccgg cacatcatac cagcaaaatt 3180
tgtccgtgga ccggttttgt ggactggctg aaaccgcgtt atgatggcat tgataaagcg 3240
aaagcattct ttacctgttt cgaaagcatc catttcaaca cgcagaaaaa ctattttgag 3300
ttcgcccttg actatgagaa atttcgcggt aacattaatc atctgccgga aggtctgaaa 3360
cgtaccagct ggacctgtg tagccataat agcctgcgtg atattgccac gaaagataaa 3420
aatggtaact ggccgtataa gcagatcaat ctgacagcag aactgcttga aattctgaaa 3480
accctgaatc cgcgtaatgg cgaaaacctg gttgaacgta ttatcgagat gaacgacaaa 3540
aagttctctg agagcctgat gtgggcactg cgtgttctgc tgcaactgcg ttatggttat 3600
atcaaacgta ataacgaggg catcatcatc gaagagggtg attatattct ggtccggtg 3660
gcaaatgaaa acgcggaatt ttttgatagc cgcaactttg tgaatatcga gaaagccgat 3720
tttccaaaag atgcagatgc aaatgggtgca tataacattg cagctaaagg tctgctgctg 3780
attgcacaga atattaacaa cgccaaaatc aacgataaag gcgagggtta atgtgacctg 3840
cagattgata aaacaacctg gtttaactgg gttcagagca aaagctaa 3888

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<210> SEQ ID NO 34

<211> LENGTH: 1191

<212> TYPE: DNA

<213> ORGANISM: Escherichia coli

<400> SEQUENCE: 34

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tatggagaaa cagtagagag ttgcgataaa aagcgtcagg taggatccgc taatcttatg 60
gataaaaatg ctatggcata gcaaaagtgt acgccgtgca aataatcaat gtggactttt 120

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ctgccgtgat tatagacact tttgttacgc gtttttgta tggetttggt cccgctttgt	180
tacagaatgc ttttaataag cgggggtacc ggtttggtta gcgagaagag ccagtaaaag	240
acgcagtgc ggcaatgtct gatgcaatat ggacaattgg tttcttctct gaatggcggg	300
agtatgaaaa gtatggctga agcgcaaaat gatccctgc tgccgggata ctggtttaat	360
gcccactctg tgccgggttt aacgccgatt gaggccaacg gttatctcga tttttttatc	420
gaccgaccgc tgggaatgaa aggttatatt ctcaatctca ccattcgcgg tcagggggtg	480
gtgaaaaatc agggacgaga atttgtttgc cgaccgggtg atattttgct gttcccgcca	540
ggagagatcc atcactacgg tcgtcatccg gaggctcgcg aatggtatca ccagtgggtt	600
tactttcgtc cgcgcgccta ctggcatgaa tggttaact ggccgtcaat atttgccaat	660
acggggttct ttcgcccgga tgaagcgac cagccgcatt tcagcgacct gtttgggcaa	720
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gagcaattgt tactcggcgc catggaagcg attaacgagt cgctccatcc accgatggat	840
aatcgggtac gcgaggcttg tcagtacatc agcgatcacc tggcagacag caattttgat	900
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cagcagttag ggattagcgt cttaagctgg cgcgaggacc aacgtatcag ccaggcgaag	1020
ctgcttttga gcaccaccgc gatgcctatc gccaccgtcg gtcgcaatgt tggttttgac	1080
gatcaactct atttctcgcg ggtatttaaa aaatgcaccg gggccagccc gagcgagttc	1140
cgtgccgggt gtgaagaaaa agtgaatgat gtagccgtca agttgtcata a	1191

<210> SEQ ID NO 35
 <211> LENGTH: 177
 <212> TYPE: DNA
 <213> ORGANISM: Escherichia coli

<400> SEQUENCE: 35

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aaggaacaac taaaggaatt gcgaataata attttttcac gttgaaaatc tccaaaaaaa	120
aaggctcaa aaggagcctt taattgtatc ggtttatcag cttgctttcg aggtgaa	177

<210> SEQ ID NO 36
 <211> LENGTH: 10216
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 36

caatcggcgt taaaccgcgc accagatggg cattaaacga gtatcccggc agcaggggat	60
cattttgcgc ttcagccata cttttcatat tcccgccatt cagagaagaa accaattgtc	120
catattgcat cagacattgc cgtcactgcg tcttttactg gctcttctcg ctaacaaaac	180
cggtaacccc gcttattaaa agcattctgt aacaaagcgg gaccaaagcc atgacaaaaa	240
cgcgtaacaa aagtgtctat aatcacggca gaaaagtcca cattgattat ttgcacggcg	300
tcacactttg ctatgccata gcatttttat ccataagatt agcggatcct acctgacgct	360
ttttatcgca actctctact gtttctccat acccgttttt tggtaaccggg cccccctcgc	420
agtttatttt agggaggcaaa aatgatcttc aacaacttca ccagaaatt cagcctgagc	480

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aaaaccctgc gttttgaact gcgtccggtt gatgccggtg gtaatgttat taccgatctg	540
accatttttcg aagaaccat caaaaacgat cagaacgct atgaagcata cctggcaatt	600
aaaccgctgg ttgatgaac ccataaacac tttattcaga ccgttctgag tggctcgaca	660
gatctgatta aatccgatga gatgaagaac tacctggaac acaaaaatct gatccgtcag	720
aaagatgtgg aagagaaagt taaaaccaa agcatcgacg tgatcaacaa aatcgaaaaa	780
gattggcgta aacgtgtgag cgatagcttt accaaacatc cgcagtacaa aaagatgttc	840
gataaaacc tggttgccga tgaaagtcg ctgtataaac tggccgaaaa tgattttcag	900
cgtagccaga ttaaaatctt cgagaaatct accggctact tcaacggctt tcatgagaat	960
cgtaaaaatc tgtatgtggc ggaaaaacag ggcacgcaa ttgcaaatcg tgtgattaat	1020
gaaaacctgc cgaaatttat cgagaacgcc aataaactga aacgtgcctt tgaaaaatat	1080
ccgaatttc tgtccaaat cagcgaggat aaaagcttcc aggcactgct gattaaaaac	1140
cagctgagcc tggaaaaact gctgcagccg ctgacctta atctgctgat ttctcagacc	1200
ggtatcgata gctataatga agtgtaggt ggttacacac cggaaaatag cgaaccgatt	1260
aaaggcctga atcaactgat taatctgtac cgccagaaaa ttaacctggc acgtaacgat	1320
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gtgtataaac ctctggaaaa tgtggcagat gtttatagca gtgtgttga actgtgtcag	1440
aacctgctga gcaaacagag cgatattaat aaatggatcg aggatatcaa catcagcagc	1500
ggtcagattt ggatctataa aagccatctg agcggctctg cagttatgct gtttggtgaa	1560
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<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 41

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gcccatgcaa cagaaactat aaaaaataca gagaatgaaa agaaacagat agatttttta      60
gttcttttagg cccgtagtct gcaaatcctt ttatgatttt ctatcaaaca aaagaggaaa      120
atagaccagt tgcaatccaa acgagagtct aatagaatga ggtagc                       165

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<210> SEQ ID NO 42

<211> LENGTH: 169

<212> TYPE: DNA

<213> ORGANISM: Escherichia coli

<400> SEQUENCE: 42

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tttgctggc ggcagtagcg cggtaggtccc acctgacccc atgccgaact cagaagtgaa      60
acgccgtagc gccgatggta gtgtgggggc tccccatgcg agagtaggga actgccaggc      120
atcaaataaa acgaaaggct cagtcgaaag actgggcctt tcgttttat                    169

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<210> SEQ ID NO 43

<211> LENGTH: 24

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 43

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gtggcggtag gttgatgcat cgcg                                     24

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<210> SEQ ID NO 44

<211> LENGTH: 5039

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 44

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gacgaaaggg cctcgtgata cgcctathtt tataggttaa tgtcatgata ataatggttt      60
cttagacgtc aggtggcact tttcggggaa atgtgcgcgg aacctctatt tgtttathtt      120
tctaaataca ttcaaatatg tatccgctca tgagacaata acctgataa atgcttcaat      180
aatattgaaa aaggaagagt atgagtattc aacatttccg tgtagccctt attccctttt      240
ttgcggcatt ttgccttctt gtttttgctc acccagaaac gctggtgaaa gtaaaagatg      300
ctgaagatca gttgggtgca cgagtggggt acatcgaact ggatctcaac agcggtaaga      360
tccttgagag ttttcgcccc gaagaacgtt ttccaatgat gagcactttt aaagttctgc      420
tatgtggcgc ggtattatcc cgtattgacg ccgggcaaga gcaactcggg cgccgcatac      480
actattctca gaatgacttg gttgagtact caccagtcac agaaaagcat cttacggatg      540
gcatgacagt aagagaatta tgcagtgtcg ccataaccat gagtataaac actgcggcca      600
acttacttct gacaacgacg ggaggaccga aggagctaac cgcttttttg cacaacatgg      660
gggatcatgt aactcgcctt gatcgttggg aaccggagct gaatgaagcc ataccaaagc      720
acgagcgtga caccacgatg cctgtagcaa tggcaacaac gttgcgcaaa ctattaactg      780
gcgaactact tactctagct tcccggaac aattaataga ctggatggag gcggataaag      840

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ttgcaggacc	acttctgcgc	tcggcccttc	cggtcggtcg	gtttattgct	gataaatctg	900
gagccggtga	gcgtgggtct	cgcggtatca	ttgcagcact	ggggccagat	ggtaagccct	960
cccgatcgt	agttatctac	acgacgggga	gtcaggcaac	tatggatgaa	cgaaatagac	1020
agatcgctga	gatagggtgcc	tacttgatta	agcattggta	actgtcagac	caagtttact	1080
catatatact	ttagattgat	ttaaaacttc	atttttaatt	taaaaggatc	taggtgaaga	1140
tcctttttga	taatctcatg	acaaaaatcc	cttaacgtga	gttttcgttc	cactgagcgt	1200
cagaccccg	agaaaagatc	aaaggatctt	cttgagatcc	ttttttctcg	cgcgtaatct	1260
gctgcttgca	aacaaaaaaa	ccaccgctac	cagcggtggt	ttgtttgccg	gatcaagagc	1320
taccaactct	ttttccgaag	gtaactggct	tcagcagagc	gcagatacca	aatactgtcc	1380
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gctggccttt	tgtcacatg	ttctttctcg	cgttatcccc	tgattctgtg	gataaccgta	1920
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cagtgcagca	ggaagcggaa	gagcgcccaa	tacgcaaacc	gcctctcccc	gcgcgttggc	2040
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acgcaattaa	tgtgagttag	ctcactcatt	aggcacccc	ggctttacac	tttatgcttc	2160
cggctcgtat	gttgtgtgga	attgtgagcg	gataacaatt	tcacacagga	aacagctatg	2220
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cccgaccgga	tcgcccttcc	caacagttgc	gcagcctgaa	aggcaggccg	ggccgtggtg	2940
gccacggcct	ctaggccaga	tccagcggca	tctgggtag	tcgagcgcgg	gccgcttccc	3000
atgtctcacc	aggcgagacc	tgtttcgcga	tctcagcatc	tgaatcttc	ccggccttgc	3060
gcttcgctgg	ggccttcccc	accgccttgg	cgggcttctt	cgggtccaaa	ctgaacaaca	3120

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gatgtgtgac cttgcgcccc gtctttcgct gcgcccactc cactgtagc gggtgtgtct	3180
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accggccttc cagttgaaac cactttcgca gctggtaaat ttctatttcg cgtggccga	3300
tgtgtccca ttgcatgagc agctcgtaa gcctgatcgc gtgggtgctg tccatcttgg	3360
ccacgtcagc caaggcgat ttggtgaact gtttggtag ttccgtcagg tacggcagca	3420
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ggacttccc cgttttcagg cgcagggccg cttctttgag ctggtttag gaagattcga	3600
tagggacacc cgccatcgtc gctatgtcct ccgcgcgcac tgaatacatc acttcacgg	3660
tgacaggctc gctcctcttc acctggctaa tacaggccag aacgatccgc tgttctgaa	3720
cactgaggcg atacgcggcc tcgaccaggg cattgctttt gtaaacatt gggggtgagg	3780
ccacgttcga cattccttgt gtataagggg acaactgtatc tgcgtccac aatacaaca	3840
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tggcccggtt ggggcgcgat ttgtgcccct gaaggggcgg gggaggcgtc tgggcaatcc	4080
ccgttttacc agtcccctat cgcgcgcctga gagggcgag gaagcgagta atcagggtat	4140
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gagcgggttg acaccaaggg gaggggctaa gaccggttta tcagtcccct ttccctcgtt	4260
tctttccaac gcgatatgcc agcaaggccg ccaccgttgc caccgtcacc ccagcaagca	4320
cagccagtgg cgtgtaattg gggctctgac ctcagtggaa cgaaaactca cgttaaggga	4380
ttttggtcat gagattatca aaaaggatct tcacctagat ccttttaaat taaaaatgaa	4440
gttttaaatc aatctaaagt atatatgagt aaacttggtc tgacagttaac caatgcttaa	4500
tcagtgaggc acctatctca gcgatctgtc tatttcgttc atccatagtt gctgactcc	4560
ccgtcgtgta gataactacg atacgggagg gcttaccatc tggcccagc gctgcaatga	4620
taccgcgaga tccacgtcga ccggctccag atttatcagc aataaaccag ccagccggaa	4680
gggcgcgagc cagaagtggc cctgcaactt tatccgcctc catccagtct attaattgtt	4740
gcgggaagc tagagtaagt agttcgccag ttaatagttt gcgcaacgtt gttgccattg	4800
ccgcctgatg cggtatcttc tccttacgca tctgtgcggt atttcacacc gcatatggtg	4860
cactctcagt acaatctgct ctgatgccgc atagttaagc cagccccgac acccgccaac	4920
acccgctgac gcgcctgac gggcttgtct gctcccgca tccgcttaca gacaagctgt	4980
gaccgtctcc gggagctgca tgtgtcagag gttttcacgc tcatcacga aacgcgcga	5039

<210> SEQ ID NO 45

<211> LENGTH: 674

<212> TYPE: DNA

<213> ORGANISM: Escherichia coli

<400> SEQUENCE: 45

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ttccatagcc tccgcccccc tgacgagcat cacaaaaatc gacgctcaag tcagagggtg	120

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cgaaacccga caggactata aagataccag gcgtttcccc ctggaagctc cctcgtgcgc	180
tctcctgttc cgaccctgcc gcttacccga tacctgtccg cctttctccc ttcgggaagc	240
gtggcgcttt ctcatactc acgctgtagg tatctcagtt cgggtgtaggt cgttcgctcc	300
aagctgggct gtgtgcacga acccccgtt cagcccgacc gctgcgcctt atccggtaac	360
tatcgtcttg agtccaaccc ggtaagacac gacttatcgc cactggcagc agccactggt	420
aacaggatta gcagagcgag gtatgtaggc ggtgctacag agttcttgaa gtgggtggct	480
aactacggct acactagaag gacagtattt ggtatctgcg ctctgctgaa gccagttacc	540
ttcggaaaaa gagttgtag ctcttgatcc ggcaaaaaa ccaccgctgg tagcggtggt	600
ttttttgttt gcaagcagca gattacgcgc agaaaaaag gatctcaaga agatcctttg	660
atcttttcta cggg	674

<210> SEQ ID NO 46

<211> LENGTH: 1884

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 46

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caacaattaa tagactggat ggaggcggat aaagttgcag gaccacttct gcgctcggcc	120
cttcggctg gctggtttat tgcgtataaa tctggagccg gtgagcgtgg atctcgcggt	180
atcattgcag cactggggcc agatggtaag cctcccgtg tcgtagttag ctacacgacg	240
gggagtcagg caactatgga tgaacgaaat agacagatcg ctgagatagg tgcctcactg	300
attaagcatt ggtaactgtc agaccaagtt tactcatata tacttttagat tgatttaaaa	360
cttcattttt aatttaaaag gatctagggt aagatccttt ttgataatct catgacccaa	420
atcccttaac gtgagttttc gttccactga gcgtcagacc ccaattacac gccactggct	480
gtgcttgctg ggggtgacgtg ggcaacggtg gcggccttgc tgggctatcg cgttggaag	540
aaacgaggga aaggggactg ataaaccggt cttagccctt ccccttggtg tccaaccgct	600
ctgtaggcct ctcagggcc cgtggtgccc ctggttgga gcccaagggtg aatccgcctc	660
gataccctga ttactcgctt cctgcgcctt ctcaggcgcc gataggggac tggtaaaacg	720
gggattgccc agacgcctcc cccgcccctt caggggcaca aatgcggccc caacggggcc	780
acgtagtgtt gcgttttttg cgtttccacc cttttcttcc ttttcccttt taaacctttt	840
aggacgtcta caggccacgt aatccgtggc ctgtagagtt taaaaaggga cggatttgtt	900
gccattaagg gacggatttg ttgtaagaa gggacggatt tgttgttgta aagggaacga	960
tttgttgat tgtgggacgc agatacagtg tcccctata cacaaggaat gtcgaacgtg	1020
gcctcaccac caatggttta caaaagcaat gccctgggtg agggccgcta tcgcctcagt	1080
gttcaggaa acgcggatcgt tctggcctgt attagccagg tgaagaggag cgagcctgtc	1140
accgatgaag tgatgtattc agtgacggcg gaggacatag cgacgatggc ggggtgctcc	1200
atcgaatctt cctacaacca gctcaaagaa gcggccctgc gcctgaaacg gcgggaagtc	1260
cggttaaccc aagagcccaa tggcaagggg aaaagaccga gtgtgatgat taccggctgg	1320
gtgcaaacaa tcactaccc ggagggtgag gcccggtgag aactcagggt caccaaaagac	1380

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atgctgccgt acctgacgga actcaccaaa cagttcacca aatacgctt ggctgacgtg	1440
gccaagatgg acagcaccca cgcgatcagg ctttacgagc tgctcatgca atgggacagc	1500
atcgccagc gcgaaataga aattgaccag ctgcgaaagt gggttcaact ggaaggccgg	1560
tatccctcga tcaaggactt caagttgcga gtgcttgatc cagccgtgac gcagatcaac	1620
gagcacagcc cgctacaggt ggagtgggcg cagcgaaaga cggggcgcaa ggtcacacat	1680
ctgttgttca gttttggacc gaagaagccc gccaaaggcg tgggtaaggc cccagcgaag	1740
cgcaaggccg ggaagatttc agatgctgag atcgcgaaac aggctcgccc tggtagaca	1800
tgggaagcgg cccgcgctcg actaaccag atgcgctgg atctggccta gaggcgctgg	1860
ccaccacggc cggcctgcc tttc	1884

<210> SEQ ID NO 47
 <211> LENGTH: 570
 <212> TYPE: DNA
 <213> ORGANISM: Streptomyces noursei

<400> SEQUENCE: 47

atgaccactc ttgacgacac ggcttaccgg taccgcacca gtgtcccggg ggacgccgag	60
gccatcgagg cactggatgg gtccttcacc accgacacgg tcttcgcgt caccgccacc	120
ggggacggct tcaccctgcg ggaggtgccg gtggaccgc cctgaccaa ggtgttcccc	180
gacgacgaat cggacgacga atcgacgcc ggggaggacg gcgaccgga ctcccgacg	240
ttcgtcgcgt acggggacga cggcgacctg gcgggcttcg tggctgtctc gtactccggc	300
tggaaaccgc ggctgaccgt cgaggacatc gaggtcgccc cggagcaccg ggggcacggg	360
gtcggggcgg cgttgatggg gctcgcgacg gagttcgccc gcgagcgggg cgcggggcac	420
ctctggctgg aggtcaccaa cgtcaacgca cggcgatcc acgcgtaccg gcggatgggg	480
ttaccctct cggcctgga caccgccctg tacgacggca ccgcctcgga cggcgagcag	540
gcgctctaca tgagcatgcc ctgccctga	570

<210> SEQ ID NO 48
 <211> LENGTH: 90
 <212> TYPE: DNA
 <213> ORGANISM: Bacillus subtilis

<400> SEQUENCE: 48

atggcgaaga cgttgtccga tattaaaaga tcgcttgatg ggaatttagg taaaaggctg	60
acgttaaaag caaacggtgg ccggatccat	90

<210> SEQ ID NO 49
 <211> LENGTH: 660
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Bacterial species

<400> SEQUENCE: 49

atggcgaaga cgttgtccga tattaaaaga tcgcttgatg ggaatttagg taaaaggctg	60
acgttaaaag caaacggtgg ccggatccat atgaccactc ttgacgacac ggcttaccgg	120
taccgcacca gtgtccgggg ggacgccgag gccatcgagg cactggatgg gtccttcacc	180
accgacacgg tcttcgcgt caccgccacc ggggacggct tcaccctgcg ggaggtgccg	240

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gtggacccgc cctgaccaa ggtgttcccc gacgacgaat cggacgacga atcggacgcc	300
ggggaggacg gcgacccgga ctcccgacg ttcgtcgcgt acggggacga cggcgacctg	360
gcgggcttcg tggctgtctc gtactccggc tggaaaccgc ggctgacctg caggacatc	420
gaggtcgccc cggagcaccg ggggcacggg gtcggggcgc cgttgatggg gctcgcgacg	480
gagttcgccc gcgagcgggg cgcggggcac ctctggctgg aggtcaccaa cgtcaacgca	540
ccggcgatcc acgcgtaccg gcggatgggg ttcacctctc gcggcctgga caccgccctg	600
tacgacggca ccgcctcgga cggcgagcag gcgctctaca tgagcatgcc ctgccctga	660

<210> SEQ ID NO 50

<211> LENGTH: 822

<212> TYPE: DNA

<213> ORGANISM: *Bacillus subtilis*

<400> SEQUENCE: 50

atgcgagcca tttggacggg ttcgatcgcc ttcgggctgg tgaacgtgcc ggtcaaggtg	60
tacagcgcta ccgcagacca cgacatcagg ttccaccagg tgcacgcca ggacaacgga	120
cgcacccggt acaagcgcgt ctgcgaggcg tgtggcgagg tggtcgacta ccgcgatctt	180
gcccggggcct acgagtccgg cgacggccaa atgggtggcg tcaccgacga cgacatcgcc	240
agcttgccctg aagaacgcag ccgggagatc gaggtgttgg agttcgtccc cgcgcgcgac	300
gtggacccga tgatgttcga ccgcagctac tttttggagc ctgattcgaa gtcgtcgaaa	360
tcgtatgtgc tgctggctaa gacactcgcc gagaccgacc ggatggcgat cgtgcatttc	420
acgtgcgca acaagaccag gctggcggcg ttgcgcgtca aggatttcgg caagcgagag	480
gtgatgatgg tgcacacgtt gctgtggccc gatgagatcc gcgacccga cttcccggtg	540
ctggaccaga aggtggagat caaacccgcg gaactcaaga tggccggcca ggtggtggac	600
tcgatggccg acgaattcaa tccggaccgc taccacgaca cctaccagga gcagttacag	660
gagctgatcg acaccaaact cgaagtgagg caggcattta ccgccgagga ccaaccgagg	720
ttgctggacg agcccgaaaga cgtctccgac ctgctcgcca agctggaggc cagcgtgaag	780
gcgcgctcga aggccaaactc aaacgtccca acgcctccgt ga	822

<210> SEQ ID NO 51

<211> LENGTH: 822

<212> TYPE: DNA

<213> ORGANISM: *Mycobacterium tuberculosis*

<400> SEQUENCE: 51

atgcgagcca tttggacggg ttcgatcgcc ttcgggctgg tgaacgtgcc ggtcaaggtg	60
tacagcgcta ccgcagacca cgacatcagg ttccaccagg tgcacgcca ggacaacgga	120
cgcacccggt acaagcgcgt ctgcgaggcg tgtggcgagg tggtcgacta ccgcgatctt	180
gcccggggcct acgagtccgg cgacggccaa atgggtggcg tcaccgacga cgacatcgcc	240
agcttgccctg aagaacgcag ccgggagatc gaggtgttgg agttcgtccc cgcgcgcgac	300
gtggacccga tgatgttcga ccgcagctac tttttggagc ctgattcgaa gtcgtcgaaa	360
tcgtatgtgc tgctggctaa gacactcgcc gagaccgacc ggatggcgat cgtgcatttc	420
acgtgcgca acaagaccag gctggcggcg ttgcgcgtca aggatttcgg caagcgagag	480
gtgatgatgg tgcacacgtt gctgtggccc gatgagatcc gcgacccga cttcccggtg	540

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ctggaccaga aggtggagat caaacccgcg gaactcaaga tggccggcca ggtggtggac	600
tcgatggccg acgacttcaa tccggaccgc taccacgaca cctaccagga gcagttacag	660
gagctgatcg acaccaaact cgaaggtggg caggcattta ccgccgagga ccaaccgagg	720
ttgctggacg agccccaaga cgtctccgac ctgctcgcca agctggaggc cagcgtgaag	780
gcgcgctcga aggccaaactc aaacgtccca acgcctccgt ga	822

<210> SEQ ID NO 52

<211> LENGTH: 2280

<212> TYPE: DNA

<213> ORGANISM: Mycobacterium tuberculosis

<400> SEQUENCE: 52

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gccaccggga	ccacaaagtc	cgatatcttc	gactactacg	ccggtgttgc	cgaagtcattg	120
ctcgccaca	tcgctggacg	gccggcgacg	cgaagcgt	ggcctaaccg	cgctgaccaa	180
cccgcttct	tcgaaaagca	gttggcggtg	tcggcgccgc	cttggtgtgc	acgtgcaacg	240
gtggcgacc	ggctcgggac	gacgacctat	ccgatcatcg	atagcgcaac	cggtggtggc	300
tggatcgccc	aacaggcggc	gctggaggtg	cacgtgccgc	agtggcggtt	tgtcgccgag	360
cccgatcag	gtgagttaaa	tccgggccc	gcaacgcgtt	tggtgttcga	cctggacccc	420
ggcgaaggcg	tgatgatggc	ccagctggcc	gaggtggcgc	gcgcggttcg	tgatcttctc	480
gcccgatatc	ggttgggtc	cttcccggtc	accagcgcca	gcaagggatt	gcattctgtac	540
acaccgctgg	atgagccggt	gagcagcagg	ggagccacgg	tgttgcccaa	gcgcgtcgcg	600
cagcgattgg	agcaggcgat	gcccgcggtg	gtcactctga	ccatgaccaa	aagcctgcgg	660
gcccgggaagg	tgtttgtgga	ctggagccag	aacagcggt	cgaagaccac	catcgcgccg	720
tactcactac	gtggccggac	gcattccgacc	gtcgcgccgc	cacgcacctg	ggcggagctc	780
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<211> LENGTH: 9652

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

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<223> OTHER INFORMATION: Bacterial species

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1. A nucleic acid molecule encoding an RNA-guided DNA endonuclease, which is

- (a) a nucleic acid molecule encoding the RNA-guided DNA endonuclease comprising or consisting of the amino acid sequence of any one of SEQ ID NOs 9, 1 to 5, 7, 8 and 10 to 15;
- (b) a nucleic acid molecule comprising or consisting of the nucleotide sequence of any one of SEQ ID NOs 24, 16 to 20, 22, 23 and 25 to 30;
- (c) a nucleic acid molecule encoding a RNA-guided DNA endonuclease the amino acid sequence of which is at least 70% identical to the amino acid sequence of (a), preferably at least 80% identical, more preferably at least 90% identical, and most preferred at least 95% identical;
- (d) a nucleic acid molecule comprising or consisting of a nucleotide sequence which is at least 70% identical to the nucleotide sequence of (b), preferably at least 80% identical, more preferably at least 90% identical, and most preferred at least 95% identical;
- (e) a nucleic acid molecule which is degenerate with respect to the nucleic acid molecule of (d); or (f) a

nucleic acid molecule corresponding to the nucleic acid molecule of any one of (a) to (d) wherein T is replaced by U.

2. The nucleic acid molecule of claim 1, wherein the nucleic acid molecule is operably linked to a promoter that is native or heterologous to the nucleic acid molecule.

3. The nucleic acid molecule of claim 1, wherein said nucleic acid molecule is codon-optimized for expression in a eukaryotic cell, preferably a plant cell or an animal cell.

4. A vector encoding the nucleic acid molecule of claim 1.

5. A host cell comprising the nucleic acid molecule of claim 1 or being transformed, transduced or transfected with a vector encoding the nucleic acid molecule of claim 1.

6. The host cell of claim 5, wherein the host cell is a eukaryotic cell or a prokaryotic cell and is preferably a plant cell or an animal cell.

7. A plant, seed or a part of a plant, said part of a plant not being a single plant cell, or an animal comprising the nucleic acid molecule of claim 1 or being transformed, transduced or transfected with a vector encoding the nucleic acid molecule of claim 1.

8. A method of producing an RNA-guided DNA endonuclease comprising culturing the host cell of claim **5** and isolating the RNA-guided DNA endonuclease produced.

9. An RNA-guided DNA endonuclease encoded by the nucleic acid molecule of claim **1**.

10. A composition comprising the nucleic acid molecule of claim **1**, a vector encoding the nucleic acid molecule of claim **1**, a host cell comprising the nucleic acid molecule of claim **1** or being transformed, transduced or transfected with a vector encoding the nucleic acid molecule of claim **1**, a plant, seed, part of a cell or animal comprising the nucleic acid molecule of claim **1** or being transformed, transduced or transfected with a vector encoding the nucleic acid of claim **1**, an RNA-guided DNA endonuclease encoded by the nucleic acid molecule of claim **1** or a combination thereof.

11. The composition of claim **10**, wherein the composition is a pharmaceutical composition or a diagnostic composition.

12. A method for treating a disease in a subject or a plant by modifying a nucleotide sequence at a target site in the genome of the subject or plant, comprising administering a nucleic acid molecule of claim **1**, a vector encoding the nucleic acid molecule of claim **1**, a host cell comprising the nucleic acid molecule of claim **1** or being transformed, transduced or transfected with a vector encoding the nucleic acid molecule of claim **1**, a plant, seed, part of a cell or animal comprising the nucleic acid molecule of claim **1** or being transformed, transduced or transfected with a vector encoding the nucleic acid molecule of claim **1**, an RNA-guided DNA endonuclease encoded by the nucleic acid molecule of claim **1** or a combination thereof, to said subject or plant.

13. A method of modifying a nucleotide sequence at a target site in the genome of a cell comprising introducing into said cell

(i) a DNA-targeting RNA or a DNA polynucleotide encoding a DNA-targeting RNA, wherein the DNA-targeting RNA comprises:

(a) a first segment comprising a nucleotide sequence that is complementary to a sequence in the target DNA; and

(b) a second segment that interacts with an RNA-guided DNA endonuclease encoded by the nucleic acid molecule of claim **1**; and

(ii) the RNA-guided DNA endonuclease encoded by the nucleic acid molecule of claim **1**, or a nucleic acid molecule encoding an RNA-guided DNA endonuclease of claim **1**, or a vector encoding the nucleic acid molecule of claim **1**, wherein the RNA-guided DNA endonuclease comprises:

(a) an RNA-binding portion that interacts with the DNA-targeting RNA; and

(b) an activity portion that exhibits site-directed enzymatic activity.

14. The method of claim **13**, wherein the cell is not the natural host of a gene encoding said RNA-guided DNA endonuclease.

15. The method of claim **13**, wherein in case the RNA-guided DNA endonuclease and the DNA-targeting RNA are directly introduced into the cell they are introduced in the form of a ribonucleoprotein complex (RNP).

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